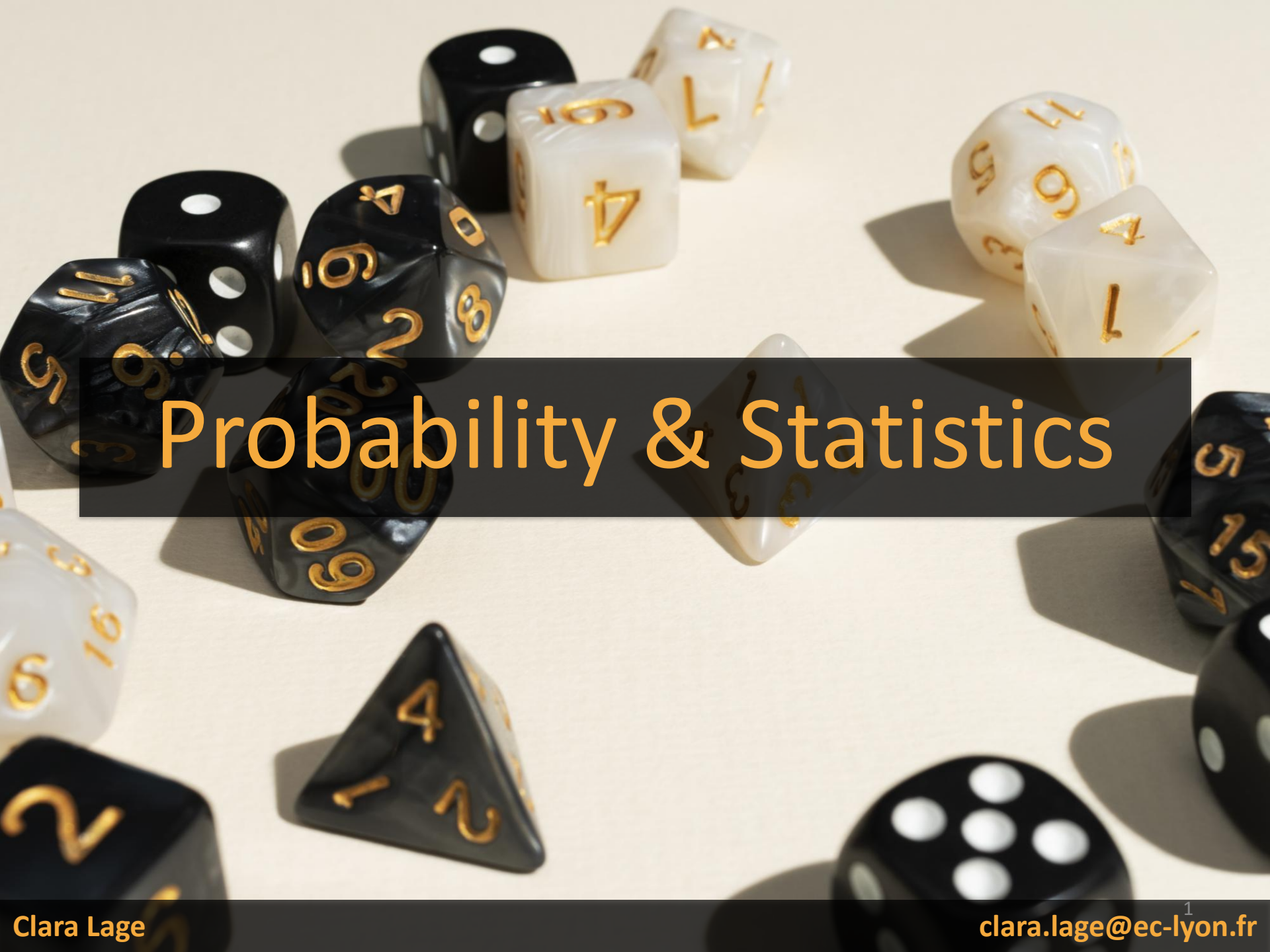


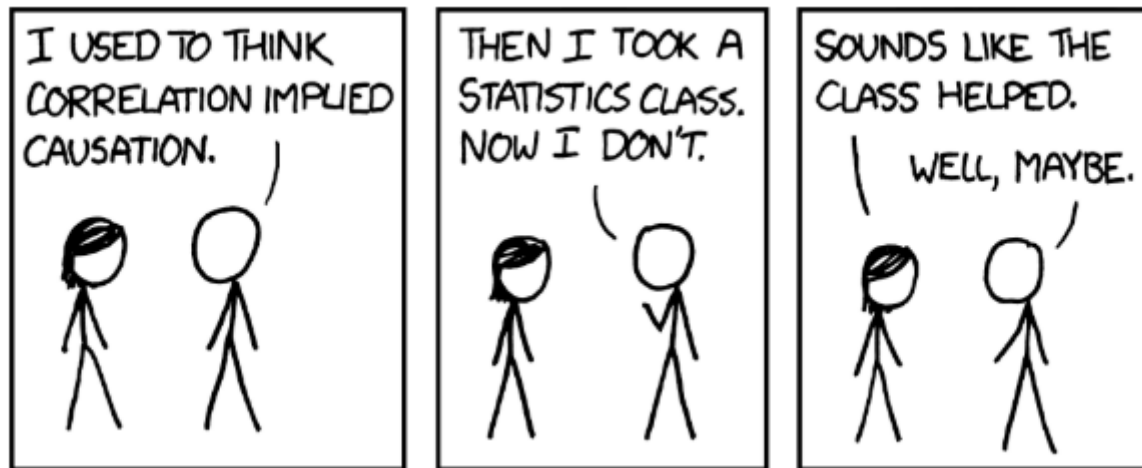


Probability & Statistics

Practical organization of the course

- Before each class, the corresponding slides will be available on Moodle.
- These slides should be used as study material for the course, along with the tutorial sheets, which contain the exercises you are expected to work on.
- Bring your computers to both the lecture and the tutorial (TD) classes, as we may use them at different moments.
- **Two exams:** a midterm exam during a tutorial session; and another at the end.
- Both exams may include a computational component carried out in Python. Therefore, it is important that you feel comfortable with this programming language throughout the course.
- During the exams, you may not use cell phones or computers, but you are allowed to bring a summary sheet containing anything from the course that you find helpful.
- **Asking questions is part of learning, so don't hesitate!**

Probability and statistics also help you understand XKCD comics ...



Descriptive Statistics

Class 1

Objective:

Processing, analyzing, and visualizing data —
and the moment data analysis turns into classical statistics

Exemples of simple data sets

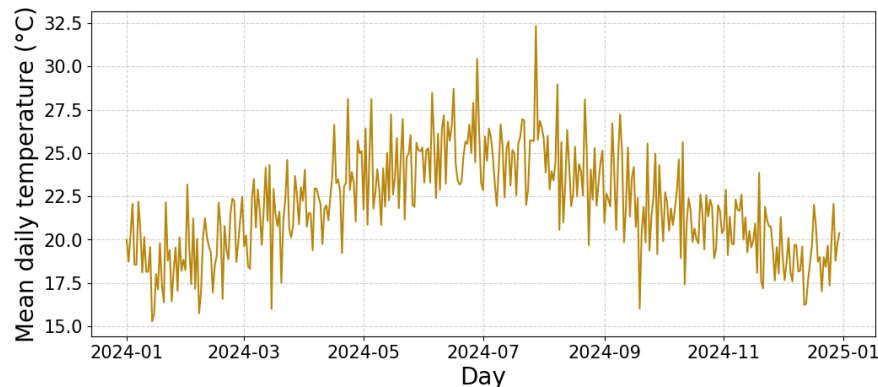
Students' grades on a test:

$$x = (x_1, \dots, x_{10}) = (7, 8, 15, 10, 12, 19, 18, 8, 6, 19)$$

Number of children per person in a workplace:

Number of Children	Frequency
1	12
2	16
3	8
4	3
5	1
6	1

Mean daily temperature in São Paulo:



Summary Statistics

Key statistical measures and visualization

If I simply give you the raw list:

17.2, 18.1, 21.5, 22.0, 19.7, 33.4, 34.1, 12.2, 11.9, 25.4, 27.1, ... (365 numbers)

You still cannot easily answer:

- Is this a hot or cold city?
- How much do temperatures vary?
- Are there extreme days?
- Is the temperature stable?
- Is there a clear seasonal pattern?

Having all the data does not mean understanding the data.

Approaches to understanding the data

Summary statistics

Data visualization

Mean and median

Mean:

Let $x_i, i \in \{1, \dots, n\}$ be the dataset. Then:

$$\bar{x} := \frac{1}{n}(x_1 + x_2 + \dots + x_n) = \frac{1}{n} \sum_{i=1}^n x_i$$

is the mean value of our dataset.

Example: The mean annual temperature in São Paulo is 22.01 °C

Median:

Consider $x_{\sigma(1)} \leq x_{\sigma(2)} \leq \dots \leq x_{\sigma(n)}$, for a permutation σ then the **median** is:

$$x_{\sigma(\frac{n+1}{2})}, \text{ if } n \text{ odd.} \quad \frac{x_{\sigma(\frac{n}{2})} + x_{\sigma(\frac{n}{2}+1)}}{2}, \text{ if } n \text{ even.}$$

Example: The median annual temperature in São Paulo is 21.92 °C

Question: When mean and median are very different? Ex: Distribution of incomes

Measures of spread

Variance:

Let $x_i, i \in \{1, \dots, n\}$ be the data set. Then:

$$\text{Var}(x) = \frac{(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + \dots + (x_n - \bar{x})^2}{n - 1} = \frac{1}{n - 1} \sum_{i=1}^n (x_i - \bar{x})^2$$

is the variance of our dataset.

Example: The variance of São Paulo temperature is 8.5

Standard deviation:

The standard deviation is given by:

$$\text{sd}(x) = \sqrt{\text{Var}(x)}$$

Example: The standard deviation of São Paulo temperature is 8.5

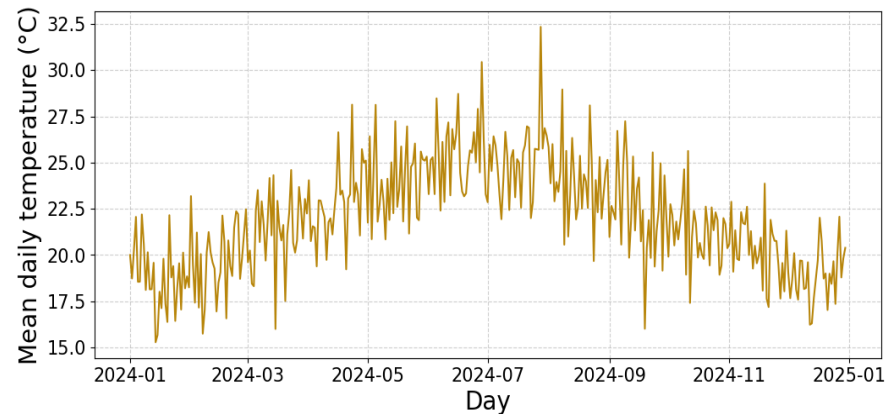
Question: If the variance equals zero, what can we conclude about the data?

Data visualization

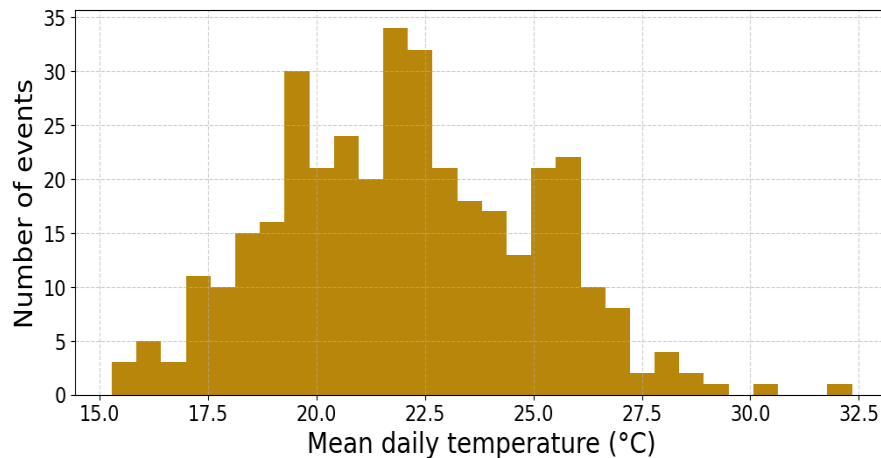
Histogram

Example dataset: Mean daily temperature in the city of São Paulo over one year.

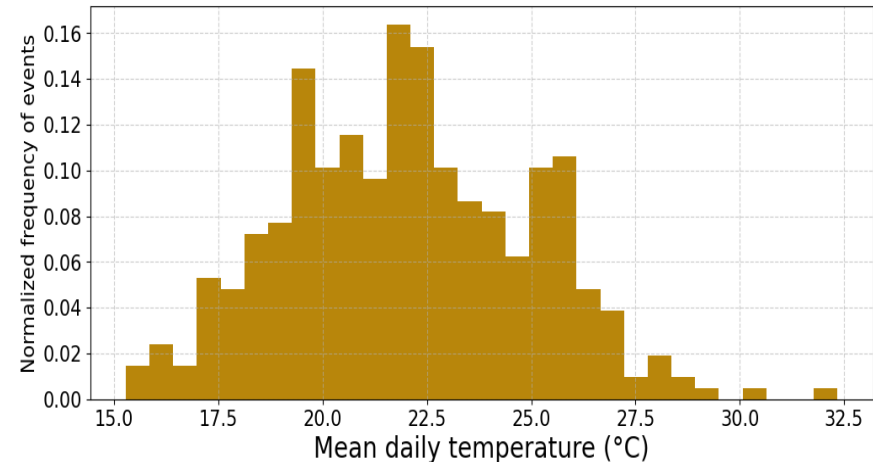
As a time-series



As a histogram



As a density histogram



Histogram

Discussion

- What is the range of values (minimum and maximum) in the time series?
- Does the histogram show *when* the values occurred? Explain.
- What information is lost when we turn a time series into a histogram?
- Is the histogram symmetric, skewed to the left, or skewed to the right?
- Are there any signs of outliers? How can they be seen in the histogram?

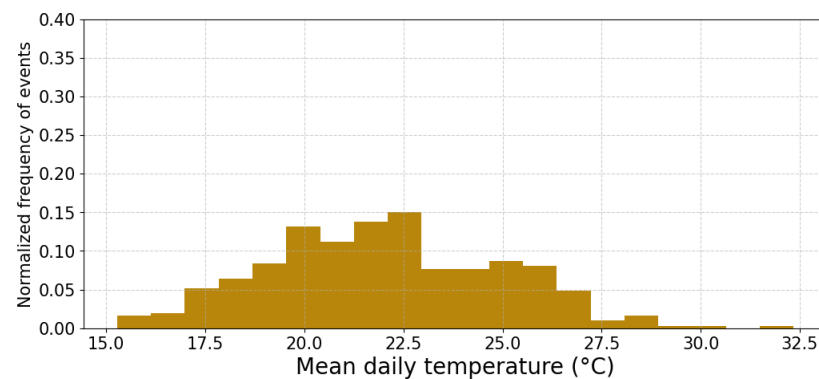
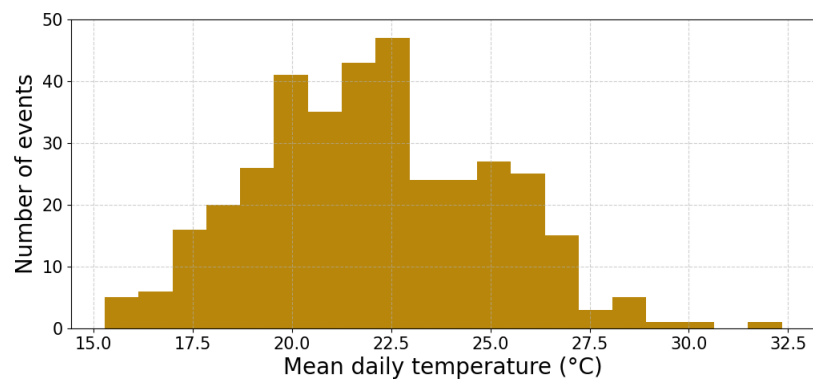
Practical Exercise: Exploring Histograms with Temperature Data

- Use Python to open a CSV file containing temperature values. `pandas.read_csv()`
- Plot a regular histogram.
- Then plot another histogram using `plt.hist()` with `density=True`.
- Apply the y-axis limit to both histograms
- Try different numbers of bins (e.g., 5, 20, 50)
- When you use fewer bins, what is the trend of the regular histogram and the density histogram? Explain your observation.

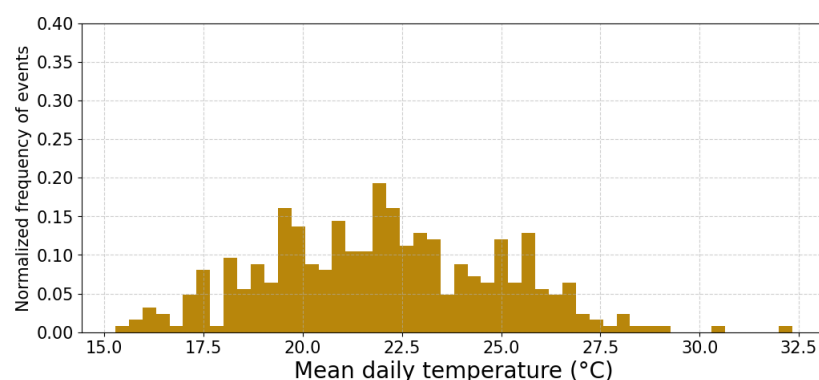
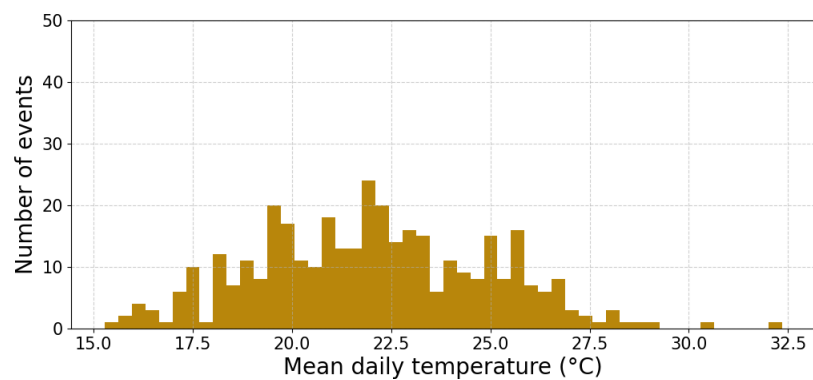
density = FALSE

density = TRUE

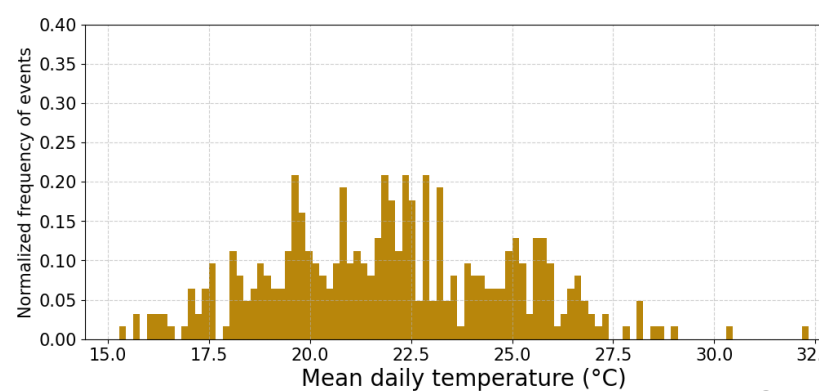
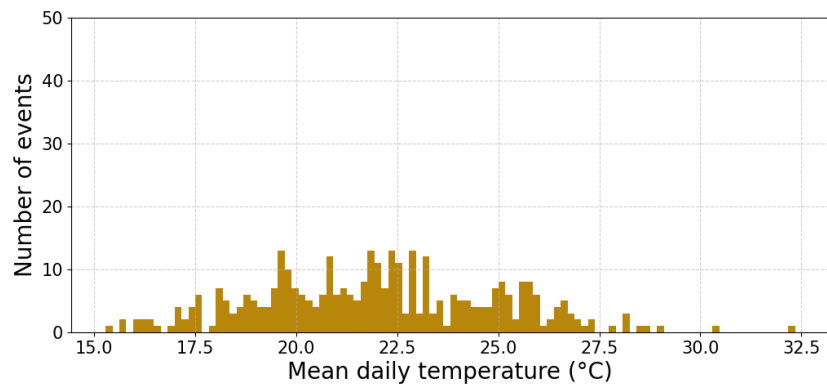
bins = 30



bins = 50



bins = 100



Back to summary statistics

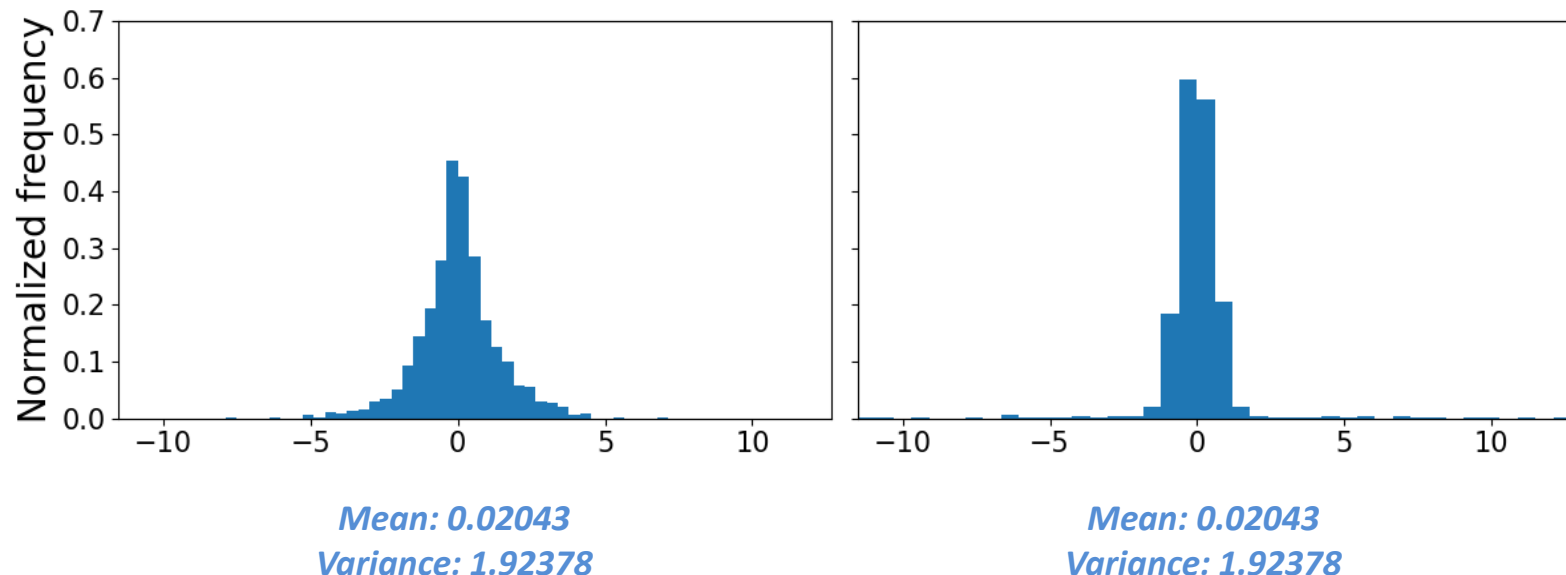
Other measures of spread

Why use other measures of spread?

Both the mean and the variance may be insufficient to represent a dataset, because:

- They are excessively affected by extreme values;
- We have no idea about the symmetry or asymmetry of the data distribution.

Example: Different datasets with the same mean and variance



Empirical Cumulative Distribution Function

Definition

Given a dataset of n observations $x_i, i \in \{1, \dots, n\}$, the **Empirical Cumulative Distribution Function (ECDF)** is defined as:

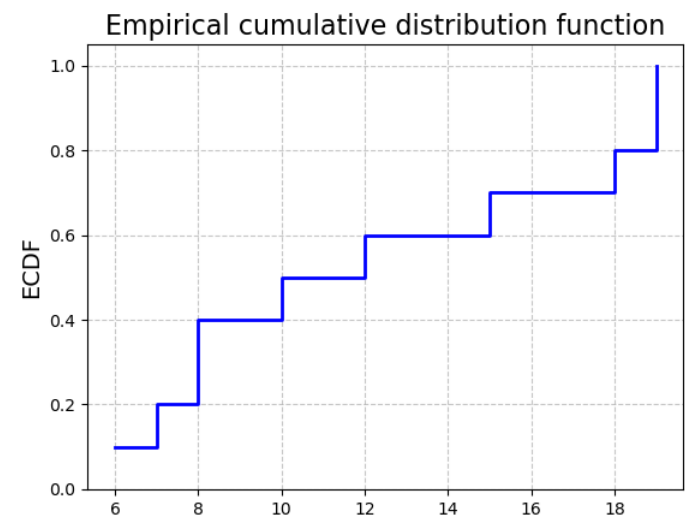
$$\hat{F}_n(z) = \frac{1}{n} \sum_{i=1}^n I(x_i \leq z)$$

where $I(\cdot)$ is the indicator function, which equals 1 if the condition is true and 0 otherwise.

The ECDF at a point x is the proportion of observations less than or equal to x .

Example: Students grades in a test

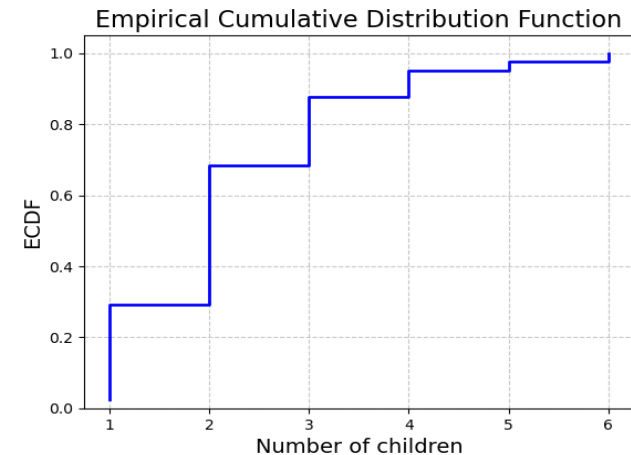
$$\begin{aligned} x &= (x_1, x_2, \dots, x_{10}) \\ &= (7, 8, 15, 10, 12, 19, 18, 8, 6, 19) \end{aligned}$$



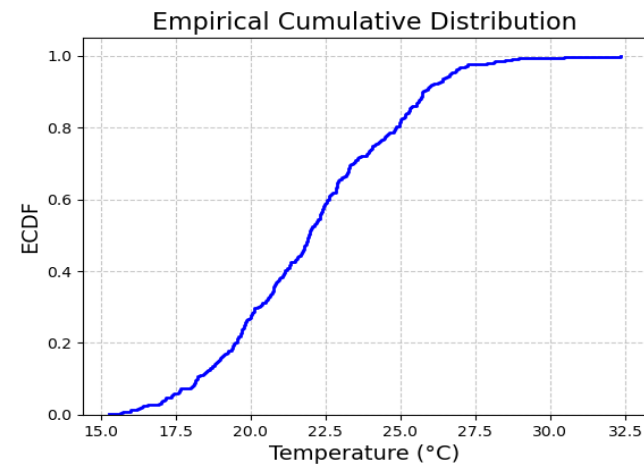
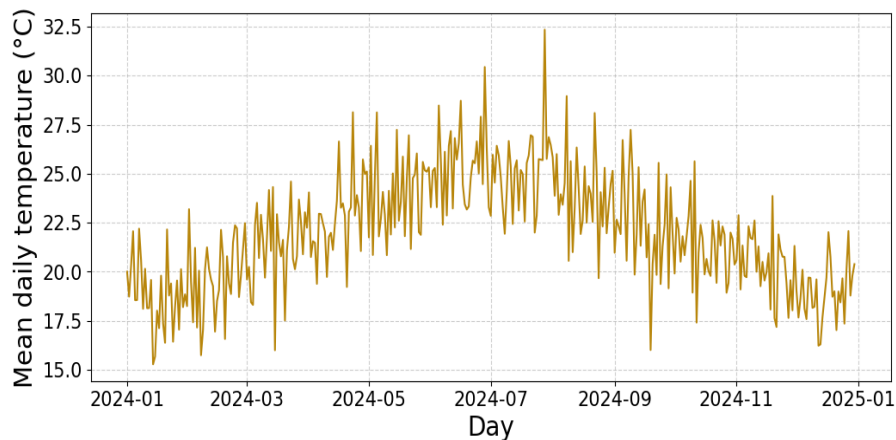
Examples

Example: Number of children per person in a workplace

Number of Children	Frequency
1	12
2	16
3	8
4	3
5	1
6	1



Example: Mean daily temperature in São Paulo



Quantiles

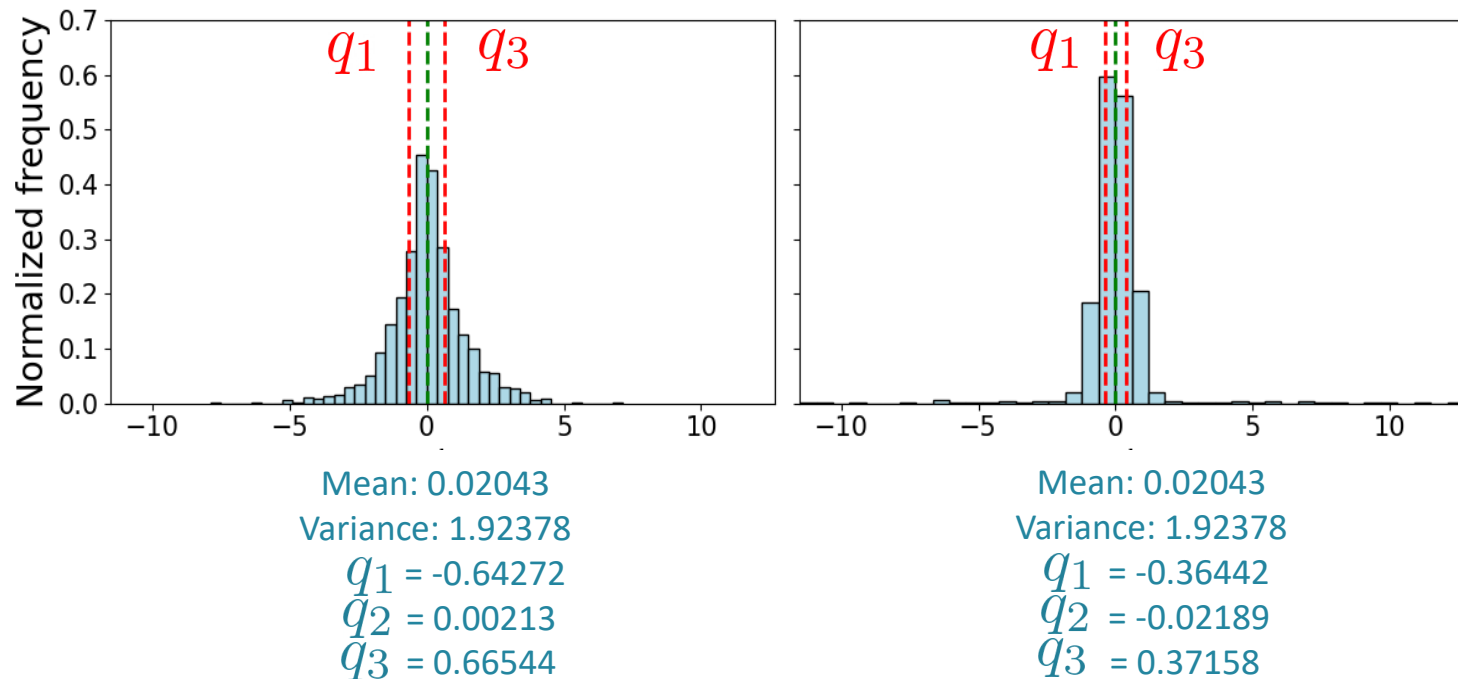
Definition

The **q-quantile** is defined as :

$$Q(q) = \inf\{z \in \mathbb{R} : \hat{F}_n(z) \geq q\}$$

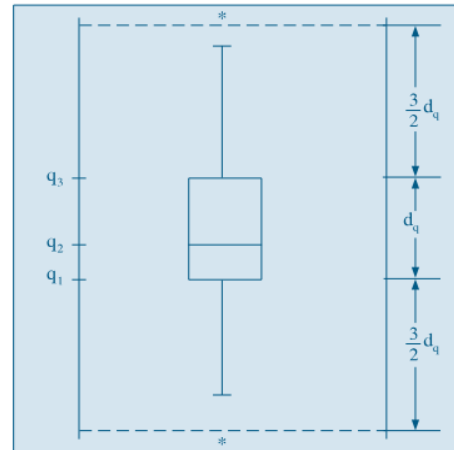
Notation: $q_1 = Q(0.25)$; $q_2 = Q(0.5)$; $q_3 = Q(0.75)$

Example



Back to data visualization

Boxplot



q_1 quantile 0.25

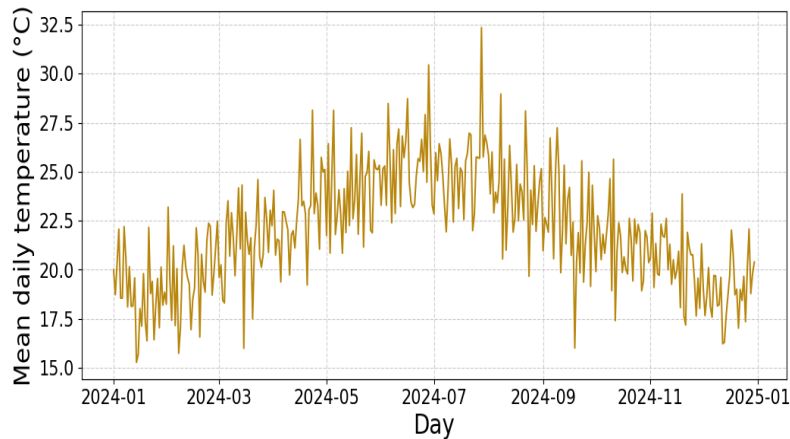
q_2 quantile 0.5

q_3 quantile 0.75

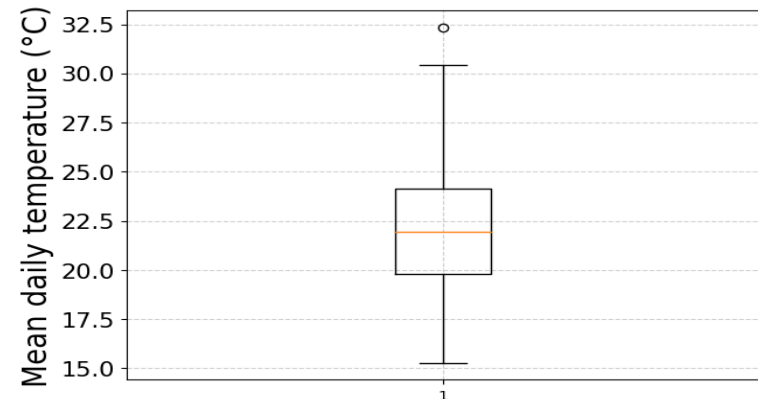
$$d_q = q_3 - q_1$$

Example : Mean daily temperature in the city of São Paulo over one year.

As a time-series



As a boxplot

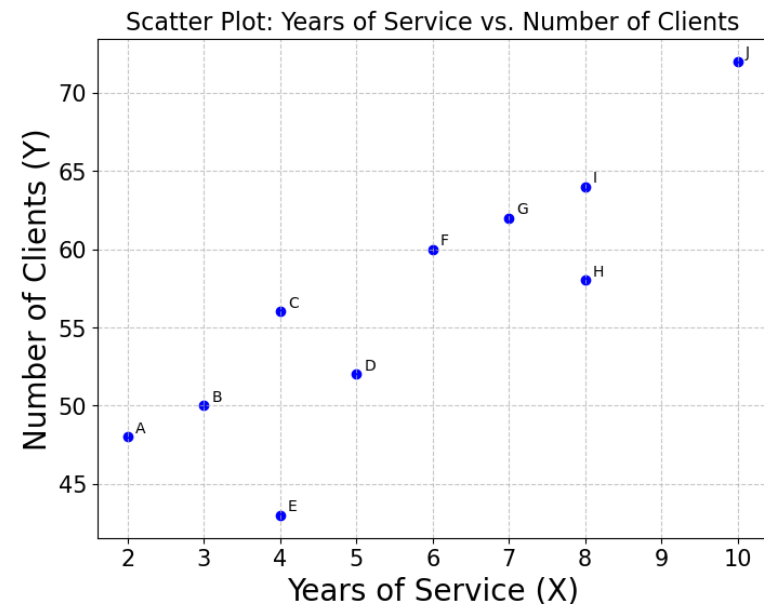


Relation between two datasets

First example

Example: Dataset (x,y) of years of service x number of clients

Agent	Years of Service (x)	Number of Clients (y)
A	2	48
B	3	50
C	4	56
D	5	52
E	4	43
F	6	60
G	7	62
H	8	58
I	8	64
J	10	72

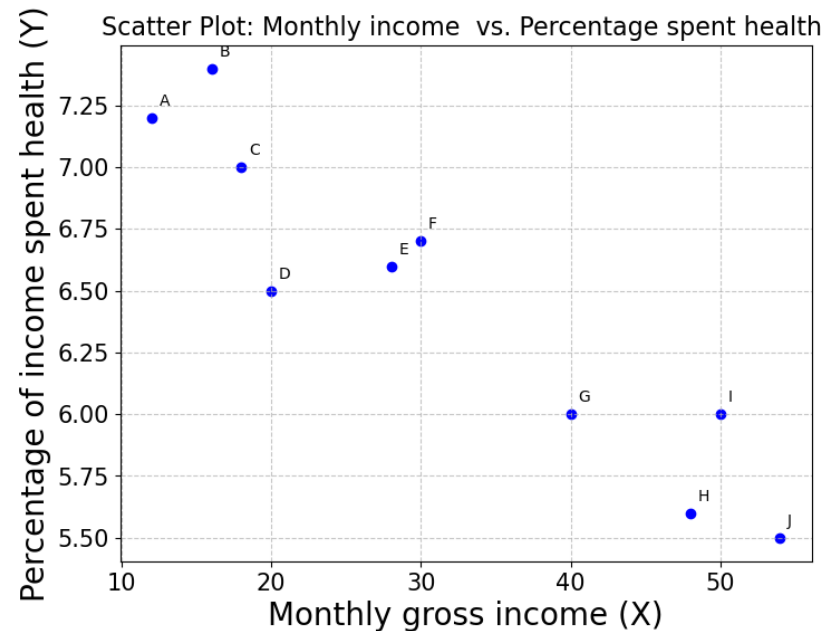


Question: How can we measure the relation between the two variables ?

Second example

Example: Monthly gross income (x) and percentage of income spent on health (y) for a set of families.

Family	x	y
A	12	7.2
B	16	7.4
C	18	7.0
D	20	6.5
E	28	6.6
F	30	6.7
G	40	6.0
H	48	5.6
I	50	6.0
J	54	5.5

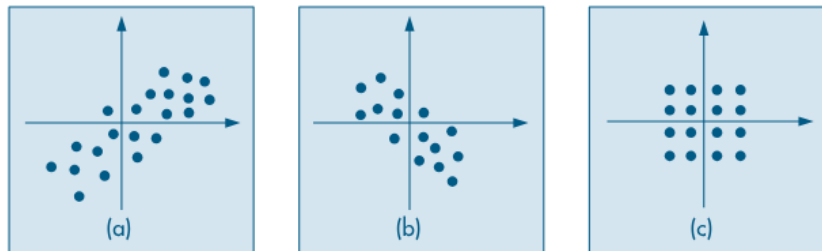


Question: How can we measure the relation between the two variables ?

Linear correlation

Factors to be considered in defining correlation

- Center the data around (0,0) using the mean: $(x - \bar{x}, y - \bar{y})$



- When $(x - \bar{x}, y - \bar{y})$ have the same sign they seem to be more linear correlated
- Divide by the dataset size (having more data should not necessarily imply more correlation)

Back to an example

Example: Dataset (x,y) of years of service x number of clients

Table 5: Calculation of standardized values and products for correlation.

Agent	x	y	$x - \bar{x}$	$y - \bar{y}$	$\frac{x - \bar{x}}{\text{sd}(x)} = z_x$	$\frac{y - \bar{y}}{\text{sd}(y)} = z_y$	$z_x \cdot z_y$
A	2	48	-3.7	-8.5	-1.54	-1.05	1.617
B	3	50	-2.7	-6.5	-1.12	-0.80	0.846
C	4	56	-1.7	-0.5	-0.71	-0.06	0.043
D	5	52	-0.7	-4.5	-0.29	-0.55	0.160
E	4	43	-1.7	-13.5	-0.71	-1.66	1.179
F	6	60	0.3	3.5	0.12	0.43	0.052
G	7	62	1.3	5.5	0.54	0.68	0.367
H	8	58	2.3	1.5	0.95	0.19	0.181
I	8	64	2.3	7.5	0.95	0.92	0.874
J	10	72	4.3	15.5	1.78	1.91	3.400
Total	57	565	0	0			8.769

$$\bar{x} = 5.7, \quad \text{sd}(x) = 2.41, \quad \bar{y} = 56.5, \quad \text{sd}(y) = 8.11$$

Linear correlation and covariance

Definition of correlation

Given a dataset $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$, the correlation coefficient is defined as:

$$\text{corr}(x, y) = \frac{1}{n-1} \sum \left(\frac{x_i - \bar{x}}{\text{sd}(X)} \right) \left(\frac{y_i - \bar{y}}{\text{sd}(Y)} \right) = \frac{\sum x_i y_i - (n-1) \bar{x} \bar{y}}{\sqrt{(\sum x_i^2 - (n-1) \bar{x}^2) (\sum y_i^2 - (n-1) \bar{y}^2)}}$$

Definition of covariance

Given a dataset $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$, the covariance is defined as:

$$\text{cov}(x, y) = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \frac{\sum_{i=1}^n x_i y_i - (n-1) \bar{x} \bar{y}}{n-1}.$$

Question: $\text{corr}(x, x) = ?$; $\text{cov}(x, x) = ?$

Back to data visualisation

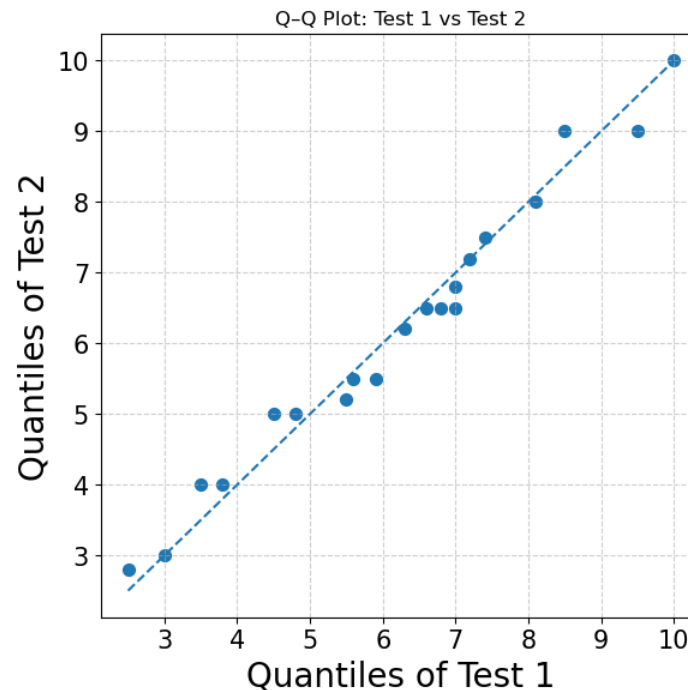
Comparing quantiles: qxq plot

Definition of q x q plot

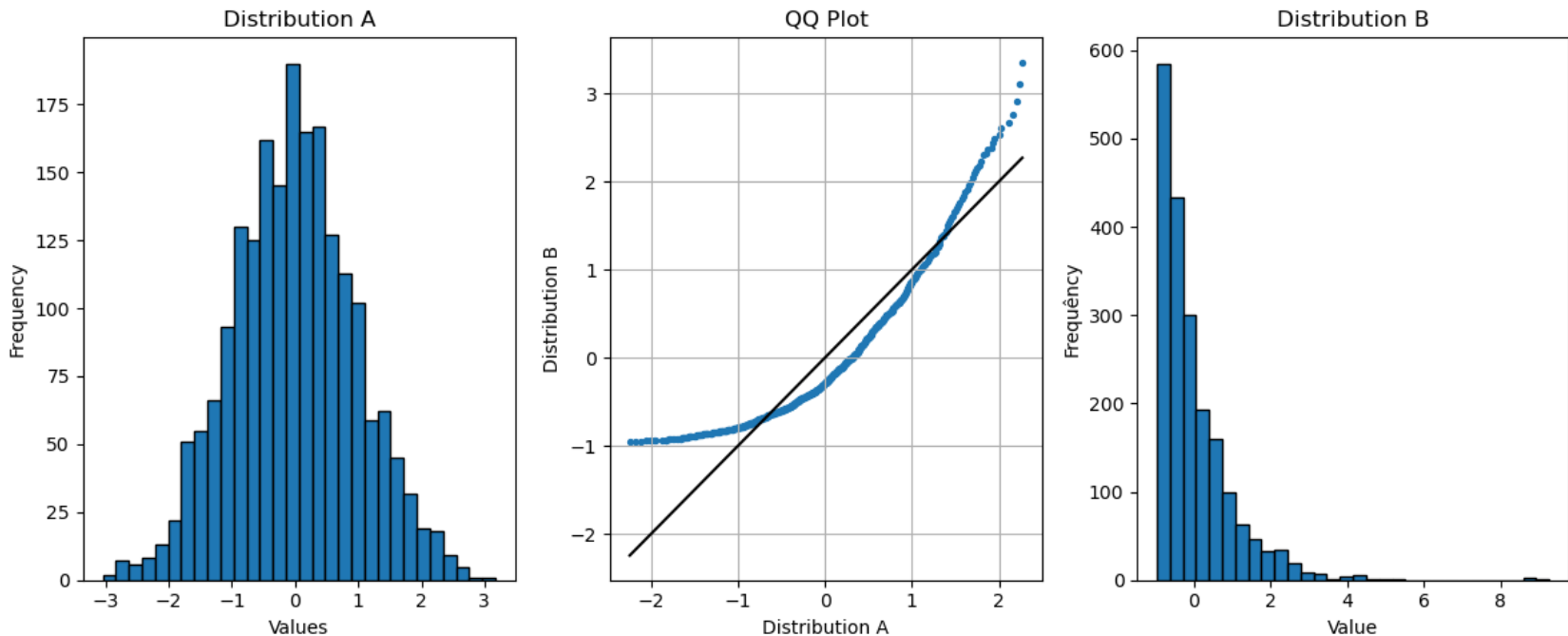
Given two datasets: $x = (x_1, \dots, x_n)$ and $y = (y_1, \dots, y_m)$, a q x q plot is a graphical tool that plots their corresponding quantiles against each other.

Example: Student's grades in two tests

Student	Test 1	Test 2
1	8.5	8.0
2	3.5	2.8
3	7.2	6.5
4	5.5	6.2
5	9.5	9.0
6	7.0	7.5
7	4.8	5.2
8	6.6	7.2
9	2.5	4.0
10	7.0	6.8
11	7.4	6.5
12	5.6	5.0
13	6.3	6.5
14	3.0	3.0
15	8.1	9.0
16	3.8	4.0
17	6.8	5.5
18	10.0	10.0
19	4.5	5.5
20	5.9	5.0

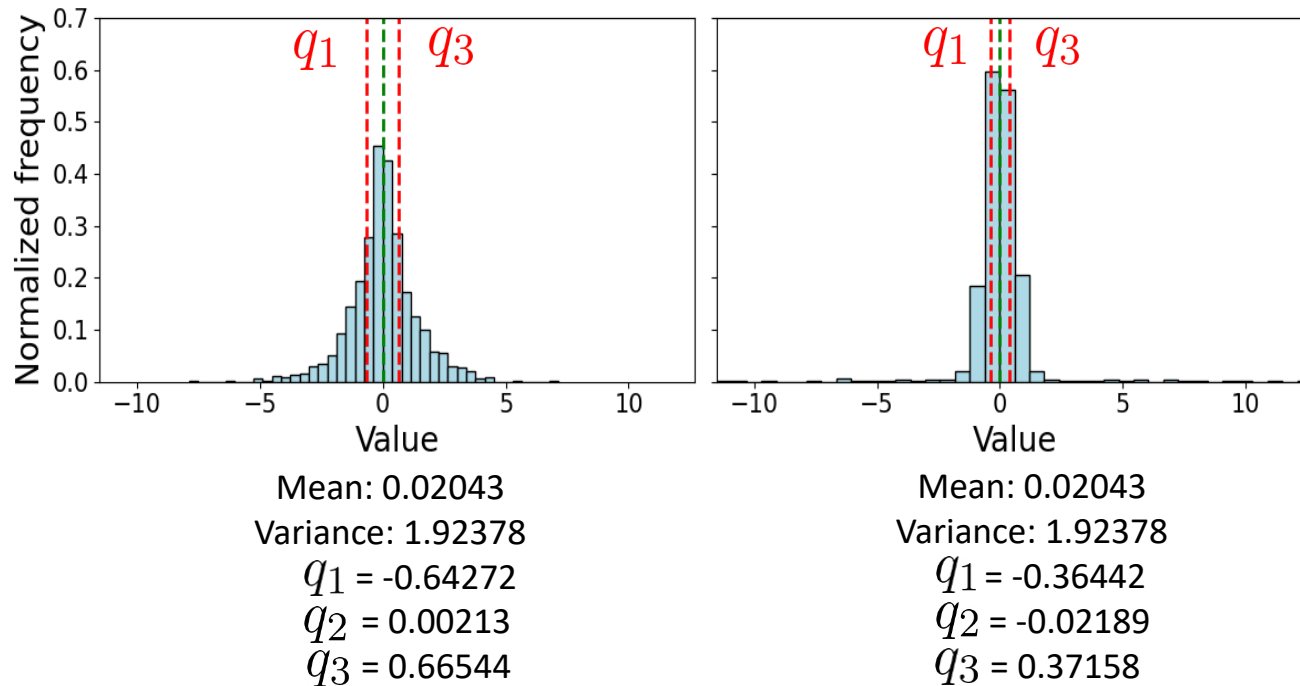


Comparing quantiles : Example

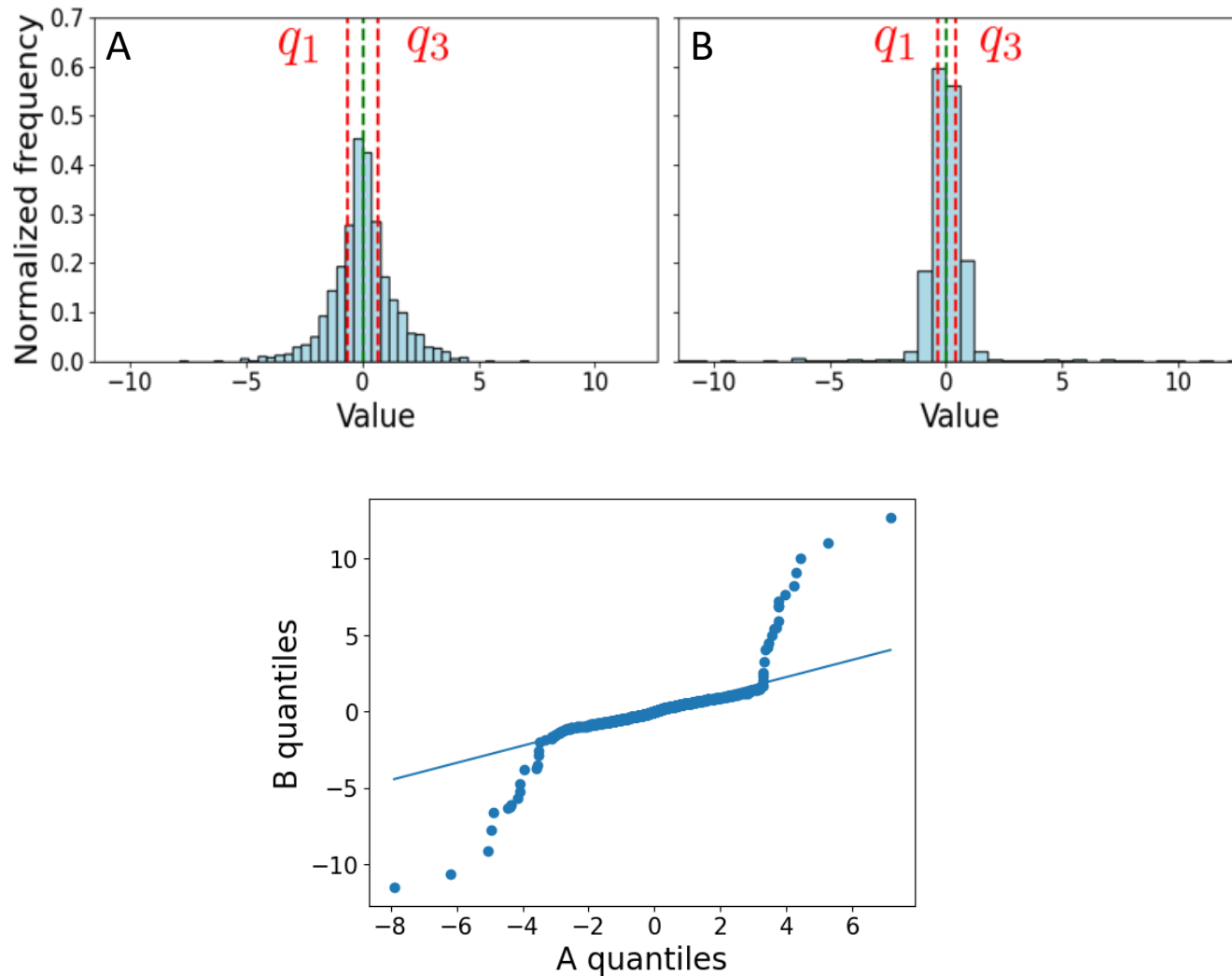


Back to out example

Example



Back to out example



When it turns to statistics

Class 1-2

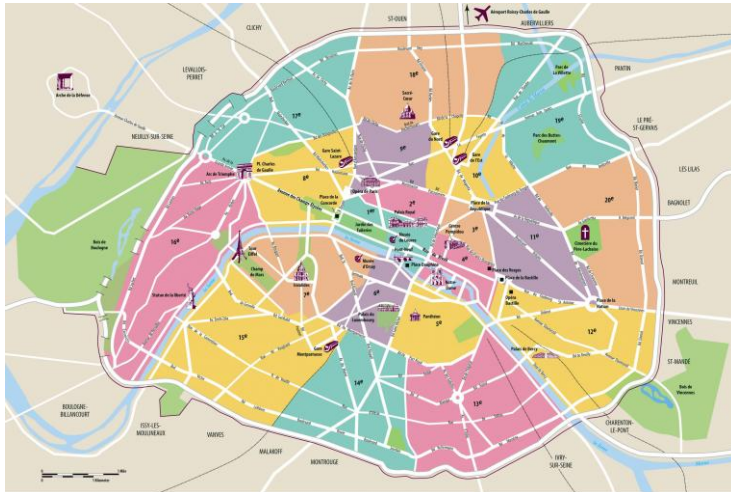
Objective

Descriptive statistics allow us to analyze existing data, but to answer certain questions, we need probabilistic models.

Election pool

A change in perspective: So far, we have visualized and analyzed some useful measures to understand data. But what if these data represent not an entire population, but only a sample?

Example: Election pool



Candidate	Number of respondents
Candidate A	362
Candidate B	638

Table 2: Election poll results with 1,000 respondents.

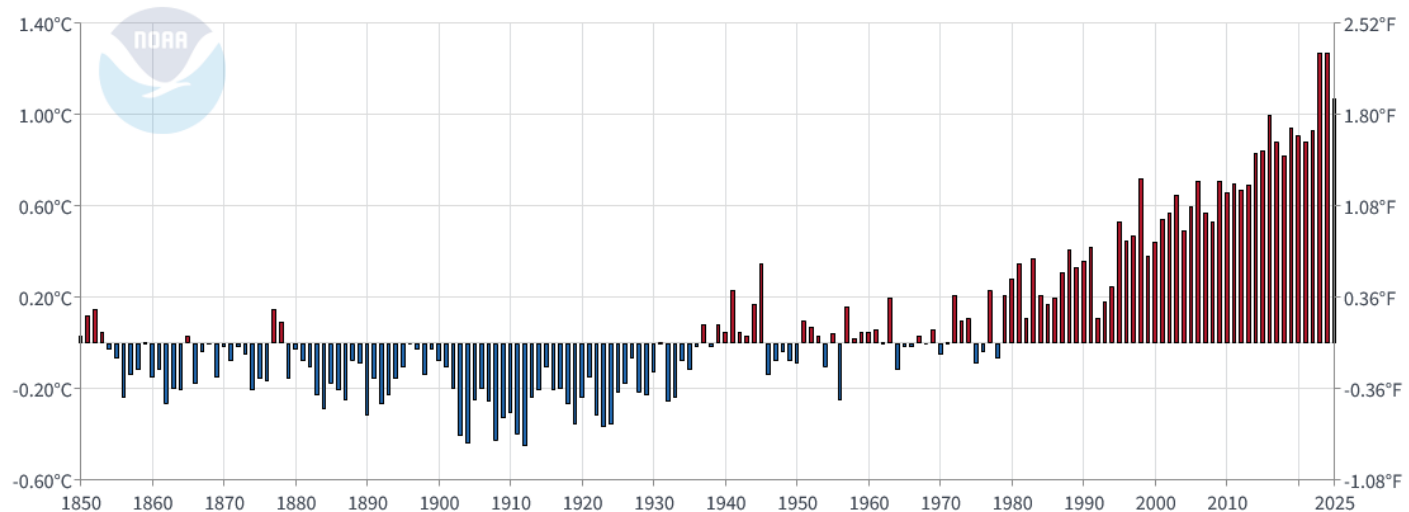
Statistical question: What is the probability that candidate B will win the election?

Climate change illustration

Graphic of mean anomaly temperature by year: $A(t) = T_{\text{obs}}(t) - \overline{T}_{\text{baseline}}$

Global Land and Ocean Average Temperature Anomalies

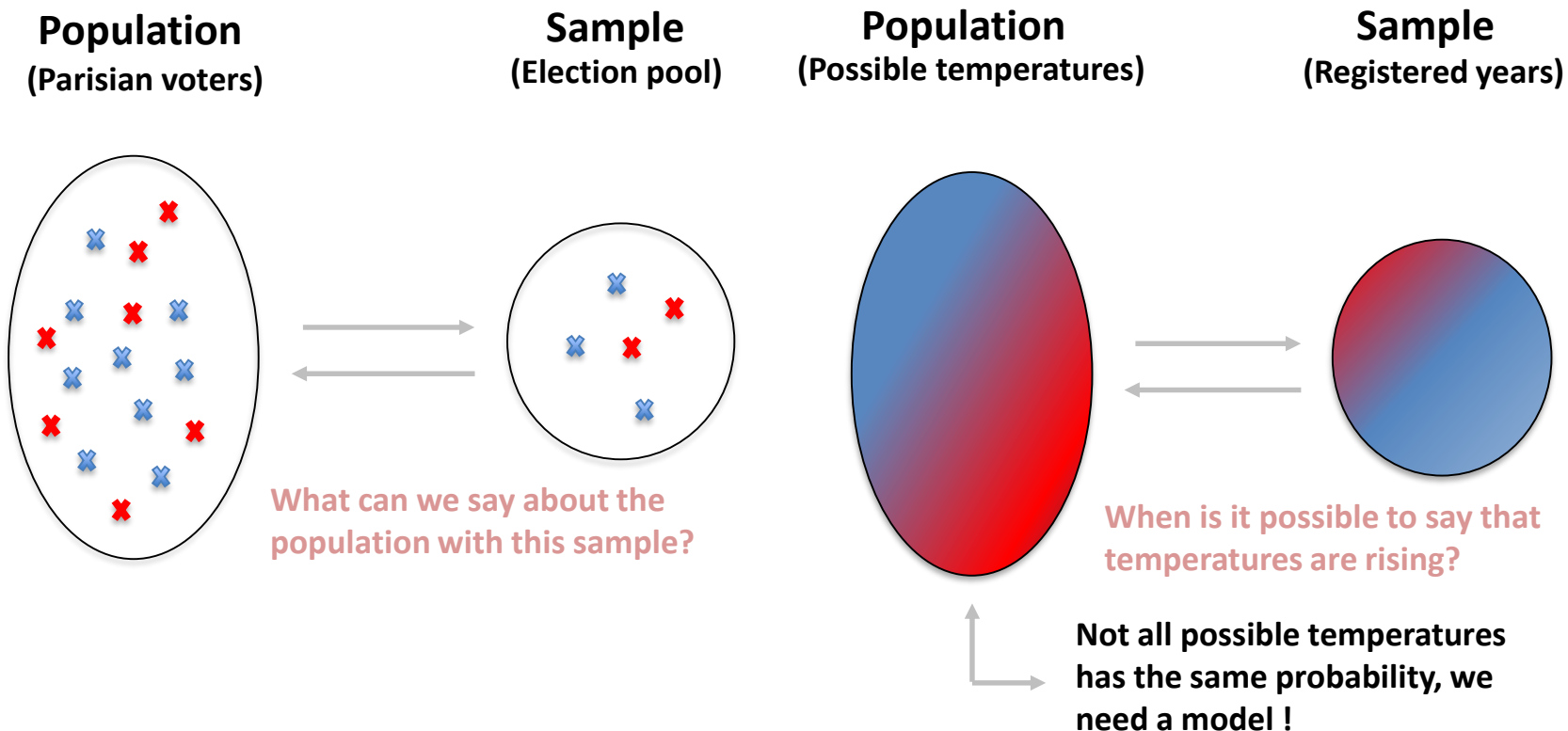
August



Statistical question: How many years we need to observe that the temperature is rising ?

When it turns to statistics

With suitable assumptions, we can build a theoretical model that reasonably reproduces the observed frequency distribution. Such models are called probabilistic models and will be the focus of this and the following chapters



Basic Probability

Class 2

Objective

Understanding the formalism of probability and how it can guide scientific reasoning and conclusions

Probability models

To understand how to ask statistical questions, we first need to understand what a *probability model* is.

What is the difference between a probability model and a data-driven model?

A **probability model** is defined *before* observing data. It is based on **assumptions** about how the world behaves.

Exemple: Rolling a die

Let de **sample space** be $S = \{1, 2, 3, 4, 5, 6\}$ and:

$$P(\{i\}) = \frac{1}{6}, i \in \{1, \dots, 6\}$$

We assume this model and use it to predict or explain outcomes.

Sample space and probability

Sample space

To build a theoretical model, the first step is to have a space containing all possible outcomes of an experiment.

- **Example:** Rolling a die once $\rightarrow S = \{1, 2, 3, 4, 5, 6\}$
- **Example:** Rolling a die twice $\rightarrow S = \{(1, 1), (1, 2), \dots, (6, 6)\}$

Probability

- A probability is a function $P : \mathcal{F} \rightarrow [0, 1]$, where the elements of $\mathcal{F}$ are subsets of S
- **Example:** Rolling a fair die twice:

$$P(\{(i, j)\}) = \frac{1}{36} \quad i, j \in \{1, \dots, 6\}$$

Remember: The probability of the sample space is always equal to 1!!

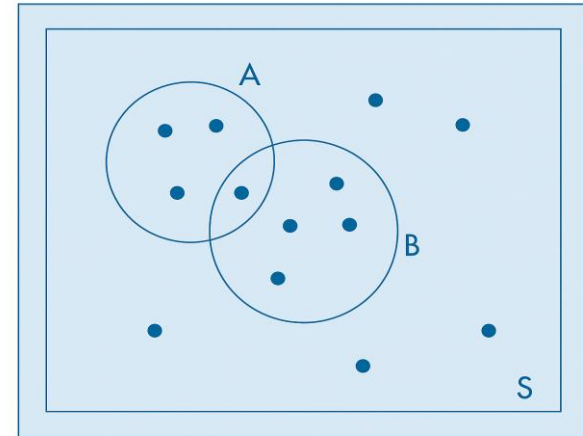
$$P(S) = 1$$

Events

Event E

Any set (or subset) of outcomes from the sample space that we are interested in observing: $E \subset S$

If S is the set of dots, A and B are different events with a nonempty intersection.



Examples of events

- $S = \{\text{all 52 cards in a standard deck}\}$
Example of Events: E_1 - drawing a heart; E_2 - drawing a face card (Jack, Queen, King).
- $S = \{\text{Candidate A, Candidate B, Candidate C}\}$
Example of Events: E_1 - the respondent chooses Candidate A ; E_2 - the respondent chooses candidate B or C.

Probability properties

Probability properties

- Probability is a function that assigns values in $[0, 1]$
- $P(S) = 1$ and $P(\emptyset) = 0$
- For any event $A \subset S$: $0 \leq P(A) \leq 1$
- For two events $A, B \subset S$: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Complementarity

A^c = 'not A', all outcomes in S that are not in A , and $P(A^c) = 1 - P(A)$

Example: $P(\text{rolling an odd number}) = 1 - P(\text{even}) = 1 - 1/2 = 1/2$

Independence of events

Definition

A and B are independent if: $P(A \cap B) = P(A) \cdot P(B)$

Example of independence: $S = \{\text{Possible diseases}\}$

A: “have pollen allergy”

B: “be myopic”

20% of people have a pollen allergy; 25% are myopic: $P(A) = 0.2, P(B) = 0.25$

If the two conditions are independent, then: $P(A \cap B) = P(A) \times P(B) = 0.05$

Only **5%** of people have **both** conditions.

Example of non-independent events: $S = \{\text{Weather configuration}\}$

A: “It is raining”

B: “The sky is completely overcast.”

Probability of rain on a typical day: $P(A) = 0.1$

Probability that the sky is fully overcast on a typical day: $P(B) = 0.2$

$$P(A \cap B) > P(A) \times P(B) = 0.02$$

Conditional probability

Probability of A given B

A occurs given that B has occurred

Example 1 — Weather

A: “It is raining”

B: “The sky is cloudy”

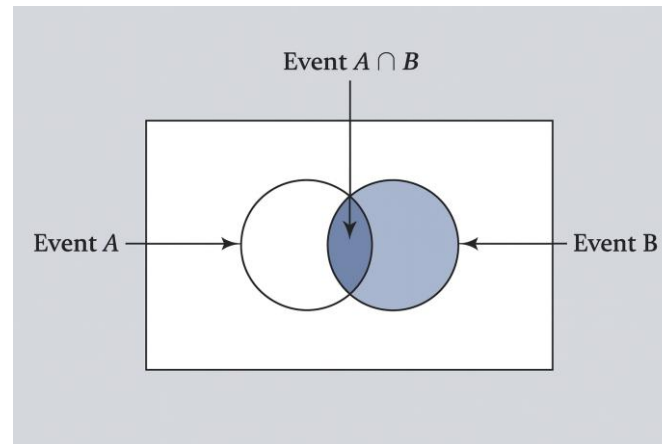
A | B: “It is raining given that the sky is cloudy.”

Example 2 — School

A: “The student passed the exam.”

B: “The student studied.”

A | B: “The student passed the exam given that they studied.”



Law of Total Probability

$$\{A_1, A_2, \dots, A_n\}$$

$$\bigcup_{i=1}^n A_i = S, \quad A_i \cap A_j = \emptyset, \quad \text{for all } i, j \in \{1, \dots, n\}$$



$$P(B) = \sum_{i=1}^n P(B | A_i) \cdot P(A_i)$$

Probability of A given B

Consider a medical test for a rare disease. Define the events:

A: “the person has the disease”

B: “the test result is positive”

What is usually **easy to know** is:

- $P(B|A)$: the probability that the test is positive given that the person has the disease.

But what the doctor **actually wants to know** is:

- $P(A|B)$: the probability that the person has the disease given that the test is positive.

Challenge: How to relate $P(B|A)$ and $P(A|B)$?

Bayes Formula

Definition:

- $P(A|B)$: probability of A given B
- $P(B|A)$: probability of B given A

Bayes formula

$$P(A | B) = \frac{P(B | A) P(A)}{P(B)}$$

Example: Medical context (given information)

- A: patient has the disease
- B: test is positive
- $P(A) = 10\%$ (0.1) — disease prevalence -> spread of disease inside population
- $P(B|A) = 99\%$ (0.99) — sensitivity -> pharmaceutical lab test
- $P(B|A^c) = 5\%$ (0.05) — false positive rate ->

Medical example

Example: Medical context (given information)

- A: patient has the disease
- B: test is positive
- $P(A) = 10\%$ (0.1) — disease prevalence
- $P(B|A) = 99\%$ (0.99) — sensitivity
- $P(B|A^C) = 5\%$ (0.05) — false positive rate

Computing the probability of having the disease with a positive test

Law of Total Probability

$$P(B) = P(B|A)P(A) + P(B|A^C)P(A^C)$$
$$P(B) = 0.99 \times 0.1 + 0.05 \times 0.9 = 0.1465$$

Bayes theorem

$$P(A|B) = \frac{0.99 \times 0.1}{0.1465} = 0.6768$$

Results

A 68% of chance of having the disease.

Medical example

Example: Medical context (given information)

- A: patient has the disease
- B: test is positive
- $P(A) = 1\%$ (0.01) — disease prevalence
- $P(B|A) = 99\%$ (0.99) — sensitivity
- $P(B|A^C) = 5\%$ (0.05) — false positive rate

Computing the probability of having the disease with a positive test

Law of Total Probability

$$P(B) = P(B|A)P(A) + P(B|A^C)P(A^C)$$

$$P(B) = 0.99 \cdot 0.01 + 0.05 \cdot 0.99 = 0.059$$

Bayes theorem

$$P(A|B) = \frac{0.99 \times 0.01}{0.059} = 0.16$$

Conclusion

Even with a very sensitive test, the actual probability is low when the disease is rare

Discrete vs. continuous sample spaces

Discrete sample set

Discrete sample spaces are those whose outcomes are finite: $S = \{s_1, \dots, s_n\}$ or can be counted: $S = \{s_1, s_2, \dots\}$. In this cases:

- The probability is a function $p : S \rightarrow [0, 1]$ that attributes a number to each $s \in S$.
- The sum of all probabilities is 1:

$$\sum_{i=1}^n P(\{s_i\}) = 1, \text{ or } \sum_{i=1}^{\infty} P(\{s_i\}) = 1$$

Another example:

- Number of tosses until the first heads appears: $S = \{1, 2, 3, \dots\}$

$$P(\{k\}) = \left(\frac{1}{2}\right)^k, \quad k = 1, 2, 3, \dots$$

- Number of mutations until a gene undergoes a specific change.

Examples of density

An entrepreneur plans to assemble a product consisting of a **sphere** and a **cylinder**, purchased from two different suppliers (A and B).

Each component (sphere or cylinder) can be classified as **good**, **long**, or **short**, depending on whether its measurement meets, exceeds, or falls below the specification. After assembly we have the following possibilities:

- If any component is short, the set is **irrecoverable**.
- If one component is long, it represents an **additional cost**.
- If two components are long, it represents an extra **additional cost**.
- If both components are good, it represents the **maximal profit**.

Product	Factory A Cylinder	Factory B Sphere
Within specifications Good (G)	0.8	0.7
Above specifications Long (L)	0.1	0.2
Below specifications Short (S)	0.1	0.1

Question: Which are S and P ?

Continuous sample set

Imagine we want to model the following experiment: choosing a real number at random in the interval $[0, 1]$ with all values equally likely.

- $S = [0, 1]$.
- Any $x \in [0, 1]$ must have probability 0.
- The probability of the interval $[0, 1/2]$ must be 0.5
- More generally, the probability of the intervals $[a, b] \subset [0, 1]$ must be $b - a$

We use a **density function** f such that:

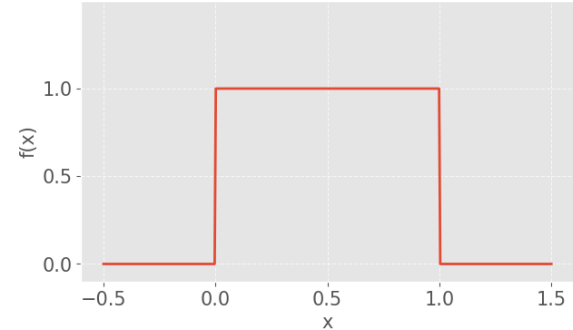
$$\text{For any event } A, \quad P(A) = \int_A f(x)dx$$

Question: Does this definition satisfy the probability properties?

Examples of density

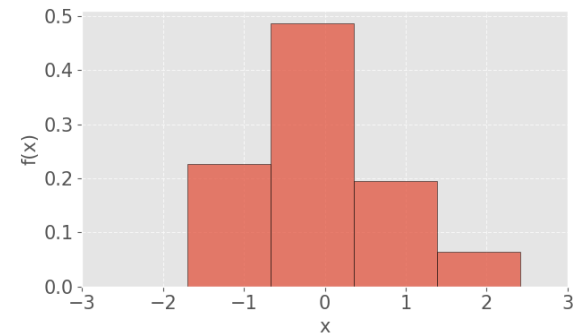
- Uniform distribution

$$f(x) = \begin{cases} 1, & \text{se } x \in [0, 1], \\ 0, & \text{otherwise.} \end{cases}$$



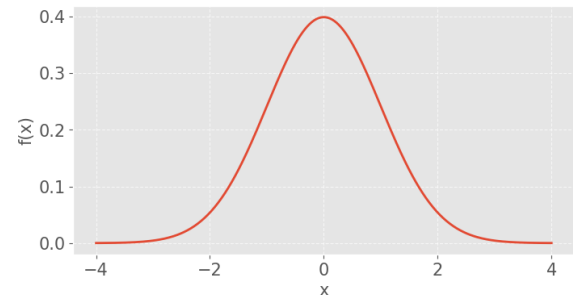
- Density histograms:

$$f(x) = \begin{cases} 0.2, & x \in [-1.7, -0.7), \\ 0.5, & x \in [-0.7, 0.4), \\ 0.2, & x \in [0.4, 1.4), \\ 0.1, & x \in [1.4, 2.4), \\ 0, & \text{Otherwise.} \end{cases}$$



- Continuous density

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$



Discrete distributions

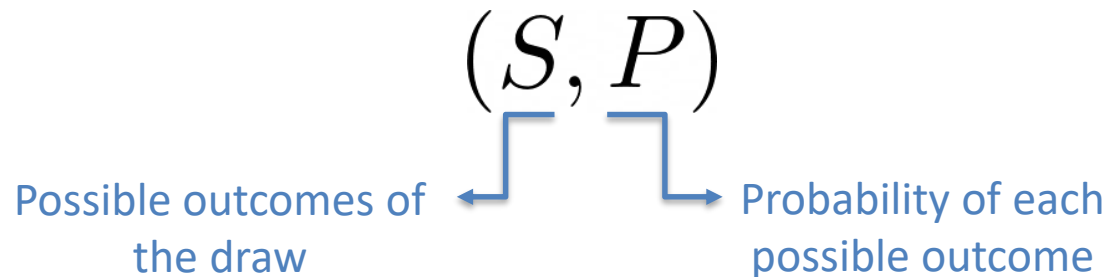
Class 3

Objective:

Understanding discrete distributions and their key metrics.
Understanding investments and risks

Random variable

Consider a random draw defined by the following sample space and probabilities:



Each possible outcome is associated with a monetary prize. If $S = \{s_1, s_2, s_3\}$:

$$X(s_1) \mapsto 25 \text{ €}$$

$$X(s_2) \mapsto 100 \text{ €}$$

$$X(s_3) \mapsto 1 \text{ €}$$

The **random variable** X maps outcomes from the sample space to a given quantity, such as the prize of the draw.

Modeling the Structure of a Game

You are invited to play a game with the following rules:

- You roll two standard six-sided dice.
- The organizer pays you an amount of money equal to the sum of the two dice.

For example, if you roll (3,4), you receive \$7.

In the exercise class you have computed:

$$P(\text{Prize} = k) = \frac{\text{Number of outcomes that sum to } k}{\text{Number of all possible outcomes}}.$$

And the expected prize:

$$\sum_{k=2}^{12} k P(\text{prize} = k)$$

The random variable behind the game

A natural way to think about this game would be by introducing a function:

$$\begin{aligned} X : S &\longrightarrow \mathbb{R} \\ (d_1, d_2) &\mapsto d_1 + d_2 \end{aligned}$$

From the player's point of view, what matters is knowing the probability of the prize:

$$P(X = k) = P(\text{prize} = k)$$

The prize also has a median ...

The prize also has a variance ...

The prize also have quantiles...

We are interested in a new *distribution*, the *distribution* of the random variable X .

Discrete random variable

Definition

Let (S, P) be a sample space with a probability P . A **random variable** X is a function:

$$X : S \rightarrow \mathbb{R}$$

that assigns to each element $s \in S$ a real number $X(s)$. The random variable is **discrete if the image of X is discrete**.

In this case $X(S) = \{x_1, x_2, \dots, x_n\}$, or $X(S) = \{x_1, x_2, \dots\}$, and its **distribution** is given by:

$$p(x_i) = P(X = x_i)$$

where:

$$p(x_i) \geq 0$$

$$\sum_i p(x_i) = 1$$

Let's play (or not)

The “Second Chance” Coin Game Rules: You pay **\$4** to play. You flip a fair coin:

- If it lands **Heads**, you **win \$6**.
- If it lands **Tails**, you get a **second chance**:
 - If the second flip is **Heads**, you **win \$4**.
 - If the second flip is **Tails**, you **win \$0**.

Which is the random variable associated with this game?

How do we decide whether to play or not?

The business owner example

An entrepreneur plans to assemble a product consisting of a **sphere** and a **cylinder**, purchased from two different suppliers (A and B).

Each component (sphere or cylinder) can be classified as **good**, **long**, or **short**, depending on whether its measurement meets, exceeds, or falls below the specification.

The suppliers provide **the unit price** (\$5.00) and **the probabilities** for each classification.

After assembly:

- **If any component is short**, the set is **irrecoverable** and will be sold for \$5.00.
- **Long components** can be reworked at an additional cost of \$5.00 each.
- The selling price of a finished unit is \$25.00.

The goal is to **determine the distribution** of the random variable X , defined as the **profit per assembled unit**, in order to assess the business feasibility.

Product	Factory A Cylinder	Factory B Sphere
Within specifications Good (G)	0.8	0.7
Above specifications Long (L)	0.1	0.2
Below specifications Short (S)	0.1	0.1

Distribution of production of suppliers A and B

Solution

Graphic representation

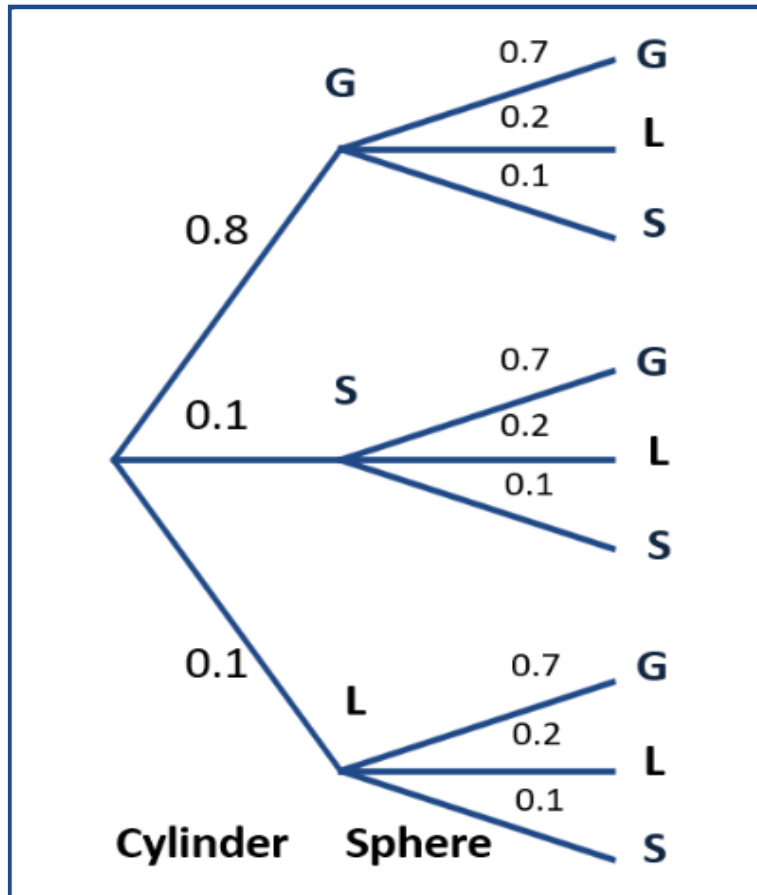


Table representation

Product	Probability	Profit per unit (X)
GG	0.56	15
GL	0.16	10
GS	0.08	-5
LG	0.07	10
LL	0.02	5
LS	0.01	-5
SG	0.07	-5
SL	0.02	-5
SS	0.01	-5

Distribution of X

x	p(x)
15	0.56
10	0.23
5	0.02
-5	0.19
Total	1.0

The discrete random variable is completely defined by the pairs $(x_i, p(x_i))$, $i = 1, \dots, n$

Summary statistics

Expected value and median

Definition

Let $\text{Im}(X) = \{x_1, x_2, \dots, x_n\}$. The **e expected value** of a **discrete** random variable is defined as:

$$\mathbb{E}[X] = \sum_{i=1}^n x_i P(\{s : X(s) = x_i\}) = \sum_i x_i P(X = x_i) = \sum_{i=1}^n x_i p(x_i)$$

When S is discrete, we have: $S = \{s_1, s_2, \dots\}$, and:

$$\mathbb{E}[X] = \sum_{j=1}^{\infty} X(s_j) P(\{s_j\})$$

The **median** of a random variable X is defined as **m** if:

$$\mathbb{P}(X \leq m) \geq \frac{1}{2} \quad \text{and} \quad \mathbb{P}(X \geq m) \geq \frac{1}{2}.$$

Business owner example:

Expected profit = $0.56 \cdot 15 + 0.23 \cdot 10 + 0.02 \cdot 5 + 0.19 \cdot -5 = 9.85$

Median profit = 15

Variance

Definition

The **variance** of a **discrete** random variable X is defined as:

$$\text{Var}[X] = \sum_{i=1}^n (x_i - \mathbb{E}[X])^2 p_i$$

The **standard deviation** is defined as:

$$\text{Sd}[X] = \sqrt{\text{Var}[X]}$$

Exercise: Compute the variance and standard deviation for the business example

Variance

Definition

The **variance** of a **discrete** random variable X is defined as:

$$\text{Var}[X] = \sum_{i=1}^n (x_i - \mathbb{E}[X])^2 p_i$$

The **standard deviation** is defined as:

$$\text{Sd}[X] = \sqrt{\text{Var}[X]}$$

Exercise: Compute the variance and standard deviation for the business example

Answer: Variance: 57.23 / SD = 7.57

Properties of EV and Var

Properties of expected value

- For any $a \in \mathbb{R} : \mathbb{E}[aX] = a\mathbb{E}[X]$
- For any random variables: $X, Y : \mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$
- If $X \leq Y$ (almost surely) $\Rightarrow \mathbb{E}[X] \leq \mathbb{E}[Y]$

Properties of variance

- For any $a \in \mathbb{R} \Rightarrow \text{Var}[aX] = a^2 \text{Var}[X]$
- Alternative definition:

$$\text{Var}[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \sum_i x_i^2 p_i - \left(\sum_i x_i p_i \right)^2$$

Why almost surely

Let $S = \{1, 2, 3\}$ with probability:

$$P(\{1\}) = \frac{1}{2}, \quad P(\{2\}) = \frac{1}{2}, \quad P(\{3\}) = 0$$

Consider two random variables:

$$X : S \longrightarrow \mathbb{R}, \quad X(1) = 1, \quad X(2) = 2, \quad X(3) = 3.$$

$$Y : S \longrightarrow \mathbb{R}, \quad Y(1) = 1, \quad Y(2) = 2, \quad Y(3) = 0.$$

Then the distributions of X and Y are the same:

$$P(X = 1) = P(Y = 1) = \frac{1}{2}, \quad P(X = 2) = P(Y = 2) = \frac{1}{2},$$

$$\text{For } k \neq 1, 2, \quad P(X = k) = P(Y = k) = 0$$

That is, the mean, variance, **and all other statistical measures coincide.**

We say that: $X = Y$ almost surely or (a.s.)

“From the player's point of view, what matters is knowing the probability of the prize”

Variance and risk

Investment A: A company offers a short-term corporate bond. At the end of one year, the bond might:

- **Perform normally (90% chance):** return **+5%**
- **Default partially (10% chance):** return **−10%**

Investment B: High-risk startup equity You can invest in a young startup. After one year:

- **High success (20% chance):** return **+20%**
- **Mild success (40% chance):** return **+5%**
- **Failure (40% chance):** return **−5%**

Studying investments

Investment A:

$$\text{Distribution of } X_A = \begin{cases} 5, & \text{with probability } 0.9, \\ -10, & \text{with probability } 0.1. \end{cases}$$

$$\mathbb{E}[X_A] = 0.9(5) + 0.1(-10) = 4.5 - 1 = 3.5.$$

Investment B:

$$\text{Distribution of } X_B = \begin{cases} 20, & \text{with probability } 0.2, \\ 5, & \text{with probability } 0.4, \\ -5, & \text{with probability } 0.4. \end{cases}$$

$$\mathbb{E}[X_B] = 0.2(20) + 0.4(5) + 0.4(-5) = 4 + 2 - 2 = 4.$$

What about the risk of these investments?

Studying the risk of investments

Investment A

$$\text{Distribution of } X_A = \begin{cases} 5, & \text{with probability 0.9,} \\ -10, & \text{with probability 0.1.} \end{cases}$$

$$\text{Var}(X_A) = \mathbb{E}[X_A^2] - (\mathbb{E}[X_A])^2 = 32.5 - (3.5)^2 = 32.5 - 12.25 = 20.25.$$

Investment B

$$\text{Distribution of } X_B = \begin{cases} 20, & \text{with probability 0.2,} \\ 5, & \text{with probability 0.4,} \\ -5, & \text{with probability 0.4.} \end{cases}$$

$$\text{Var}(X_B) = \mathbb{E}[X_B^2] - (\mathbb{E}[X_B])^2 = 100 - (4)^2 = 100 - 16 = 84.$$

Conclusion: Investment B offers a higher expected payoff, but at a much higher level of risk!

The cumulative distribution

Definition

The cumulative distribution function (c.d.f) is defined as:

$$F(z) = P(X \leq z)$$

Question: What are the domain and image of F ?

Back to the entrepreneur's problem :

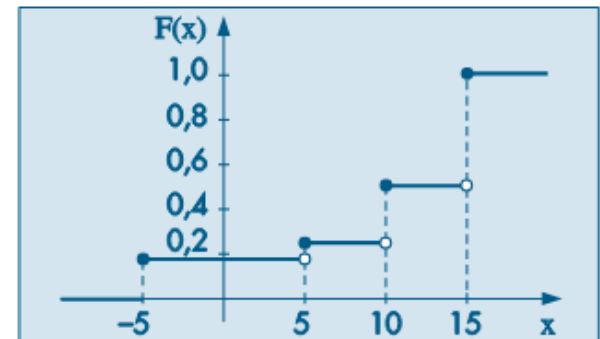
Distribution of X

x	p(x)
15	0.56
10	0.23
5	0.02
-5	0.19
Total	1.0

c.d.f of X

$$F(x) = \begin{cases} 0, & \text{if } x < -5, \\ 0.19, & \text{if } -5 \leq x < 5, \\ 0.21, & \text{if } 5 \leq x < 10, \\ 0.44, & \text{if } 10 \leq x < 15, \\ 1, & \text{if } x \geq 15. \end{cases}$$

graphic of the c.d.f



Focusing on the tails of the distribution

The price of **asset A** is \$100, future price has the following distribution in one year:

State of the economy	Probability	Price of the asset A in one year
Recession	0.05	70
Weak	0.15	90
Normal	0.50	110
Strong	0.25	130
Euphoric	0.05	200

Derivative financial products depend on an underlying asset A to determine their value. In this case, if you invest \$100, in one year your payoff will be:

- \$100 + 80% of the stock's gain, if $P \geq 100$;
- \$100, if $P < 100$ (*principal protection*);
- \$100 + 80% of the stock's gain + A fixed bonus of \$5 if $P \geq 130$

If we want to know what return we might have in 95% of the cases, what could we use?

Using quantiles to measure risk

In one year, the gain of the investor will be:

$$X(P) = \begin{cases} 0, & P \leq 100, \\ 0.8(P - 100), & 100 < P < 130, \\ 0.8(P - 100) + 5, & P \geq 130. \end{cases}$$

In one year, the gain of the investor will be:

Price of A (P)	Probability	Value of X(P)
70	0.05	0
90	0.15	0
110	0.50	8
130	0.25	29
200	0.05	85

The 95% quantile is defined by:

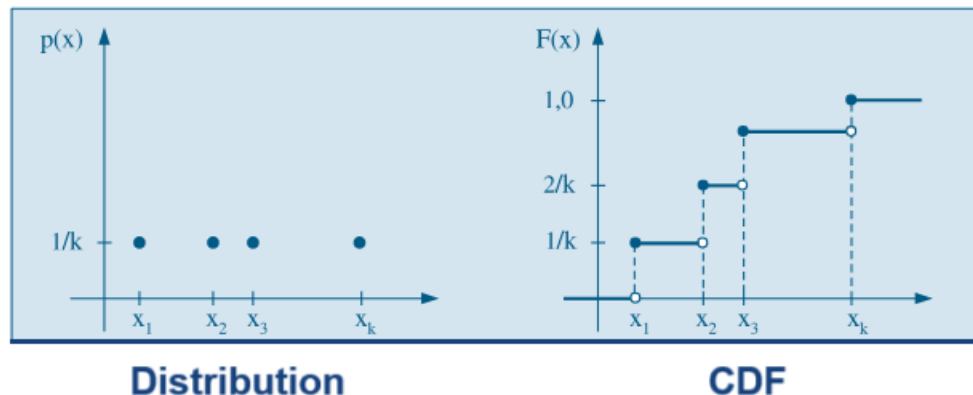
$$Q(0.95) = \inf\{z : F(z) \geq 0.95\}$$

In 95% of the cases, the investor earns at most \$29.

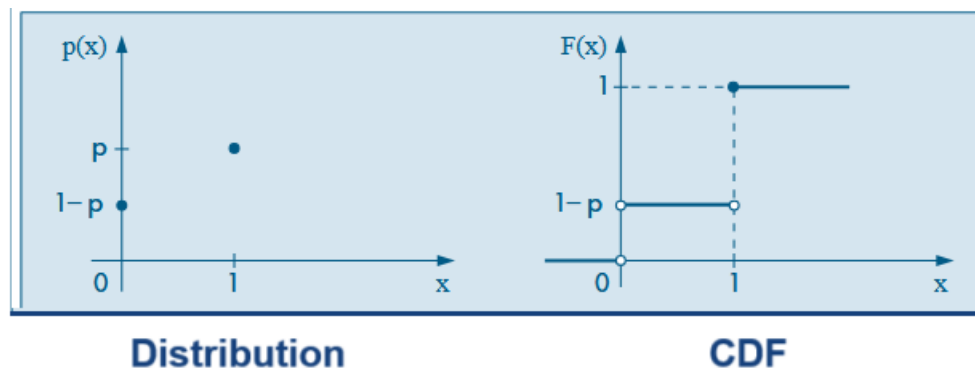
Important discrete distributions

Uniform and Bernoulli

- Uniform:** X assume the values $\{x_1, \dots, x_n\}$ and $P(X = x_i) = \frac{1}{n}$, $i \in \{1, \dots, n\}$



- Bernoulli:** X assume the values $\{0, 1\}$ and $P(X = 1) = p$, $P(X = 0) = 1 - p$



What is the expected value and variance of a Bernoulli random variable?

Examples of Bernoulli random variable

System working or failing

- A server can be up or down.
- $X = 1$ if the system is working, $X = 0$ if it fails.

Sensor detection

- A motion sensor may or may not detect movement.
- $X = 1$ if movement is detected, $X = 0$ otherwise.

Medical test: positive or negative

- $X = 1$ if the test result is positive
- $X = 0$ if negative.

If p is the probability of having 1 in these examples, we denote

$$X \sim \text{Bernoulli}(p)$$

Basketball game

A player will take **10 shots**.

Each shot is **independent**, and the probability of making a basket is **$p=0.3$** .

For every successful shot, the player earns **\$1**.

For every miss, the player earns nothing.

What is the distribution of this game?

Basketball game

A player will take **10 shots**.

Each shot is **independent**, and the probability of making a basket is **$p=0.3$** .

For every successful shot, the player earns **\$5**.

For every miss, the player earns nothing.

What is the distribution of this game?

$$X = X_1 + X_2 + \dots + X_{10}$$

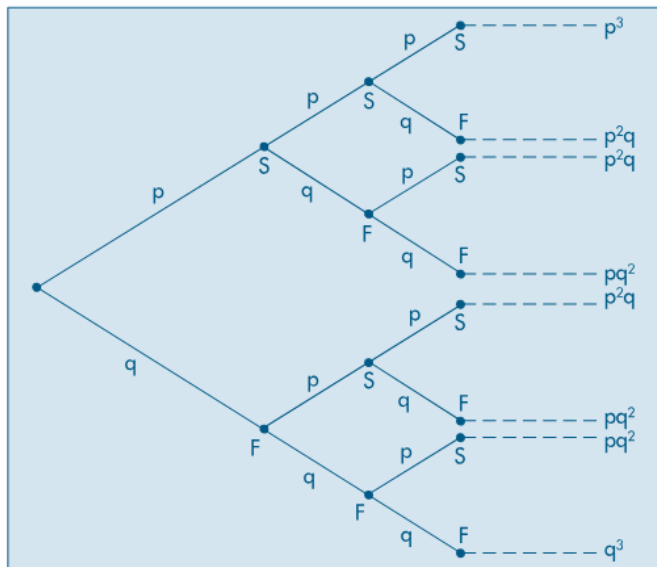
$$X_i \sim \text{Bernoulli}(0.3), i = 1, \dots, 10$$



$$X \sim \text{Binomial}(n = 10, p = 0.3)$$

Binomial

Binomial: Distribution of the repetition of a Bernoulli variable n times. For example, let $q = 1-p$ and $n = 3$:



Number of successes	Probabilities	$p = \frac{1}{2}$
0	q^3	$1/8$
1	$3pq^2$	$3/8$
2	$3p^2q$	$3/8$
3	p^3	$1/8$

The Binomial distribution has two parameters (p,n) and in general:

For $q = 1 - p$:

$$P(X = k) = \binom{n}{k} p^k q^{n-k}, \quad k = 0, 1, \dots, n,$$

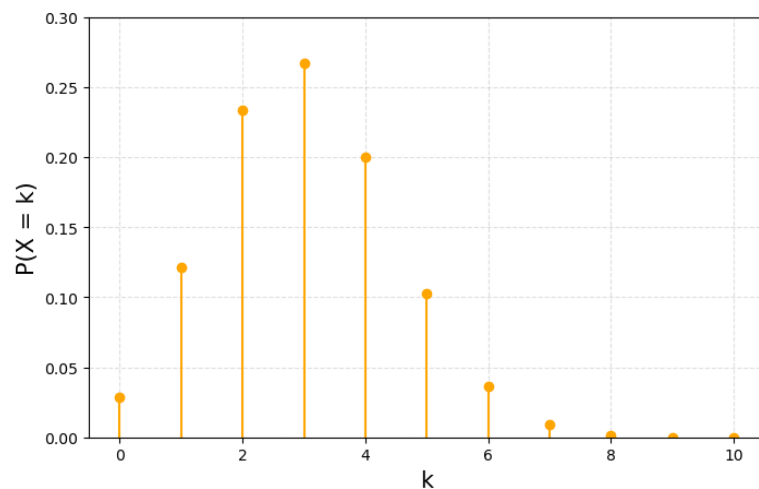
Binomial

The **Binomial distribution** has two parameters (p,n) and in general:

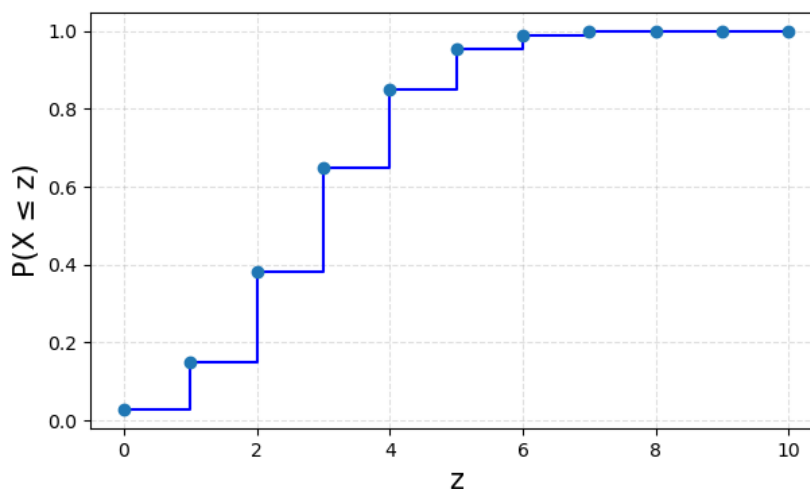
$$P(X = k) = \binom{n}{k} p^k q^{n-k}, \quad k = 0, 1, \dots, n,$$

Illustration for $n = 10$, $p = 0.3$:

Distribution



Cumulative distribution



Statistics of the binomial distribution

How to compute the expected value and variance of a binomial ?

Statistics of the binomial distribution

How to compute the expected value and variance of a binomial ?

$$X = X_1 + X_2 + \dots + X_{10} \quad X_i \sim \text{Bernoulli}(0.3), i = 1, \dots, 10$$

Then:

$$\mathbb{E}[X_1 + X_2 + \dots + X_{10}] = 10 \times 0.3 = 3.$$

More generally we have:

$$X \sim \text{Bin}(n, p) \Rightarrow \mathbb{E}[X] = np.$$

Statistics of the binomial distribution

How to compute the expected value and variance of a binomial ?

$$X = X_1 + X_2 + \dots + X_{10} \quad X_i \sim \text{Bernoulli}(0.3), i = 1, \dots, 10$$

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As for the variance, it is a bit more involved. What is the difference between:

$$X = X_1 + X_2 + \dots + X_{10} \quad \text{and} \quad Y = 10X_1 ?$$

Statistics of the binomial distribution

How to compute the expected value and variance of a binomial ?

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Then:

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As for the variance, it is a bit more involved. What is the difference between:

$$X = X_1 + X_2 + \dots + X_{10} \quad \text{and} \quad Y = 10X_1 ?$$

If $X_1, \dots, X_{10}$ are **independent**, then:

$$\text{Var}[X_1 + \dots + X_{10}] = n \text{Var}[X_1] = n \times p(1 - p)$$

Poisson

Poisson distribution: Describes the **number of events** occurring in a **fixed time interval**.

- Number of calls received by a telephone during five minutes;
- Occurrence of days with extreme rainfall in a year.

A Poisson distribution that occurs on average λ times per time interval:

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}, \quad k = 0, 1, 2, \dots$$

the parameter λ is the average number of events in the fixed time.

You can use Poisson distribution when:

- You know the average rate at which the event occurs, and it is constant (equal to λ)
- One event does not affect the occurrence of other events
- The event can occur at any moment in time.

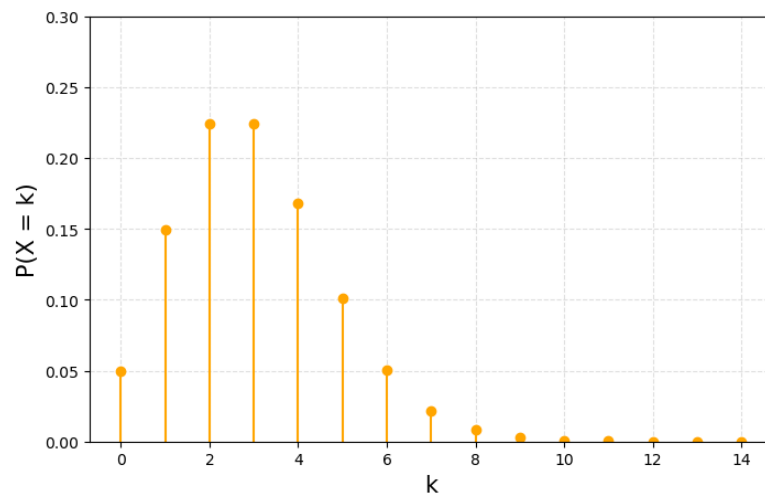
Poisson

Distribution of Poisson random variable for a parameter λ :

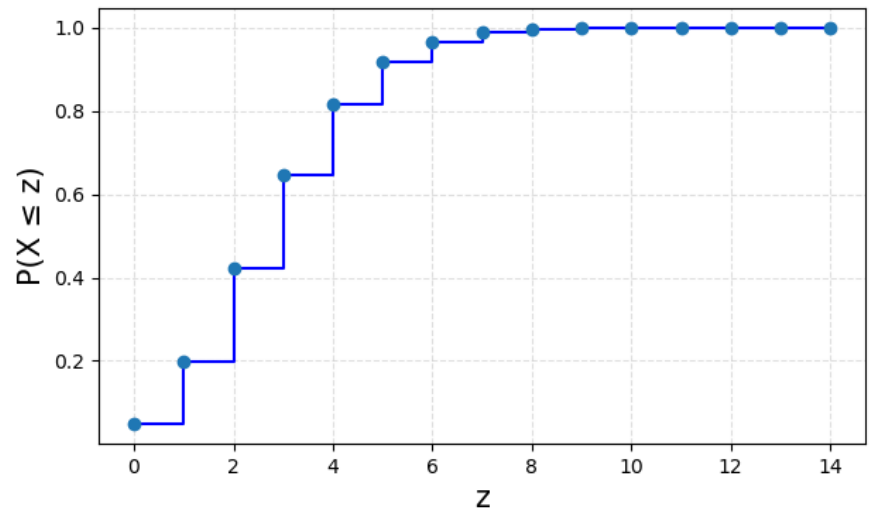
$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}, \quad k = 0, 1, 2, \dots$$

Illustration for the case $\lambda = 3$

Distribution



Cumulative distribution



Example of Poisson

The number of mutations in a given gene per cell division is on average 0.4

Question: What is the probability that, in one cell division, **exactly one mutation** occurs in that gene?

Example of Poisson

The number of mutations in a given gene per cell division is on average 0.4

Question: What is the probability that, in one cell division, **exactly one mutation** occurs in that gene?

Can we use Poisson?

- ✓ We know the average mutation rate.
- ✓ Mutations occur independently of one another
- ✓ Mutations can occur at any time.

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}, \quad k = 0, 1, 2, \dots$$

In our example $\lambda = 0.4$.

$$P(X = 1) = \frac{0.4^1 e^{-0.4}}{1!} = 0.4e^{-0.4} \approx 0.268$$

Statistic metrics for Poisson

$$X \sim \text{Poisson}(\lambda) \iff P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad k = 0, 1, 2, \dots$$

Computing the expected value:

$$\mathbb{E}[X] = \sum_{k=0}^{\infty} k P(X = k) = \sum_{k=0}^{\infty} k \frac{\lambda^k e^{-\lambda}}{k!} = \lambda \frac{\lambda^{k-1}}{(k-1)!} e^{-\lambda} = \lambda e^{-\lambda} \sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!}$$

then letting $j = k - 1$:

$$\mathbb{E}[X] = \lambda e^{-\lambda} \sum_{j=0}^{\infty} \frac{\lambda^j}{j!}.$$

But it is known by Taylor approximation that:

$$\sum_{j=0}^{\infty} \frac{\lambda^j}{j!} = e^{\lambda}$$

We conclude that:

$$\mathbb{E}[X] = \lambda e^{-\lambda} e^{\lambda} = \lambda.$$

Statistic metrics for Poisson

$$X \sim \text{Poisson}(\lambda) \iff P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad k = 0, 1, 2, \dots$$

Variance:

$$\text{Var}(X) = \lambda$$

Cumulative distribution function:

$$F_X(x) = \sum_{k=0}^{\lfloor x \rfloor} \frac{\lambda^k e^{-\lambda}}{k!}$$

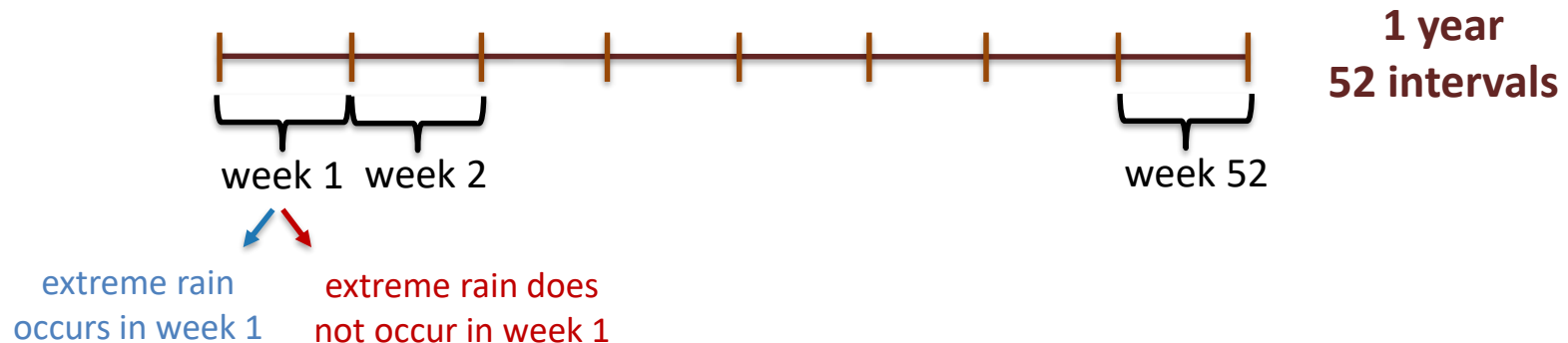
Why Poisson?

- It is natural to think in terms of the **average number of occurrences**.
- Example: The **average number of extreme rainfall** in one year is λ .

We aim to model the **distribution** of:

X = Number of extreme rainfalls in a year

Let approximate X by discretizing the year into **small intervals** (n intervals):



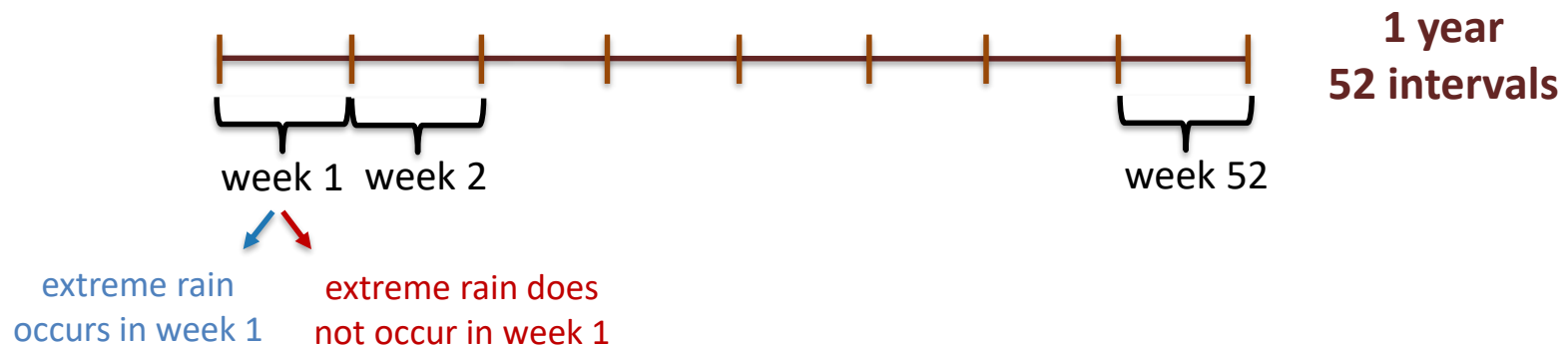
Why Poisson?

- It is natural to think in terms of the **average number of occurrences**.
- Example: The **average number of extreme rainfall** in one year is λ .

We aim to model the **distribution** of:

X = Number of extreme rainfalls in a year

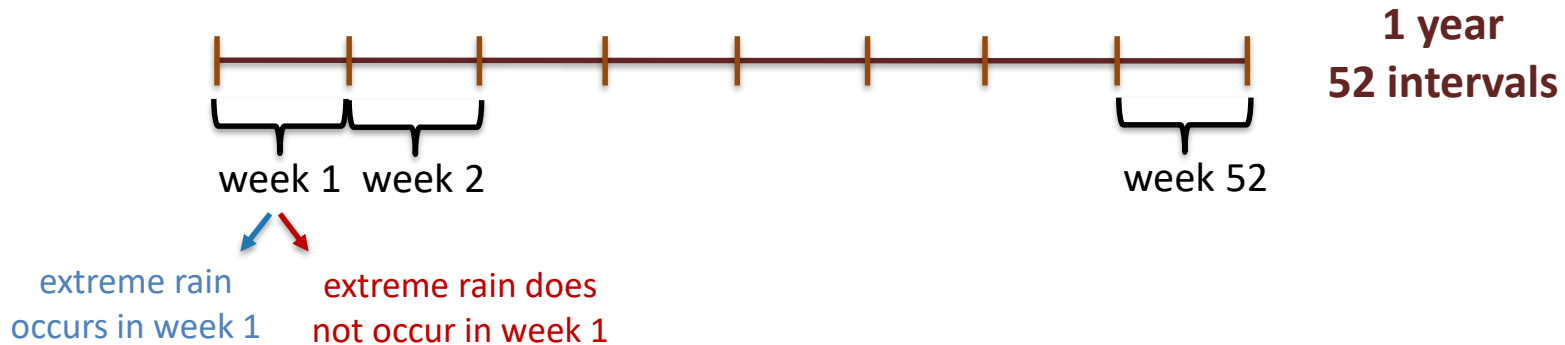
Let approximate X by discretizing the year into **small intervals** (n intervals):



$$Y \sim \text{Bernoulli}(p)$$

$$Y = \begin{cases} 1, & \text{if extreme rainfall occurs} \\ 0, & \text{otherwise} \end{cases}$$

Why Poisson?



$$Y \sim \text{Bernoulli}(p)$$

Since one event does not affect the occurrence of other events, approximately:

$$Y_1 + Y_2 + \dots + Y_{52} \approx \text{Number of rainfalls in one year} = X$$

Why it is just an approximation ?

Since Y is 0 or 1, we *limit weekly events*, but our model allows *multiple rainfalls* per week (The event can occur at any moment in time)

Large values of n $\longleftrightarrow$ Better approximation of continuous time

Why Poisson?



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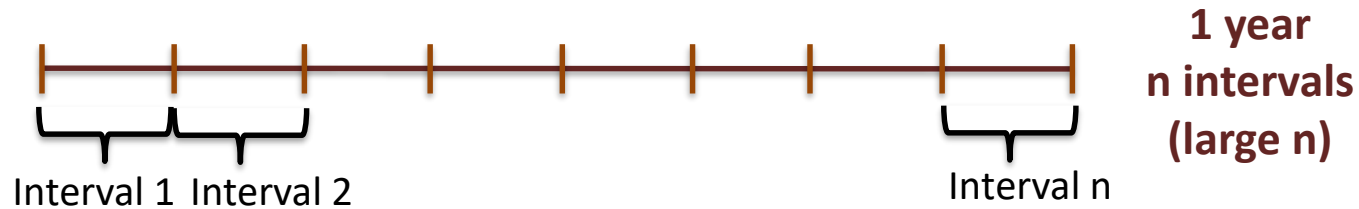
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Large values of n $\longleftrightarrow$ Better approximation of continuous time

Why Poisson?



$$\underbrace{Y_1 + Y_2 + \dots + Y_n}_{\text{Binomial}(n, p)} \approx \text{Number of rainfalls in one year} = X$$

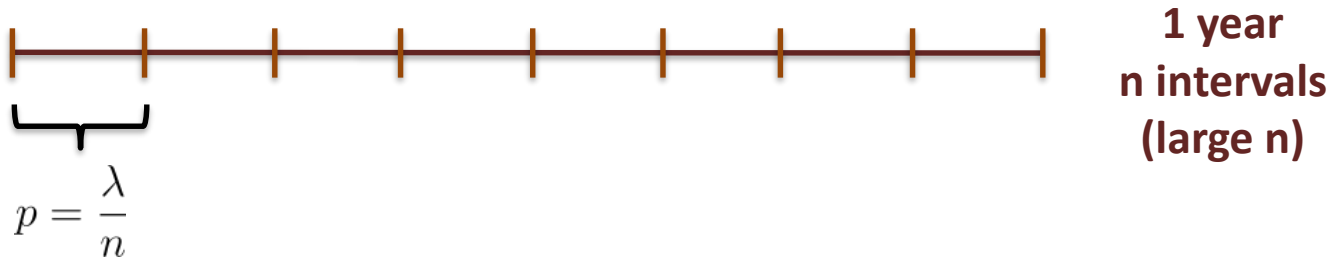
Taking the expected value on both sides:

$$\mathbb{E}[\text{Binomial}(n, p)] = np \approx \lambda$$

$$\Rightarrow p = \frac{\lambda}{n}$$

Remember : Our approximation gets better when we increase n !

Why Poisson?



X counts **how many** extreme rainfall event occur in one year, then:

$$P(X = k) \approx \underbrace{\binom{n}{k} p^k (1 - p)^{n-k}}_{\text{Bin}(n, p)} = \binom{n}{k} \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k}$$

The approximation gets better for larger values of n , then:

$$P(X = k) = \lim_{n \rightarrow \infty} \frac{n!}{k!(n-k)!} \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k} = \boxed{\frac{e^{-\lambda} \lambda^k}{k!}}$$

the Poisson distribution!

Independence and continuous distributions

Class 4

Objective:

Understanding continuous distributions and their key metrics
An application to defining meteorological alert levels.

Independence of random variables

Thinking about independence

Example 1: Two-coin flips

X_1 - 1 if first coin flip is Heads and 0 if it is Tails

X_2 - 1 if the second coin flip is Heads and 0 if it is Tails

We have:

$$P(X_1 = 0) = \frac{1}{2} \text{ and } P(X_2 = 0) = \frac{1}{2} \quad P(X_1 = 1) = \frac{1}{2} \text{ and } P(X_2 = 1) = \frac{1}{2}$$

And:

$$P(X_1 = 0, X_2 = 0) = P(X_1 = 0)P(X_2 = 0) = \frac{1}{4}$$

More generally:

$$P(X_1 = i, X_2 = j) = P(X_1 = i)P(X_2 = j) = \frac{1}{4}, \text{ For all } i, j \in \{0, 1\}$$

Independence of random variables

Example 2: Two-coin flips

Y_1 - 1 if first coin flip is Heads and 0 if it is Tails

Y_2 - $(1 - Y_1)$

We have:

$$P(Y_1 = 0) = \frac{1}{2} \text{ and } P(Y_2 = 0) = \frac{1}{2} \quad P(Y_1 = 1) = \frac{1}{2} \text{ and } P(Y_2 = 1) = \frac{1}{2}$$

However:

$$P(Y_1 = 1, Y_2 = 1) = 0 \neq P(Y_1 = 1)P(Y_2 = 1)$$

Independence of random variables

Example 2: Two-coin flips

Y_1 - 1 if first coin flip is Heads and 0 if it is Tails

Y_2 - $(1 - Y_1)$

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$$P(Y_1 = 0) = \frac{1}{2} \text{ and } P(Y_2 = 0) = \frac{1}{2} \quad P(Y_1 = 1) = \frac{1}{2} \text{ and } P(Y_2 = 1) = \frac{1}{2}$$

However:

$$P(Y_1 = 1, Y_2 = 1) = 0 \neq P(Y_1 = 1)P(Y_2 = 1)$$

Definition

The random variables X and Y are independent if:

$$P(X \in A, Y \in B) = P(X \in A)P(Y \in B)$$

for any $A, B \subset \text{Im}(X)$

Continuous distributions

Continuous random variable

Definition

Let (S, P) be a sample space with a probability P . A **random variable** X is a function:

$$X : S \rightarrow \mathbb{R}$$

that assigns to each element $s \in S$ a real number $X(s)$. The random variable is continuous if the image of X is continuous.

Examples of continuous distributions:

- Distribution of salaries across a country
- Distribution of temperatures over a year
- The arrival time of a customer at a store

Question: Give an example where S is continuous, but X is discrete.

Continuous random variable

In the case of a continuous distribution, it makes more sense to analyze probabilities over intervals rather than at single, exact values.

Example: Consider X the Temperature at a given day in Paris

It is natural to consider the probability that the temperature falls within a certain range:

$$P(15 \leq X \leq 20) = ?$$

rather than the probability that it is exactly a specific number.

Density of a random variable

Density of a random variable

The density f of a random variable X has the following property:

$$P(a \leq X \leq b) = \int_a^b f(x), \forall a, b \in \mathbb{R}$$

Consequently:

$$\int_{-\infty}^{\infty} f(x) = 1 \quad \text{and} \quad f \geq 0$$

Question: What is the density of a discrete random variable ?

Question: What is the relation between the cumulative distribution function and the density f ?

Cumulative distribution and density

Theorem

Let f be a continuous density defining a probability P with cumulative distribution F
Then:

$$F'(x) = f(x)$$

Why?

Let X be a random variable with density f . By fundamental theorem of calculus:

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(x)dx \Rightarrow F'(x) = f(x)$$

Statistical measures

Expected value and variance of a random variable

We have seen that for a **discrete** random variable, the **expected value** is defined as:

$$E(X) = \sum x_i P(X = x_i) = \sum x_i p_i$$

The natural extension for a **continuous** random variable is :

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx$$

The variance of a continuous random variable is defined as:

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$$

That is:

$$\text{Var}(X) = \int_{-\infty}^{\infty} (x - \mathbb{E}[X])^2 f(x) dx = \int_{-\infty}^{\infty} x^2 f(x) dx - \left(\int_{-\infty}^{\infty} x f(x) dx \right)^2.$$

Important: Not every distribution has a finite expectation or variance. Example:

$$f(x) = \frac{1}{x^2}, \quad x \geq 1.$$

Important Continuous distributions

Uniform distribution

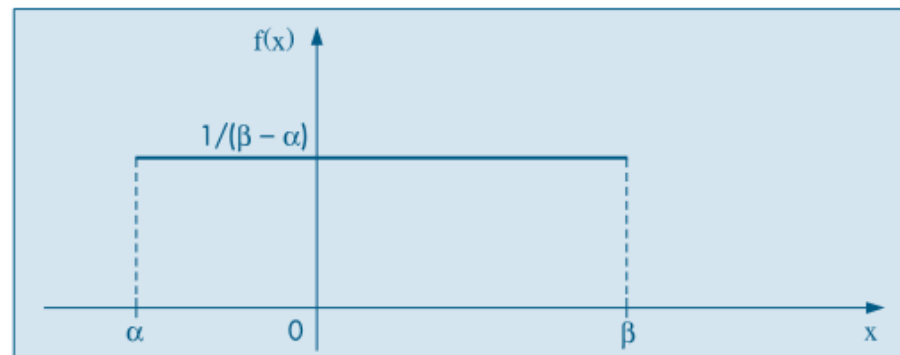
Definition

A random variable X has a **uniform distribution** on the interval $[\alpha, \beta]$ if its probability density function (p.d.f.) is given by :

$$f_X(x) = \begin{cases} \frac{1}{\beta - \alpha}, & \alpha \leq x \leq \beta, \\ 0, & \text{otherwise.} \end{cases}$$

Notation: $X \sim \text{Uniform}([a, b])$

Graphic of the p.d.f :



Question: What are the expected value, variance and the cumulative density function for the uniform distribution ?

Uniform distribution

Definition

A random variable X has a **uniform distribution** on the interval $[\alpha, \beta]$ if its probability density function (p.d.f.) is given by :

$$f_X(x) = \begin{cases} \frac{1}{\beta - \alpha}, & \alpha \leq x \leq \beta, \\ 0, & \text{otherwise.} \end{cases}$$

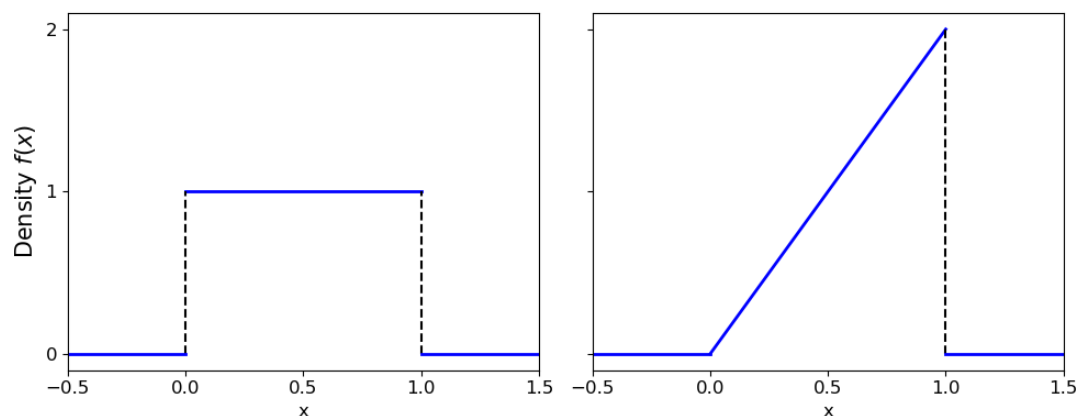
Notation: $X \sim \text{Uniform}([a, b])$

$$\mathbb{E}[X] = \frac{\alpha + \beta}{2}$$

$$\text{Var}(X) = \frac{(\beta - \alpha)^2}{12}$$

Uniform vs Beta distribution

$$f_X(x) = \begin{cases} 1, & \text{if } 0 \leq x \leq 1, \\ 0, & \text{otherwise.} \end{cases} \quad f_Y(x) = \begin{cases} 2x, & 0 \leq x \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$



Note that:

$$\mathbb{E}[Y] = \int_{-\infty}^{\infty} x f_Y(x) = \int_0^1 2x^2 = \left. \frac{2x^3}{3} \right|_0^1 = \frac{2}{3}$$

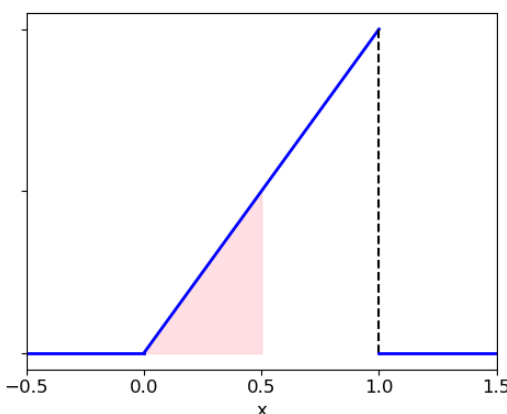
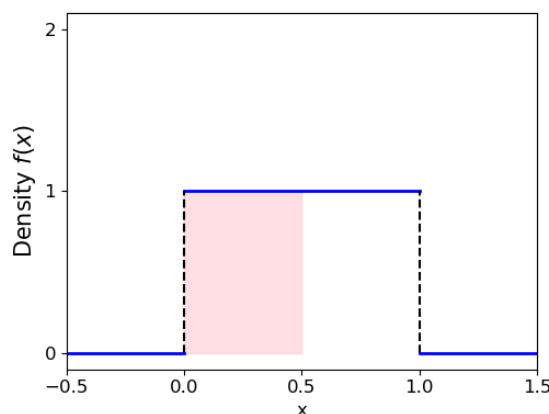
Y has higher expected value than the uniform distribution.

Uniform vs Beta distribution

Consider the event: $E = [0, 1/2]$. The probabilities $P(X \in E)$ and $P(Y \in E)$ are the area below the graphic of each distribution function.

$$f_X(x) = \begin{cases} 1, & \text{if } 0 \leq x \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

$$f_Y(x) = \begin{cases} 2x, & 0 \leq x \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$



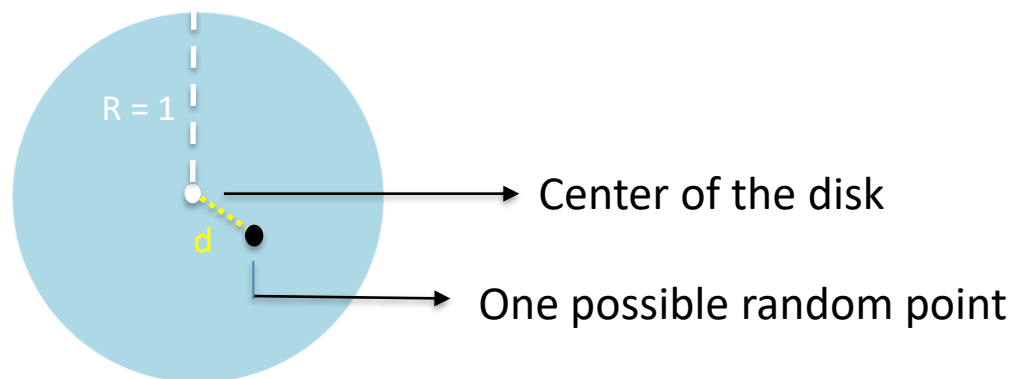
Question : Which are the probability of E in these two distributions?

Question: Can you imagine a context where f_Y is used?

Beta distribution

Choose a random (uniformly) point in a disk of radius 1, and the random variable:

X = Distance between the point and the center of the disk.

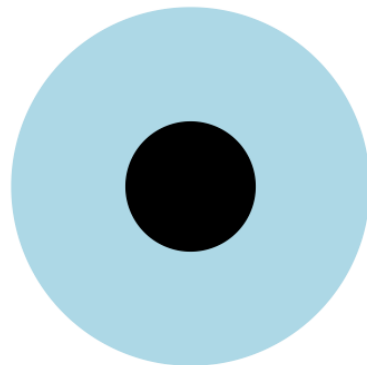


What is the strategy for computing:

$$P(X \leq r) \text{ for any } r \in [0, 1]$$

Beta distribution

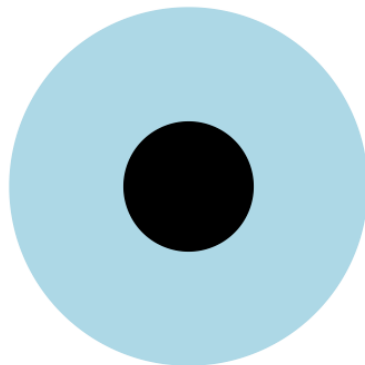
Fix a value of $r \in [0, 1]$, and consider the disk of radius r :



If the dot is in the black disk then $X \leq r$.

Beta distribution

Fix a value of $r \in [0, 1]$, and consider the disk of radius r :



Then:

$$P(X \leq r) = \frac{\text{Area (small disk)}}{\text{Area (big disk)}} = \frac{\pi r^2}{\pi} = r^2, \quad 0 \leq r \leq 1.$$

While for $Y \sim \text{Triangular}([0, 1])$:

$$P(Y \leq r) = \int_0^r 2x = r^2$$

Implying:

X and Y have the same distribution.

Double-Sensor Startup Time

Suppose two independent sensors are triggered by the same physical event, but each has an internal random delay before activation. The delay of sensor A is

$$X_A \sim \text{Unofrm}([0, 1])$$

and the delay of sensor B is:

$$X_B \sim \text{Unofrm}([0, 1])$$

The system begins recording only when **both sensors are active**, so the effective start time is:

$$M = \max(X_A, X_B)$$

Which is the distribution of M?

Hint: Compute the cumulative distribution function and use independency

Exponential distribution

The exponential distribution models the waiting time until the first occurrence of a random event, for example:

- **Time between phone calls** arriving at a call center
- **Time until a radioactive particle decays**
- **Time until the next extreme rainfall**

We use the exponential distribution when:

- The event happens at any moment in time
- The event happens with a constant average rate
- One event does not affect the occurrence of other events

This distribution has one parameter λ that represents how many times, on average, the event happens per unit of time.

Exponential distribution

Definition

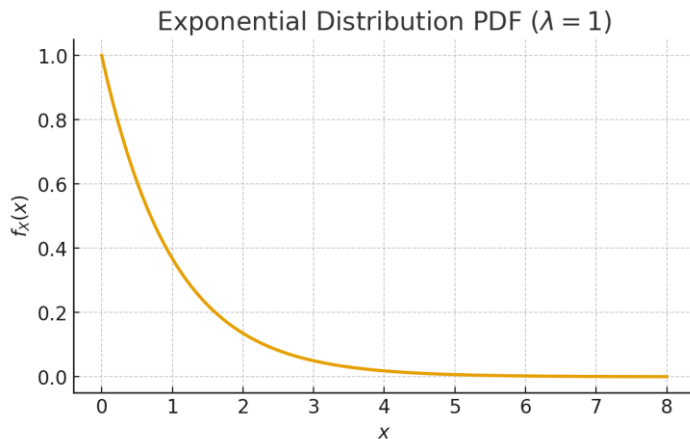
A random variable X has a **exponential distribution** with parameter λ if its probability density function (p.d.f) is given by:

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0, \\ 0, & x < 0. \end{cases}$$

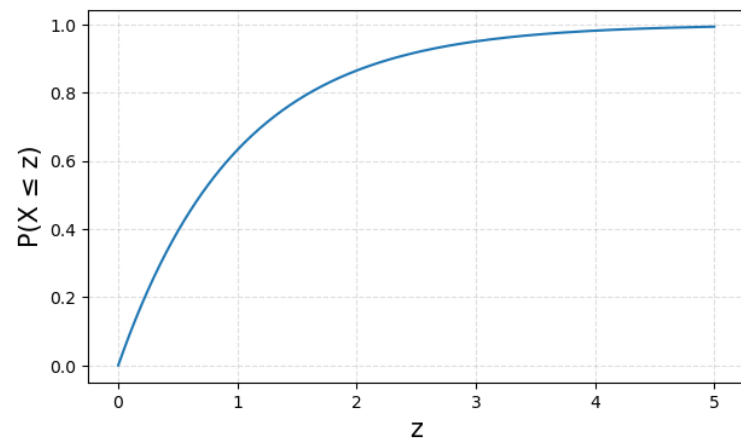
And we denote $X \sim \text{Exp}(\lambda)$.

Case $\lambda = 1$

Density



Cumulative distribution



Example exponential distribution

Customers arrive at a supermarket with rate 12 customers per hour.

Question: What is the probability that the next customer arrives **more than 10 minutes** from now?

Example exponential distribution

Customers arrive at a supermarket with rate 12 customers per hour.

Question: What is the probability that the next customer arrives **more than 10 minutes** from now?

- Can we use Poisson distribution to model the number of customers?
 - ✓ The average rate at which the event occurs, and it is constant
 - ✓ One event does not affect the occurrence of other events
 - ✓ The event can occur at any moment in time.
- Can we use exponential distribution?
 - ✓ The event happens at any moment in time

Then:

$$P(T > 10 \text{ minutes}) = P\left(T > \frac{1}{6} \text{ hours}\right) = \int_{\frac{1}{6}}^{\infty} 12e^{-12t} dt = e^{-\frac{12}{6}} = e^{-2} \approx 0.1353.$$

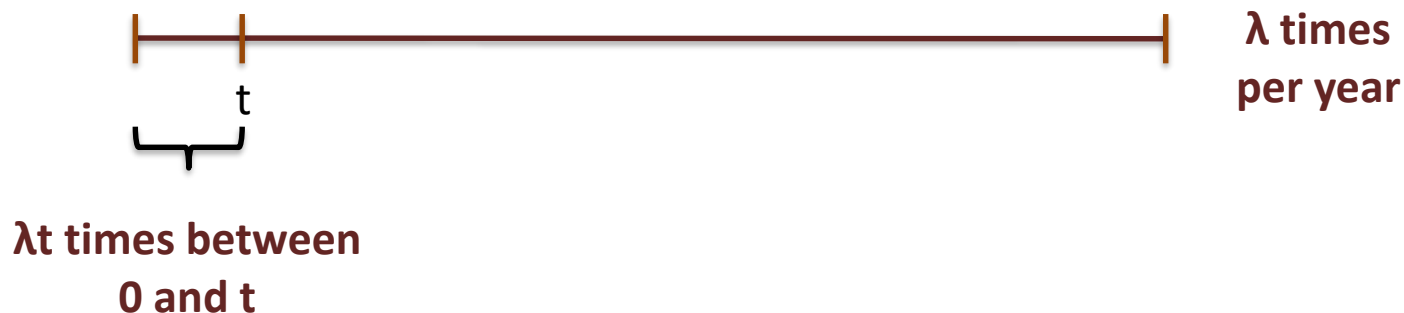
Why exponential?

- Both the exponential and Poisson distributions describe the same type of events.
- The Poisson distribution models the **number of events** occurring in a fixed time
- The Exponential distribution models the **waiting time until the first event occurs**

Then let $X \sim \text{Exp}(\lambda)$:

$P(X > t) =$ The probability that any event happens until time t

If an event occurs on average λ times in 1 year, then it occurs on average λt times between time 0 and t :



Why exponential?

Then let Y be a Poisson random variable that describes the number of events between 0 and t . Since the mean number of events is λt :

$$Y \sim \text{Poisson}(\lambda t) \iff P(Y = k) = \frac{(\lambda t)^k e^{-\lambda t}}{k!}, \quad k = 1, 2, \dots$$

The probability of waiting **more than t** for the next event is the same as the probability that **no events occur up to time t** ; that is:

$$P(X > t) = \mathbb{P}(Y = 0) = e^{-\lambda t}, \quad t \geq 0$$

then we have our **cumulative distribution function**:

$$F_X(t) = P(X \leq t) = 1 - e^{-\lambda t}$$

Question: How to obtain the density function ?

Question: What should be the expected value of an exponential distribution with parameter λ ?

Statistics for exponential

Expected Value and Variance for the exponential distribution:

$$\mathbb{E}[X] = \frac{1}{\lambda}, \quad \text{Var}(X) = \frac{1}{\lambda^2}$$

Memoryless property

The exponential distribution is memoryless: the probability of waiting an additional time t does not depend on how long has already been waited.

$$\Pr(X > s + t \mid X > s) = \Pr(X > t), \quad s, t \geq 0.$$

Proof: Let $X \sim \text{Exp}(\lambda)$, $\forall t, s > 0$:

Statistics for exponential

Expected Value and Variance for the exponential distribution:

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Proof: Let $X \sim \text{Exp}(\lambda)$, $\forall t, s > 0$:

$$P(X > s + t \mid X > s) = \frac{P(X > s + t)}{P(X > s)} = \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t} = P(X > t).$$

Memoryless property

Quiz — Memoryless Property (Exponential Distribution)

For each situation below, indicate whether the memoryless property applies or does not apply.

- The time until the next customer arrives at a service counter, assuming arrivals follow a Poisson process.
- The remaining lifetime of a brand-new electronic component that fails due to random external shocks.
- The remaining time until a student finishes writing an exam, given that they have already been working on it for 30 minutes.
- The remaining lifetime of a car engine that has already been used for 10 years.

Memoryless property

Quiz — Memoryless Property (Exponential Distribution)

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- The time until the next customer arrives at a service counter, assuming arrivals follow a Poisson process.

Memoryless applies

- The remaining lifetime of a brand-new electronic component that fails due to random external shocks.

Memoryless applies

- The remaining time until a student finishes writing an exam, given that they have already been working on it for 30 minutes.

Memoryless does NOT apply

- The remaining lifetime of a car engine that has already been used for 10 years.

Memoryless does NOT apply

Next pollution event with exponential

In a certain city, episodes of high air pollution (when the concentration of fine particles exceeds a critical level) occur randomly over time.

Suppose that the time X (in days) between two consecutive high-pollution episodes follows an exponential distribution with rate $\lambda = 0.05$:

$$X \sim \text{Exp}(0.05)$$

- Compute the probability that the next high-pollution episode occurs only after 40 days.
- Determine the mean and the median of X . State which one is larger and briefly explain why.
- The city has already gone 30 days without any high-pollution episode. Compute the probability that it will last at least 20 more days without such an episode.

Extra question: Intuitively, why the cumulative distribution of a continuous random variable is continuous ?

Next pollution event with exponential

- We have that: $P(X > t) = e^{-\lambda t}$, then: $P(X > 40) = e^{-0.05 \times 40} = e^{-2} \approx 0.1353$.
- The median m satisfies $P(X \leq m) = 0.5$ or equivalently $P(X > m) = 0.5$.

Thus:

$$e^{-0.05m} = 0.5 \implies m = \frac{\ln 2}{0.05} \approx 13.86 \text{ days.}$$

$$\text{While: } \mathbb{E}[X] = \frac{1}{0.05} = 20 \text{ days.}$$

Hence, the mean is larger than the median. This is due to the fact that exponential distribution has a long right tail that pulls the mean to the right.

- Given that no high-pollution episode has occurred during the last 30 days, we compute $P(X > 50 \mid X > 30)$. Therefore because of the memoryless property:

$$P(X > 20) = e^{-0.05 \times 20} = e^{-1} \approx 0.3679.$$

Normal distribution

Definition

A random variable X has a **normal distribution** with mean μ and standard deviation σ if its probability density function (p.d.f.) is given by :

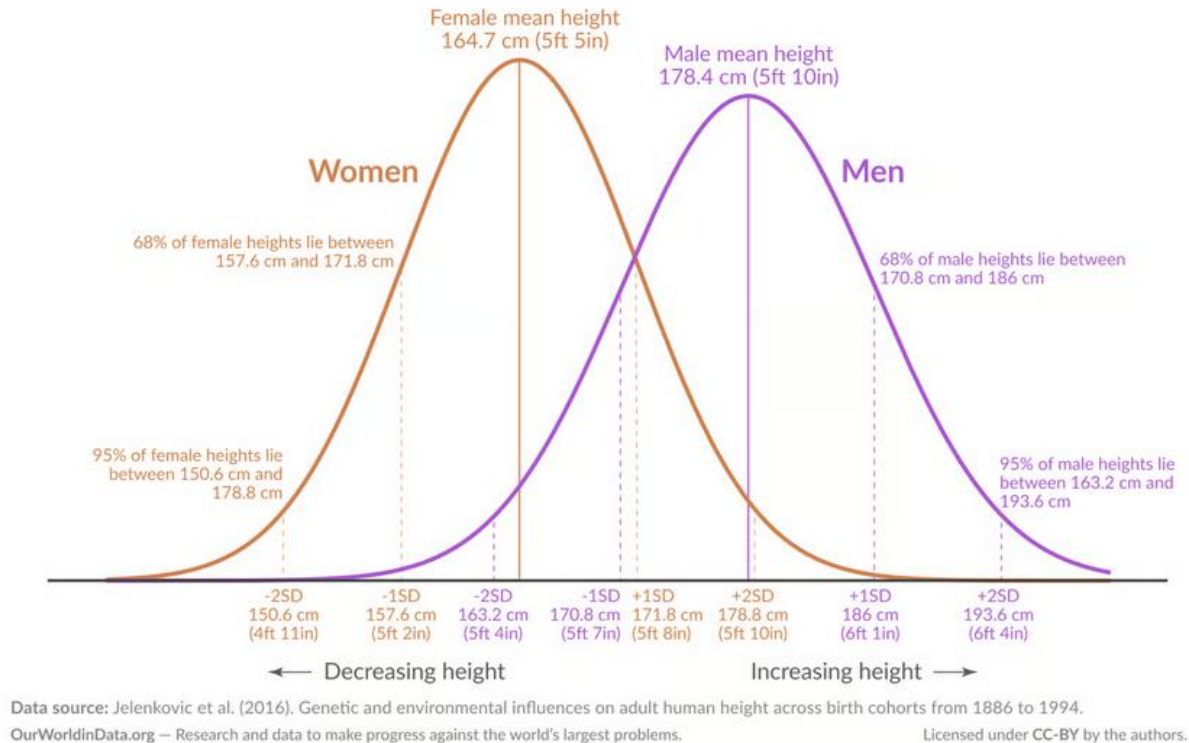
$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}.$$

And we denote $X \sim \mathcal{N}(\mu, \sigma^2)$.

Examples of phenomena modeled by the normal distribution:

- Measurement errors
- Daily temperature fluctuations
- Human height

Real data of human height

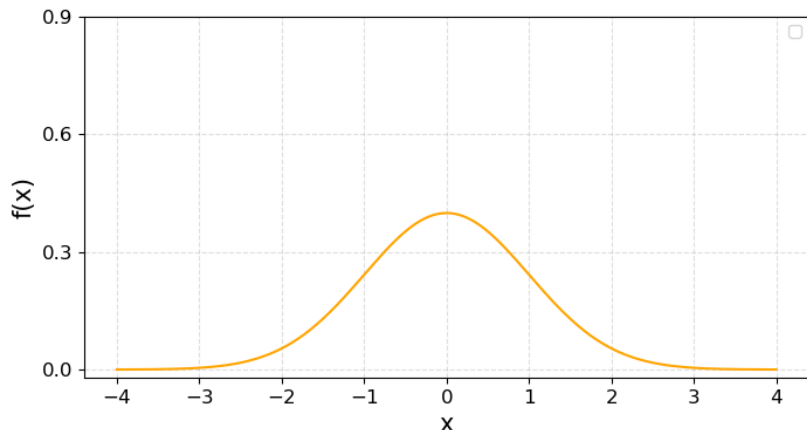


source: ourworldindata.org

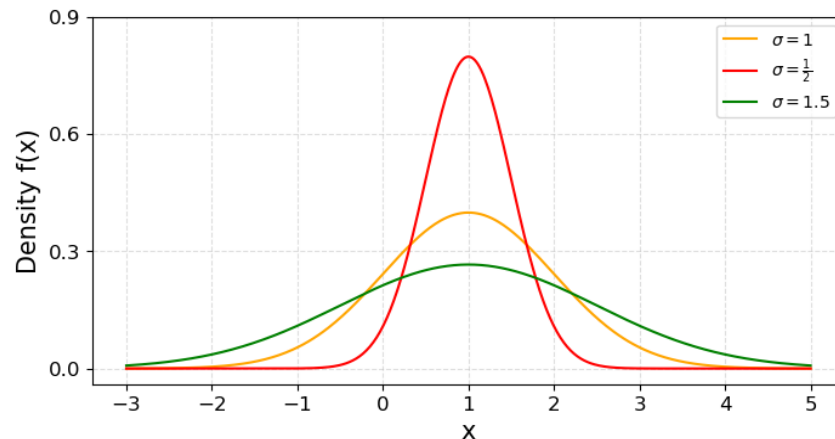
Question : Between men and women, which group has the larger standard deviation?

Properties of normal distribution

Density for $\mu = 0, \sigma = 1$:



Density for $\mu = 1$:



Important properties $X \sim N(\mu, \sigma)$:

- Expected value and variance: $\mathbb{E}[X] = \mu$, $\text{Var}[X] = \sigma^2$
- The p.d.f. is symmetric.
- If $Z = \frac{X - \mu}{\sigma}$, then: $Z \sim N(0, 1)$. Which means:

$$\text{For any } a, b \in \mathbb{R} : P(X \in [a, b]) = P\left(\frac{X - \mu}{\sigma} \in \left[\frac{a - \mu}{\sigma}, \frac{b - \mu}{\sigma}\right]\right) = P\left(Z \in \underbrace{\left[\frac{a - \mu}{\sigma}, \frac{b - \mu}{\sigma}\right]}_{\text{Standard normal probability}}\right)$$

- If $X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2)$ and $X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$ are independent, then:

$$X_1 + X_2 \sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$$

Properties of normal distribution

Consider $X \sim \mathcal{N}(\mu, \sigma^2)$ and:

$$Z = \frac{X - \mu}{\sigma}$$

Show that:

1. We have $\mathbb{E}[Z] = 0$ and $\text{Var}(Z) = 1$
2. Is 1 enough to say that: $Z \sim \mathcal{N}(0, 1)$?
3. How to prove that $Z \sim \mathcal{N}(0, 1)$?

Properties of normal distribution

Consider $X \sim \mathcal{N}(\mu, \sigma^2)$ and:

$$Z = \frac{X - \mu}{\sigma}$$

Show that:

1. We have $\mathbb{E}[Z] = \mu$ and $\text{Var}(Z) = 1$
2. Is 1 enough to say that: $Z \sim \mathcal{N}(0, 1)$?
3. How to prove that $Z \sim \mathcal{N}(0, 1)$?

Two continuous distributions are the same **if and only if** their densities are the same. Then we need to show that:

$$f_Z(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \iff P(a \leq Z \leq b) = \frac{1}{\sqrt{2\pi}} \int_a^b e^{-\frac{x^2}{2}} dx$$

Idea of the proof:

$$P(Z \in [a, b]) = P(X \in [a\sigma + \mu, b\sigma + \mu]) = \underbrace{\int_{a\sigma + \mu}^{b\sigma + \mu} f_X(x) dx}_{\text{change of variables}} = \int_a^b f_X(\mu + \sigma z) \sigma dz = \int_a^b \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{z^2}{2}\right)$$

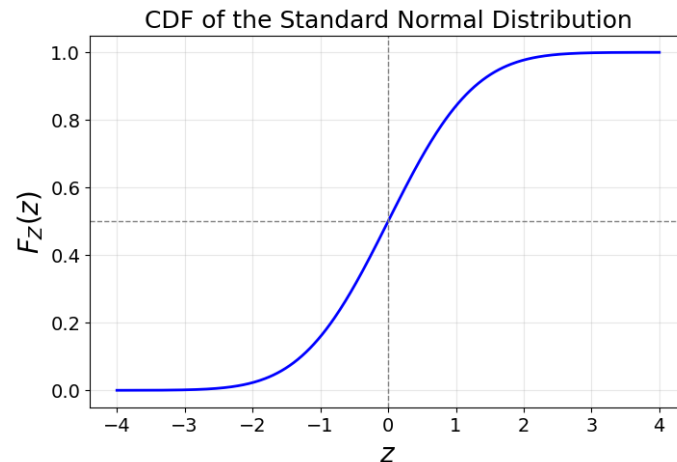
change of variables: $x = \mu + \sigma z$, $dx = \sigma dz$.

Normal distribution

Note that for the normal distribution:

- The cumulative distribution for $\mu = 0, \sigma = 1$ is:

$$F_X(z) = \mathbb{P}(X \leq z) = \int_{-\infty}^z f_X(x) dx = \int_{-\infty}^z \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right) dx$$



The normal distribution is fundamental because it naturally arises in many real-world phenomena and underlies the Central Limit Theorem.

Temperatures with normal distribution

In a certain city, the **daily maximum winter temperature** (in °C) is modeled by a normal distribution with mean $\mu=10$ and standard deviation $\sigma=3$

$$T \sim \mathcal{N}(10, 3^2)$$

- What is the probability that the maximum temperature exceeds **13°C** on a randomly chosen winter day?
- What is the probability that the maximum temperature is **below 7°C**?
- Using the **symmetry of the normal distribution**, explain why the answers to questions (1) and (2) must be equal.
- Using this fact, compute the probability that the maximum temperature lies **between 7°C and 13°C**.

Hint: Transform T in a standard normal variable and apply:

$$F_Z(1) = P(Z \leq 1) \approx 0.84, \quad Z \sim \mathcal{N}(0, 1)$$

Temperatures with normal distribution

Define the standard normal variable:

$$Z = \frac{X - \mu}{\sigma} = \frac{X - 10}{3}, \quad Z \sim \mathcal{N}(0, 1).$$

- **Note that:** $P(X > 13) = P\left(\frac{X - 10}{3} > \frac{13 - 10}{3}\right) = P(Z > 1) = 1 - \Phi(1) \approx 1 - 0.8413 = 0.1587$
- **For the same reason:**

$$P(X < 7) = P\left(\frac{X - 10}{3} < \frac{7 - 10}{3}\right) = P(Z < -1) = \Phi(-1) \approx 0.1587$$

- The normal distribution is symmetric about its mean $\mu = 10$. And the values 13 and 7 are equally distant from the mean:

$$13 - \mu = 3, \quad \mu - 7 = 3.$$

Therefore: $P(X > 13) = P(X < 7)$.

- Using the last point:

$$P(7 \leq X \leq 13) = 1 - P(X < 7) - P(X > 13) = 1 - 2(0.1587) = 0.6826.$$

Temperatures with normal distribution

Meteorologists want to define a **weather alert threshold** for unusually warm winter days in this city. They decide to issue an alert whenever the maximum temperature exceeds a value that is only exceeded on **5% of winter days**.

Using the model:

$$T \sim \mathcal{N}(10, 3^2)$$

find the temperature threshold t_{alert} such that:

$$P(T > t_{\text{alert}}) = 0.05$$

Temperatures with normal distribution

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Using the model:

$$T \sim \mathcal{N}(10, 3^2)$$

find the temperature threshold t_{alert} such that:

$$P(T > t_{\text{alert}}) = 0.05$$

Equivalently:

$$P(T \leq t_{\text{alert}}) = 0.95 \Rightarrow P\left(\frac{T - 10}{3} \leq \frac{t_{\text{alert}} - 10}{3}\right) = 0.95.$$

For Z the normal random variable:

$$Q(0.95) = \inf\{z : P(Z \leq z) \geq 0.95\} = 1.645$$

Then (**why?**):

$$\frac{t_{\text{alert}} - 10}{3} = 1.6545 \Rightarrow t_{\text{alert}} \approx 14.94^\circ\text{C}$$

Adding distributions

The sum of two independent normal random variables is still normally distributed.

Do you know an example of distribution for which the sum of two independent random variables with the same distribution does *not* belong to the same family?

Adding distributions

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Do you know an example of distribution for which the sum of two independent random variables with the same distribution does *not* belong to the same family?

Let:

$$X \sim \text{Uniform}([0, 1]), \quad Y \sim \text{Uniform}([0, 1])$$

independent. Then $X + Y$ does not follow a uniform distribution. In fact, its density is triangular:

Adding continuous distributions

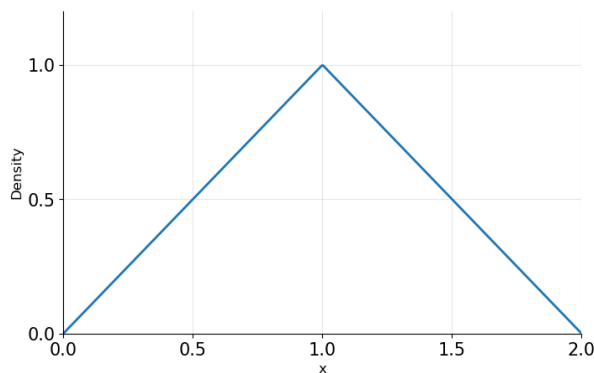
We have just seen that the sum of two independent normal random variables is still normally distributed. Do you know an example of a continuous distribution for which the sum of two independent random variables with the same distribution does *not* belong to the same family?

Let:

$$X \sim \text{Uniform}([0, 1]), \quad Y \sim \text{Uniform}([0, 1])$$

independent. Then $X + Y$ does not follow a uniform distribution. In fact, its density is triangular:

$$f_{X+Y}(t) = \begin{cases} t, & 0 < t < 1, \\ 2 - t, & 1 \leq t < 2, \\ 0, & \text{otherwise.} \end{cases}$$



Sum of random variables

In general, it is not easy to categorize the distribution of a sum of random variables.

Proposition:

If X and Y are independent and discrete with probabilities p_X, p_Y , then the sum $Z = X + Y$ has distribution given by:

$$p_Z(k) = \sum_j p_X(j) p_Y(k - j)$$

In the case of a continuous variables with densities f_X, f_Y the density of Z is:

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z - x) dx.$$

Important case:

Normal random variable: If $X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2)$, $X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$ are independent, then:

$$X_1 + X_2 \sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$$

Simulation, modeling, estimators and convergence

Class 5

Objective:

Understand what a simulation is, practice probabilistic models,
and reflect on the convergence of estimators.

Simulating random variables

Simulating random variables

A computer simulation generates values that follow the probability distribution of the random variable, allowing us to approximate its behavior through repeated sampling.

Generating binomial random variables

```
import numpy as np

p = 0.3

# one sample
x = np.random.binomial(n=1, p=p)

# many samples
samples = np.random.binomial(n=1, p=p, size=10_000)
```

Generating exponential random variables

```
lam = 0.5 # rate parameter

# one sample
x = np.random.exponential(scale=1/lam)

# many samples
samples = np.random.exponential(scale=1/lam, size=10_000)
```


Simulating random variables

Generating normal random variables

```
mu, sigma = 10.0, 2.0

# one sample
x = np.random.normal(loc=mu, scale=sigma)

# many samples
samples = np.random.normal(loc=mu, scale=sigma, size=10_000)
```

Generating poisson random variables

```
lam = 4.0

# one sample
x = np.random.poisson(lam=lam)

# many samples
samples = np.random.poisson(lam=lam, size=10_000)
```

Simulating random variables

When you simulate a normal random variable, you expect the proportion of simulated values falling in any given interval to approximate the true probability of that interval.

For example, if:

$$X \sim \mathcal{N}(0, 1)$$

the theoretical probability that X lies between $[-1, 1]$:

$$P(-1 \leq X \leq 1) = \int_{-1}^1 \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx \approx 0.6826.$$

So, if you generate many simulations in Python, you should observe that roughly **68%** of the samples fall inside the interval.

As the number of simulations increases, the empirical proportion should get closer to the theoretical value.

Question: Is it really possible to generate random samples ?

Randomness

Can an algorithm generate true randomness?

Randomness

Can an algorithm generate true randomness?

No, an algorithm cannot generate true randomness.

Algorithms are deterministic: if you know the algorithm and the initial input, the output is predictable. They can only generate **pseudo-randomness**, which looks random but is not truly unpredictable.

But be careful: each time you run the simulation, you will obtain a different result!

An interesting curiosity: In practice, the basic source of randomness in algorithms is a uniform pseudo-random distribution. Let:

$$U \sim \text{Uniform}([0, 1])$$

and let F_X be the cumulative distribution of a continuous random variable X .

Let

$$Y = F_X^{-1}(U)$$

Then:

Y has the same distribution as X .

Example of simulation

If you want to obtain always the same run, you can fix the seed parameter

```
# Fix seed for reproducibility  
np.random.seed(8)
```

Example of simulation and notation

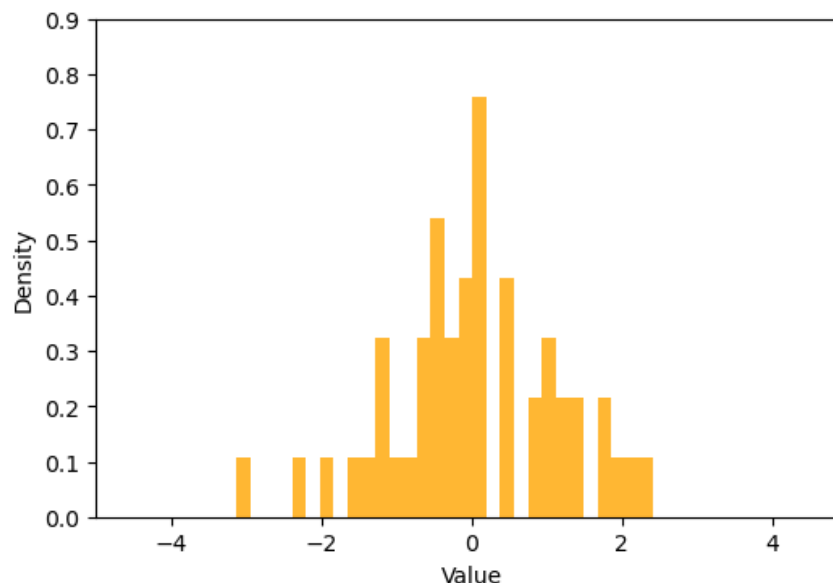
Simulation of normal distribution, density histogram for $n = 50$

Consider the sequence of independent random variables: $(X_1, X_2, \dots, X_{50})$

When we simulate this sequence, we obtain a data base: $(x_1, x_2, \dots, x_{50})$

that is a **realization** or **outcome** of the random vector: $(X_1, X_2, \dots, X_{50})$

Plot for $X_i \sim \text{Normal}(0, 1)$, for all $i \in \{1, \dots, 50\}$



Example of simulation

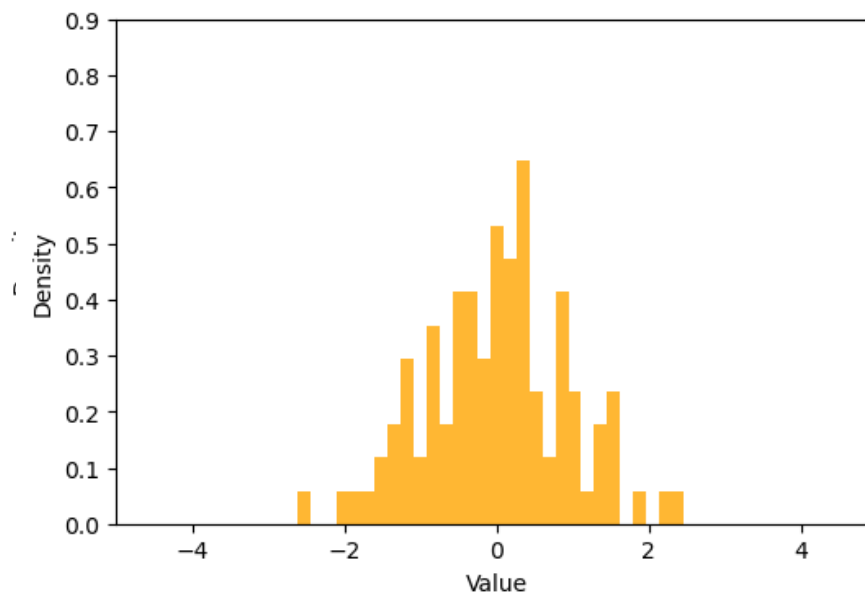
Simulation of normal distribution, density histogram for $n = 100$

Consider the sequence of independent random variables : $(X_1, X_2, \dots, X_{100})$

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that is a **realization** of the random vector: $(X_1, X_2, \dots, X_{100})$

Plot for $X_i \sim \text{Normal}(0, 1)$, for all $i \in \{1, \dots, 100\}$



Example of simulation

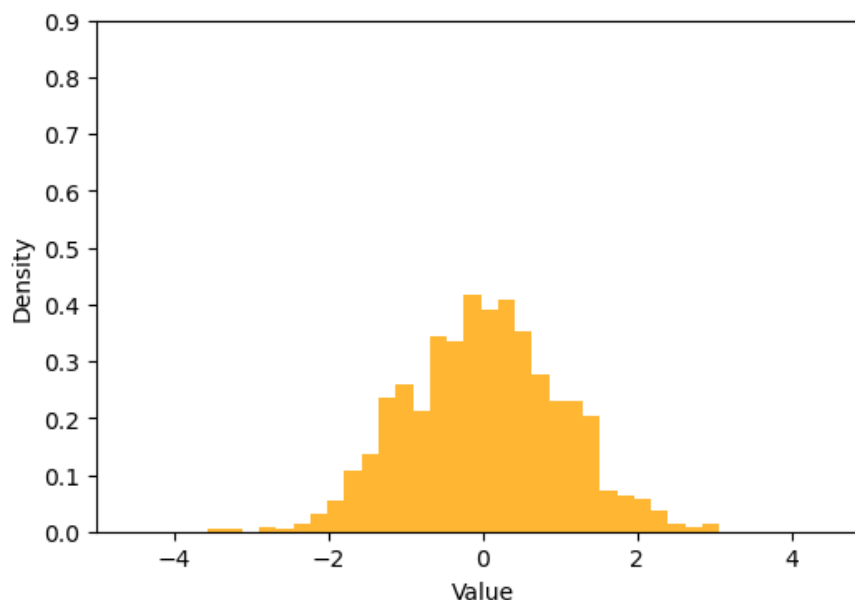
Simulation of normal distribution, density histogram for $n = 1000$

Consider the sequence of independent random variables : $(X_1, X_2, \dots, X_{1000})$

When we simulate this vector, we obtain a data base: $(x_1, x_2, \dots, x_{1000})$

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Plot for $X_i \sim \text{Normal}(0, 1)$, for all $i \in \{1, \dots, 1000\}$



Example of simulation

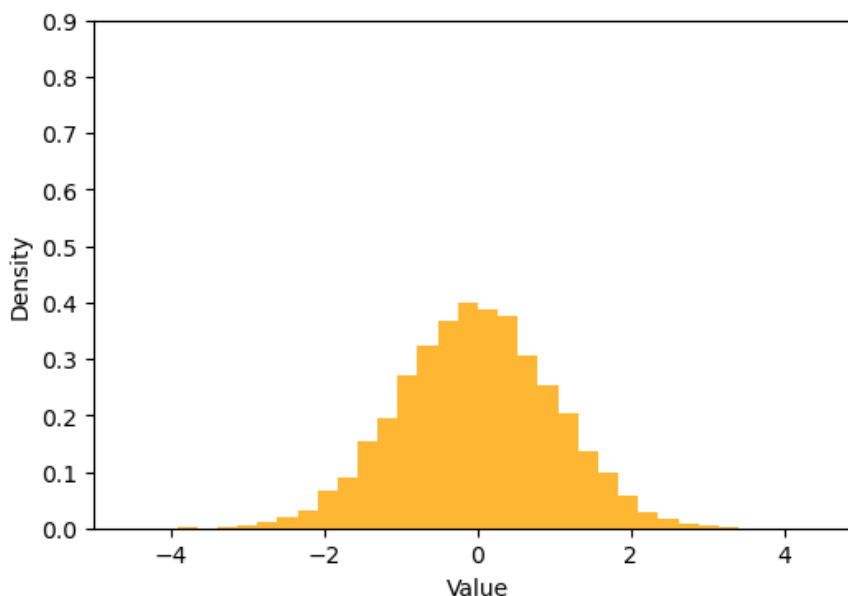
Simulation of normal distribution, density histogram for $n = 10000$

Consider the sequence of independent random variables : $(X_1, X_2, \dots, X_{10000})$

When we simulate this vector, we obtain a data base: $(x_1, x_2, \dots, x_{10000})$

that is a **realization** of the random vector: $(X_1, X_2, \dots, X_{10000})$

Plot for $X_i \sim \text{Normal}(0, 1)$, for all $i \in \{1, \dots, 10000\}$

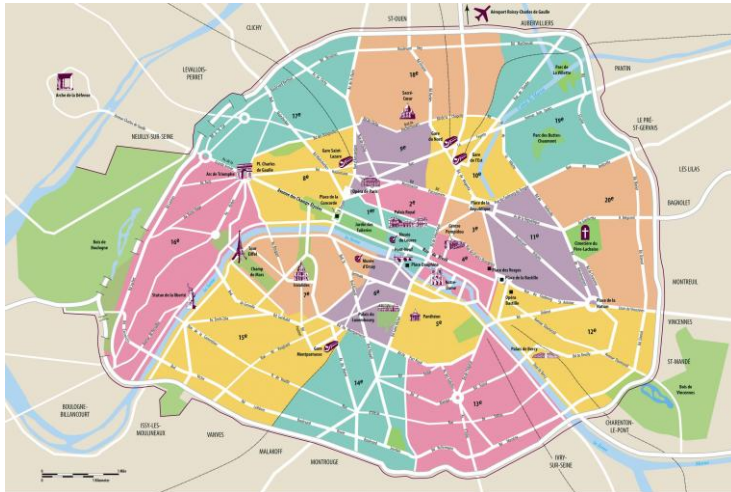


Back to the motivation

Election pool

A change in perspective: So far, we have visualized and analyzed some useful measures to understand data. But what if these data represent not an entire population, but only a sample?

Example: Election pool



Candidate	Number of respondents
Candidate A	362
Candidate B	638

Table 2: Election poll results with 1,000 respondents.

Statistical question: What is the probability that candidate B will win the election?

Model for election pool

Probabilistic model

Let p be the **true** (unknown) proportion of people who vote for candidate A, and $1-p$ the proportion who vote for candidate B. Based on the survey, we aim to **estimate** p .

Therefore, we have a Bernoulli random variable with parameter p :

$$X = \begin{cases} 1, & \text{if the person votes for candidate A,} \\ 0, & \text{if the person votes for candidate B.} \end{cases}$$

Data

The survey is viewed as a **realization** or **outcome** $(x_1, x_2, \dots, x_{1000})$ of the random vector $(X_1, X_2, \dots, X_{1000})$, where $X_i \sim \text{Bernoulli}(p), i \in \{1, \dots, 1000\}$

What are we assuming with this model?

- All respondents are independent
- The value of p is fixed during the survey
- Each respondent is chosen at random

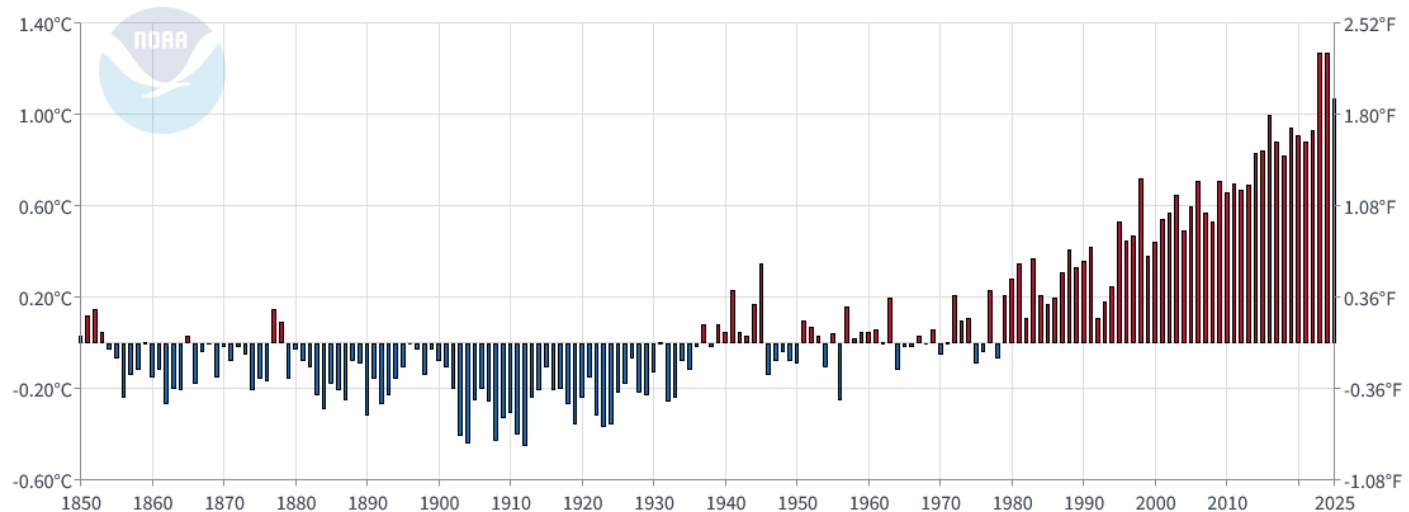


Climate change illustration

Graphic of mean anomaly temperature by year: $A(t) = T_{\text{obs}}(t) - \overline{T}_{\text{baseline}}$

Global Land and Ocean Average Temperature Anomalies

August



Statistical question: How many years we need to observe that the temperature is rising ?

Model for climate change

Probabilistic model

Let μ and σ denote the true (unknown) mean and standard deviation of the annual temperature anomalies. Our goal is to estimate μ .

Therefore, we have a normal random variable with parameters μ and σ :

$$X = \text{Temperature anomaly of a given year} \sim \mathcal{N}(\mu, \sigma)$$

Data

The survey is viewed as a realization $(x_1, x_2, \dots, x_{145})$ of the random vector $(X_1, X_2, \dots, X_{145})$ where $X_i \sim \mathcal{N}(\mu, \sigma), i \in \{1, \dots, 145\}$.

What are we assuming with this model?

- Temperature anomalies have normal distribution
- The parameters are fixed in 145 years



Statistics procedure

- Based on a real situation, we propose a **model**
 - We **collect data**
 - We **estimate** the parameters of interest in our model
-

→ It is common for probabilistic models to assume independence among the random variables involved.

Recalling the concept of independence

Definition:

We say that random variables X_1, X_2 are independent random variables if for any events A, B :

$$P(X \in A, Y \in B) = P(X \in A)P(X \in B)$$

Or, equivalently:

$$F_{X,Y}(x, y) = P(X \leq x, Y \leq y) = F_X(x) F_Y(y), \quad \forall x, y \in \mathbb{R}.$$

Independence quiz:

- Let X be the outcome of the first roll of a die, and Y the outcome of the second roll.
- Let X be the outcome of a single die roll, and Y be the indicator that the die shows an even number ($Y=1$ if even, 0 otherwise)
- Let X represent the future daily average temperature in a city and Y the corresponding future daily electricity consumption.
- Let X and Y be independent random variables and $Z = X + Y$. Are Z and X independent?

Estimators

Estimators

We can use the random vector: $(X_1, X_2, \dots, X_n)$ to estimate things:

Examples:

Mean estimator

$$\bar{X}_n := \sum_{i=1}^n X_i$$

Max estimator

$$M_n = \max(X_1, \dots, X_n)$$

Variance estimator

$$\bar{S}_n := \frac{1}{n-1} \sum_{i=1}^{n-1} (X_i - \bar{X}_n)^2$$

Observations:

- Estimators have distributions!
- These distributions are usually very difficult to compute

Mean estimator

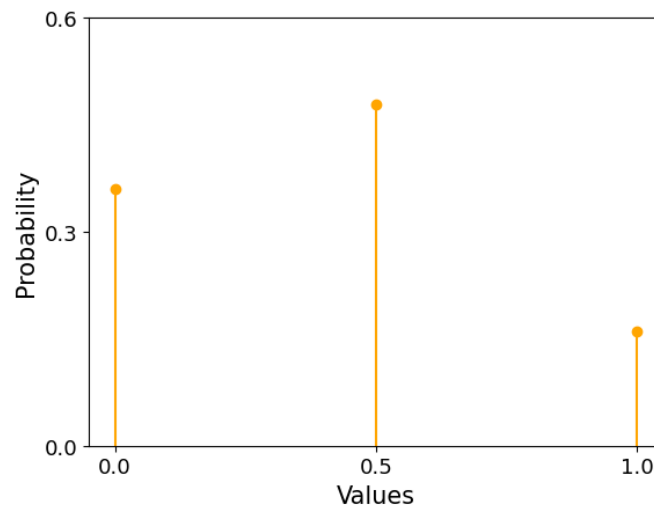
Mean estimator:

$$\bar{X}_n = \frac{X_1 + X_2 + \dots + X_n}{n}$$

- What does the mean estimator represent in the context of **tossing a coin**?

$$X \sim \text{Bernoulli}(0.4)$$

n = 2



Mean estimator

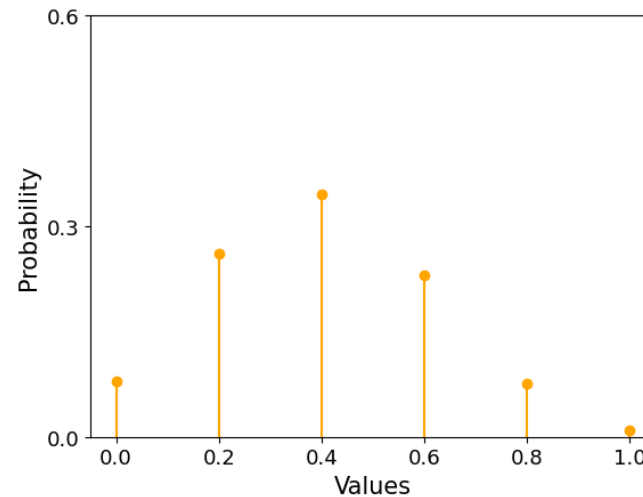
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Mean estimator

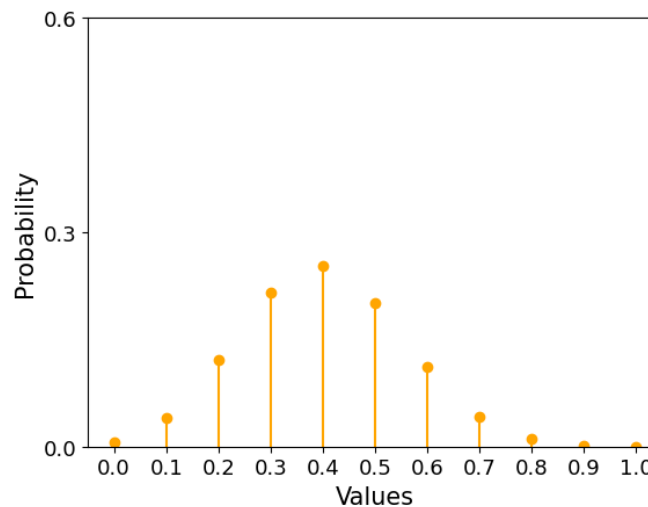
Mean estimator:

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- What does the mean estimator represent in the context of **tossing a coin**?

$$X \sim \text{Bernoulli}(0.4)$$

n = 10



Mean estimator

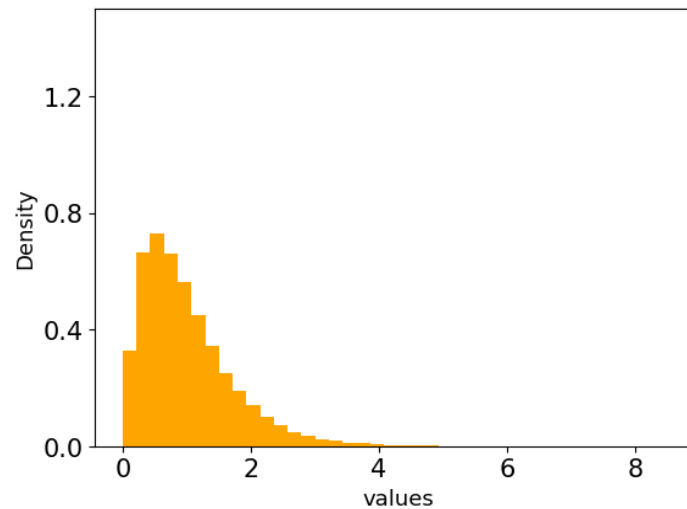
Mean estimator:

$$\bar{X}_n = \frac{X_1 + X_2 + \dots + X_n}{n}$$

- What does the mean estimator represent in the context of **waiting time**?

$$X \sim \text{Exponential}(1)$$

n = 2



Mean estimator

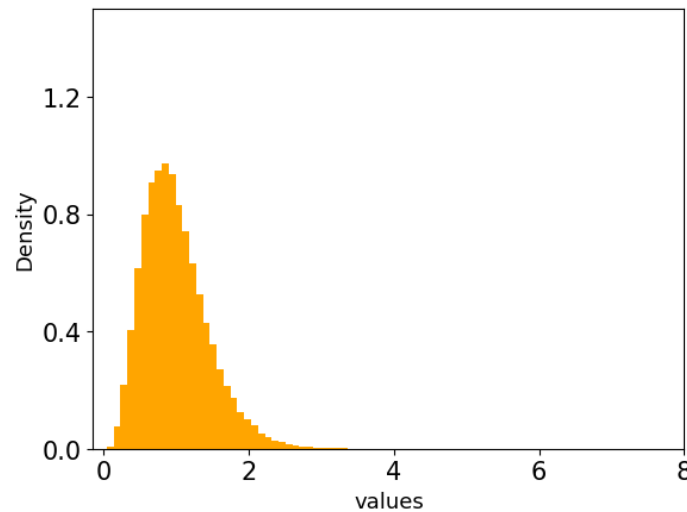
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Mean estimator

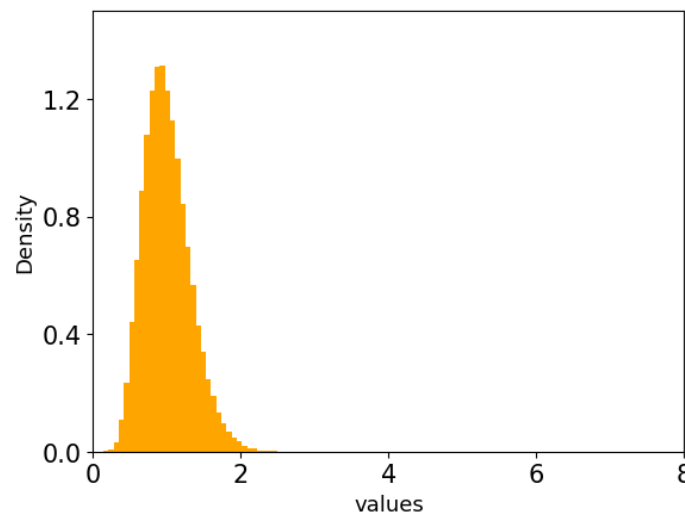
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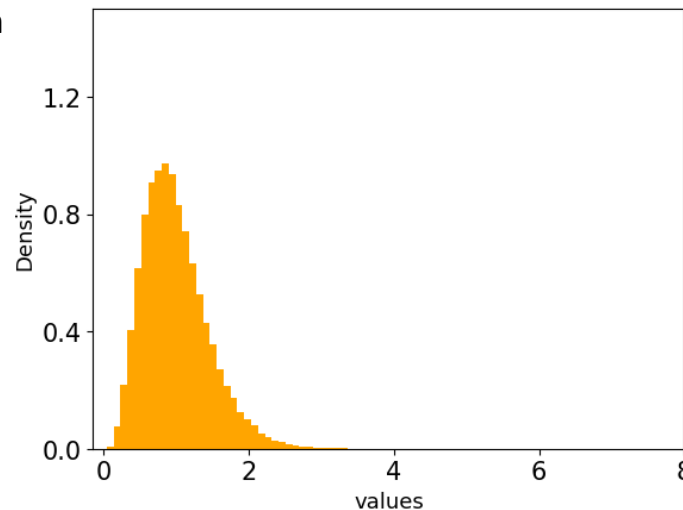
Mean estimator

Mean estimator:

$$\bar{X}_n = \frac{X_1 + X_2 + \dots + X_n}{n}$$

- It is easy to see that: $\mathbb{E} [\bar{X}] = \mu$
- **Question:** How do you think the distribution of the mean changes as n becomes large?

Hint: Compute the varian



Law of large numbers

Deriving the Law of Large Numbers

- The variance decreases with n and converges to zero
- The values concentrate around the expected value
- The expected value is constant in n and is equals to μ
- We would like to formulate:

$$\underbrace{\bar{X}_n}_{\text{Random variable / Function}} \quad \xrightarrow{\quad} \quad \underbrace{\mu}_{\text{Expected value / Number}}$$

|

when n goes to infinity

Law of large numbers

Theorem (strong law of large numbers):

Let $X_1, X_2, \dots$ be a sequence of independent and **identically distributed** random variables with $\mathbb{E}[|X_1|] < \infty$, $\mu = \mathbb{E}[X_1]$ and $\text{Var}(X_1) = \sigma^2$. Then :

$\bar{X}_n \xrightarrow{\text{a.s.}} \mu$ as $n \rightarrow \infty$, that is: $P\left(\lim_{n \rightarrow \infty} \bar{X}_n = \mu\right) = 1. \rightarrow$ Almost surely convergence

And for any $\epsilon > 0$:

$$P(|\bar{X}_n - \mu| \geq \epsilon) \leq \frac{\sigma^2}{n\epsilon^2} \rightarrow \text{Probability convergence}$$

$P(|\bar{X}_n - \mu| \geq \epsilon)$ reads:

When we look at all possible outcomes of the random sample mean ($\bar{X}_n$), the probability that the it differs from the true mean (μ) by more than ϵ can be bounded above by a quantity that depends on the variance (σ^2), on the sample size n , and on ϵ .

Law of large numbers

Intuitive explanation from theory:

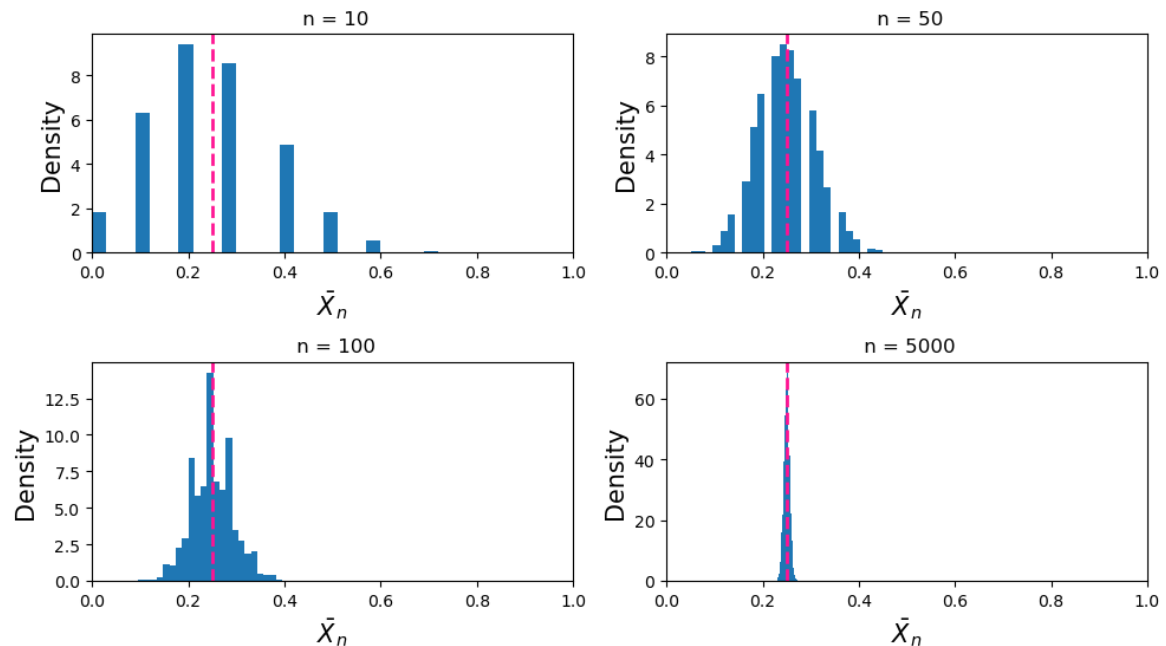
Note that $\mathbb{E}[\bar{X}_n] = \mu$. Suppose: $\text{Var}(X_i) = \sigma^2 < \infty$, then:

$$\text{Var}(\bar{X}_n) = \text{Var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n^2} \text{Var}\left(\sum_{i=1}^n X_i\right) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}.$$

The variance decreases as n increases.

Numerical intuition:

Bernoulli($p=1/4$): distribution of the sample mean



Back to the election pool

Election pool

A change in perspective: So far, we have visualized and analyzed some useful measures to understand data. But what if these data represent not an entire population, but only a sample?

Example: Election pool



Candidate	Number of respondents
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Therefore, we have a Bernoulli random variable with parameter p :

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Data

The survey is viewed as a **realization** or **outcome** $(x_1, x_2, \dots, x_{1000})$ of the random vector $(X_1, X_2, \dots, X_{1000})$, where $X_i \sim \text{Bernoulli}(p), i \in \{1, \dots, 1000\}$

What are we assuming with this model?

- All respondents are independent
- The value of p is fixed during the survey
- Each respondent is chosen at random



Estimating p

How to estimate p?

Consider the **estimator**:

$$\bar{X}_{1000} = \frac{X_1 + X_2 + \dots + X_{1000}}{1000}$$

We have access to **one realization** of this random variable:

$$\bar{x}_{1000} = \frac{x_1 + x_2 + \dots + x_{1000}}{1000} = \frac{\# \{ \text{Number of people voting for candidate A} \}}{1000} = \frac{362}{1000}$$

Because of the Law of Large Numbers:

- We know that, regardless of which individuals are interviewed, as the sample size becomes large, the sample mean $\bar{X}_n$ converges to p and:

$$P(|\bar{X}_n - p| \geq \epsilon) \leq \frac{p(1-p)}{1000\epsilon^2} \leq \frac{1}{4 \times 1000\epsilon^2}$$

What does it means ?

Consequences of LLN theorem (election)

Question: Given the result of the survey, what is the probability that candidate A wins the election?

$$\begin{array}{c} \text{Candidate A wins} \\ \text{the election} \end{array} \Leftrightarrow \{p \geq 0.5\}$$

Consider the event:

$$A = \{p \geq 0.5 \text{ and } \bar{X}_n = 0.362\}$$

We want to estimate the $P(A)$. Note that:

$$A \subset \{|\bar{X}_n - p| \geq \epsilon = 0.138\}$$

Implying that:

$$P(A) \leq P(|\bar{X}_n - p| \geq \epsilon = 0.138) \leq \frac{1}{4 \times 1000 \times 0.138^2} = \frac{1}{76.17} \approx 0.01$$

Resulting in:

The probability that candidate A wins the election given the result of the survey is at most 1%

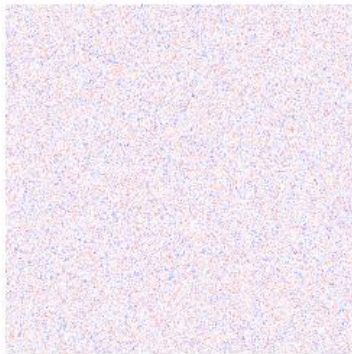
A visual example

An image:



Sampling noise with normal distribution with mean 0:

x_1 sampled from $N(0, \sigma^2)$



x_2 sampled from $N(0, \sigma^2)$



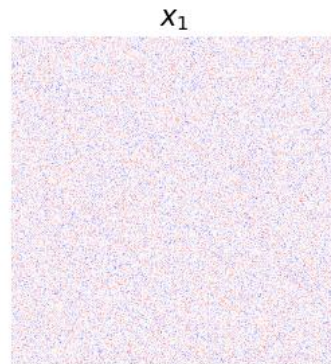
x_3 sampled from $N(0, \sigma^2)$



The colors are a form to illustrate noise: white – close to zero; blue – positive; red - negative

A visual example

Noisy image:



According to the Law of Large Numbers, the average of the noises should approach zero as n increases, and the image combined with the average noise should become clearer.

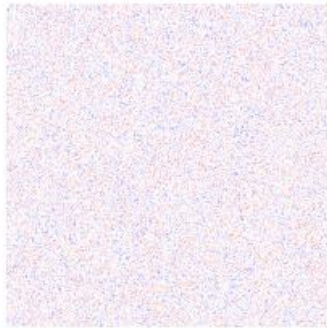
A visual example

Mean of noise samples and resulting noisy image:

$\bar{X}_1$



$\bar{X}_5$



$\bar{X}_{30}$



$\bar{X}_{100}$



Image + $\bar{X}_1$



Image + $\bar{X}_5$



Image + $\bar{X}_{30}$



Image + $\bar{X}_{100}$



Exercise

A hospital is studying the waiting times of patients in the emergency department. From previous studies, it is known that the waiting time (in minutes) can be modeled by an exponential distribution.

$$X \sim \text{Exponential}\left(\frac{1}{\lambda}\right)$$

The hospital is sure that the mean waiting time is at most 1 minute. To estimate the true mean waiting time, the hospital records the waiting times of n independent patients:

$$X_1, X_2, \dots, X_n.$$

They use the mean estimator:

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

as an estimator of the true mean.

The hospital requires that probability that the estimation error exceeds 1 minute is at most 0.05.

Using the Law of Large numbers determine how large n must be in order to this requirement hold.

Central limit theorem and confidence intervals

Class 6

Objective:

Understand the Central limit theorem and one of its most useful implications: confidence intervals

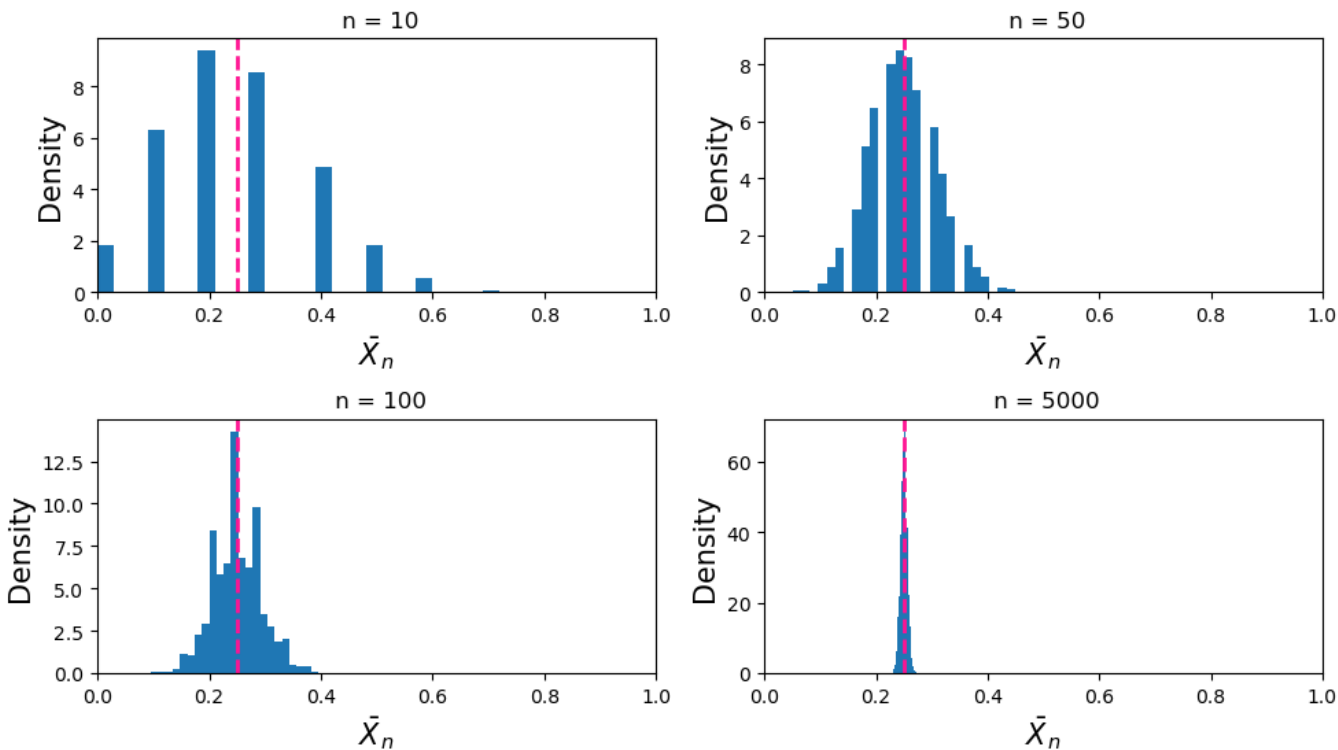
Central limit theorem

Examining the convergence of the mean

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

$X_i \sim \text{Bernoulli}(0.25)$, i.i.d

Bernoulli(p=1/4): distribution of the sample mean

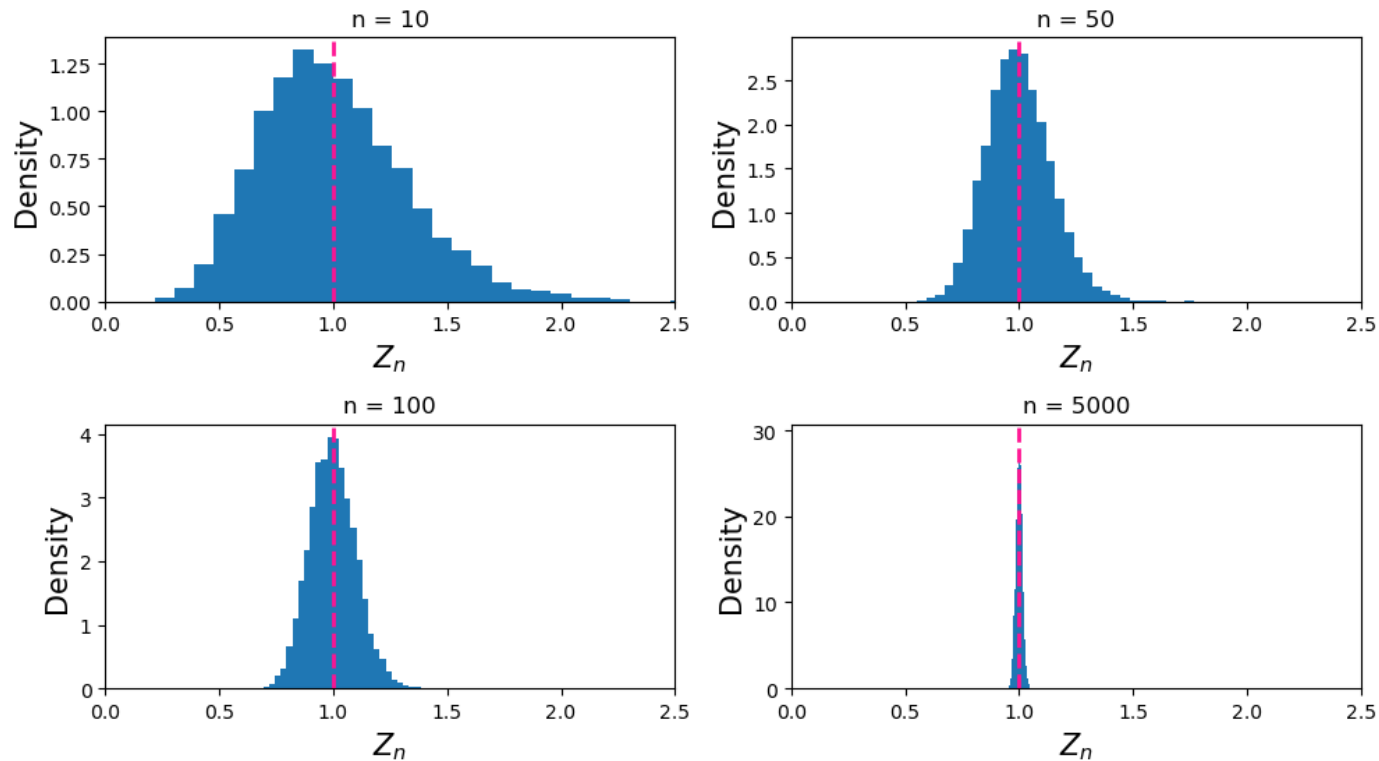


Examining the convergence of the mean

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

$$X \sim \text{Exponential}(1)$$

Exponential(lambda=1): distribution of the sample mean



Normalizing the mean estimator

Let $X_i, i \in \{1, \dots, n\}$ be identically distributed random variables and $\mathbb{E}[X_1] = \mu$
 $\text{Var}(X_1) = \sigma^2$. Then:

$$\mathbb{E}[\bar{X}_n] = \mu \quad \text{Var}(\bar{X}_n) = \frac{\sigma^2}{n}$$

Consider the random variable:

$$Z_n = \frac{\bar{X}_n - \mu}{\sqrt{\frac{\sigma^2}{n}}} \quad \text{Then: } \mathbb{E}[Z_n] = 0 \text{ and } \text{Var}(Z_n) = 1$$

Z_n can be written as:

$$Z_n = \frac{\frac{X_1 + X_2 + \dots + X_n}{n} - \mu}{\sqrt{\frac{\sigma^2}{n}}} = \frac{\frac{X_1 + X_2 + \dots + X_n - n\mu}{n}}{\frac{\sigma}{\sqrt{n}}} = \frac{X_1 + X_2 + \dots + X_n - n\mu}{\sigma\sqrt{n}}$$

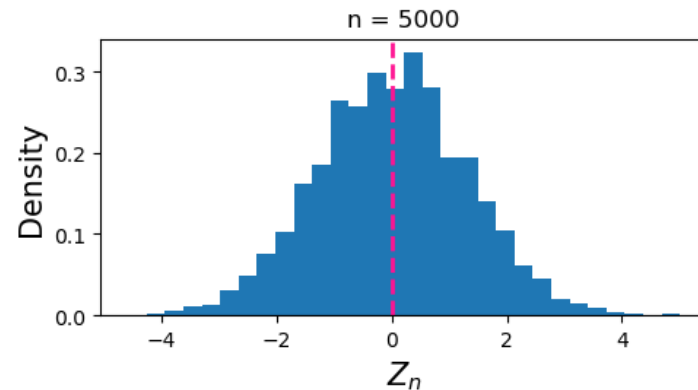
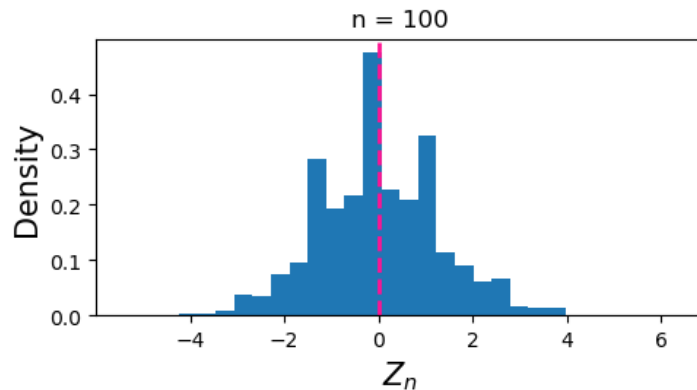
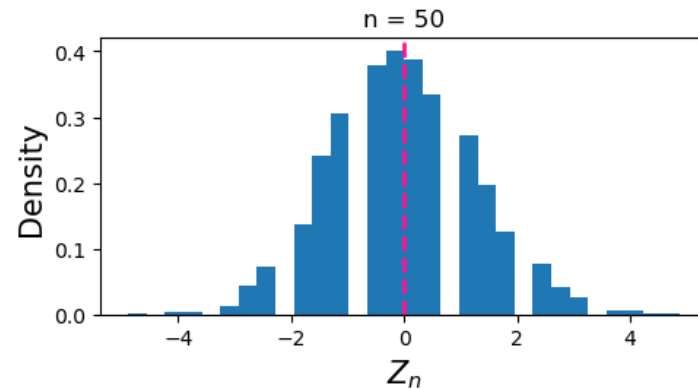
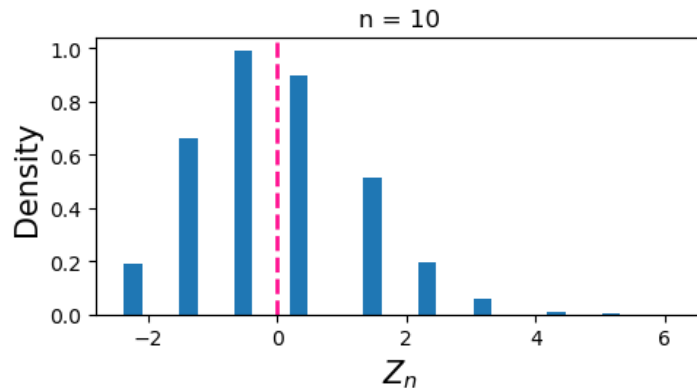
$$Z_n = \frac{\sum_{i=1}^n X_i - n\mu}{\sigma\sqrt{n}}$$

Visual normalization of the mean

$X_i \sim \text{Bernoulli}(0.25)$, i.i.d

$$Z_n = \frac{\sum_{i=1}^n X_i - n\mu}{\sigma\sqrt{n}}$$

Bernoulli(p=1/4): distribution of the sample mean

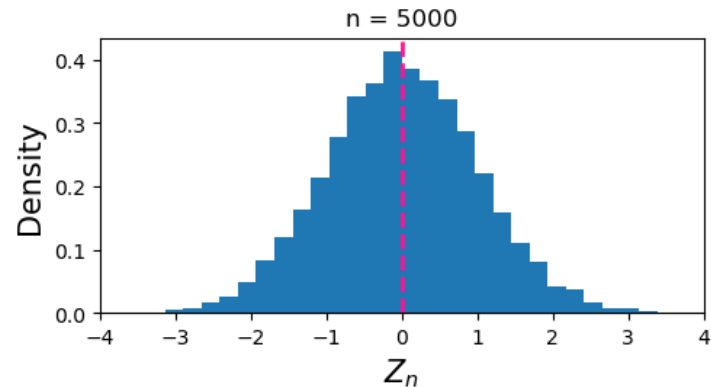
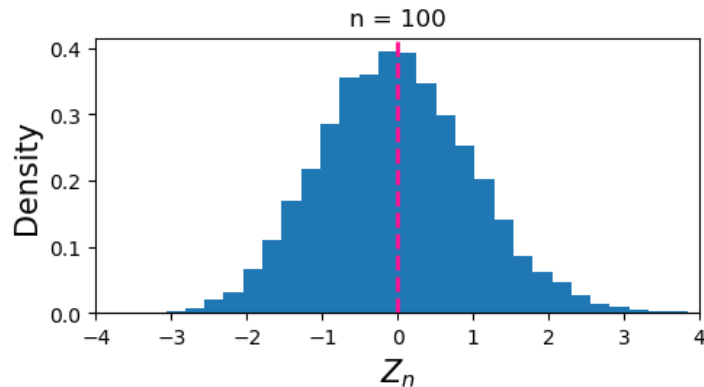
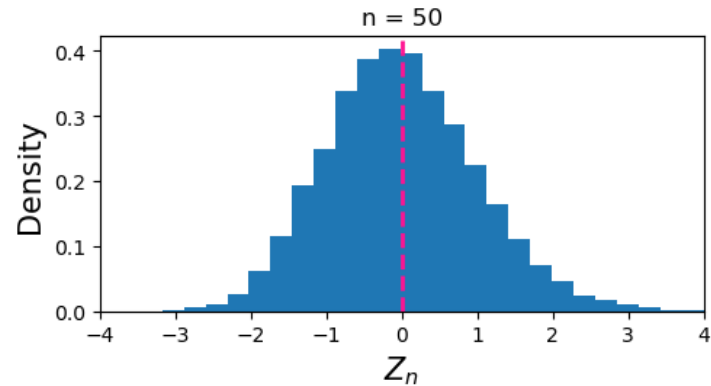
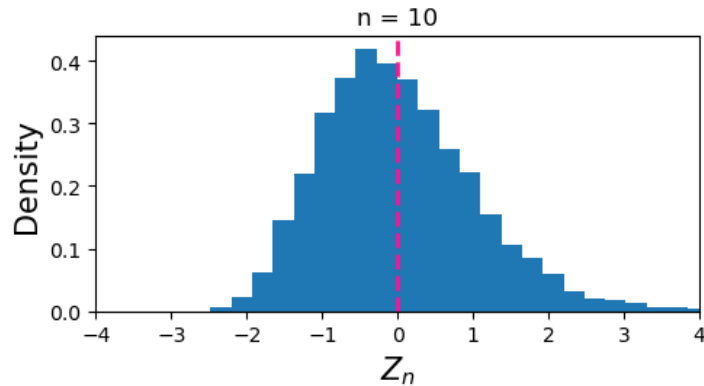


Visual normalization of the mean

$$X \sim \text{Exponential}(1)$$

$$Z_n = \frac{\sum_{i=1}^n X_i - n\mu}{\sigma\sqrt{n}}$$

Exponential(lambda = 1): distribution of the sample mean



Central limit theorem

The theorem:

Let $X_1, X_2, \dots$ be independent and identically distributed random variables with mean $\mu := \mathbb{E}[X_1]$ and variance $\sigma^2 = \text{Var}[X_1] < \infty$. Define the standardized sum:

$$Z_n = \frac{\sum_{i=1}^n X_i - n\mu}{\sigma\sqrt{n}}$$

Then, as $n \rightarrow \infty$, the distribution of Z_n converges to the standard normal distribution:

$$Z_n \xrightarrow{d} \mathcal{N}(0, 1)$$

Discussion:

- What should it mean the distribution convergence $\xrightarrow{d}$?

Convergence in distribution

Definition:

The sequence of random variables Z_n converges in distribution to $Z \sim \mathcal{N}(0, 1)$, if:

$$\lim_{n \rightarrow \infty} F_{Z_n}(t) = F_Z(t) \quad \forall t \in \mathbb{R}$$

In this case for all set $A \subset \mathbb{R}$ « sufficiently regular »:

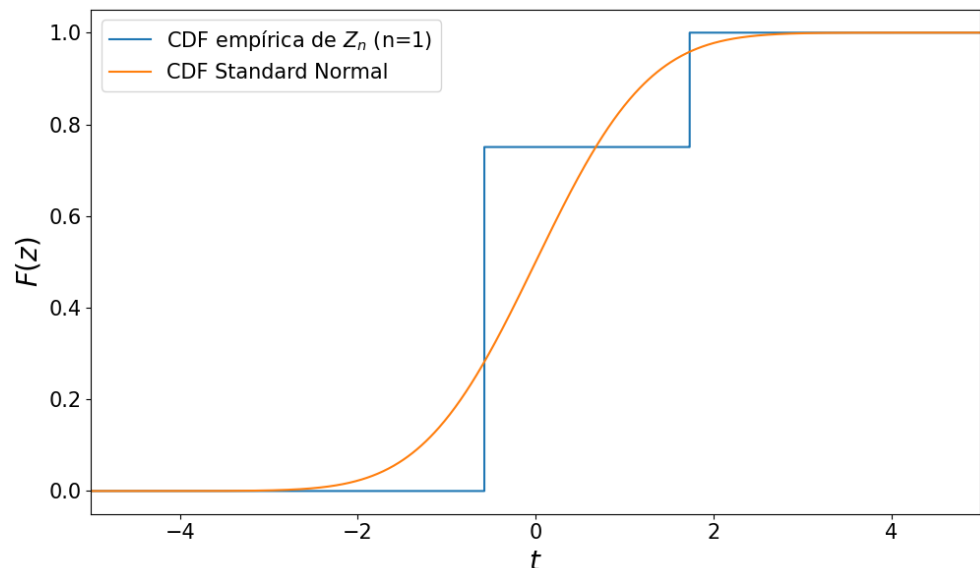
$$P(Z_n \in A) \rightarrow P(Z \in A) \rightarrow A \text{ can be any interval}$$

We denote: $Z_n \xrightarrow{d} Z$

Exemple:

$X_i \sim \text{Bernoulli}(0.25)$, i.i.d

$n = 1$



Convergence in distribution

Definition:

The sequence of random variables Z_n converges in distribution to $Z \sim \mathcal{N}(0, 1)$, if:

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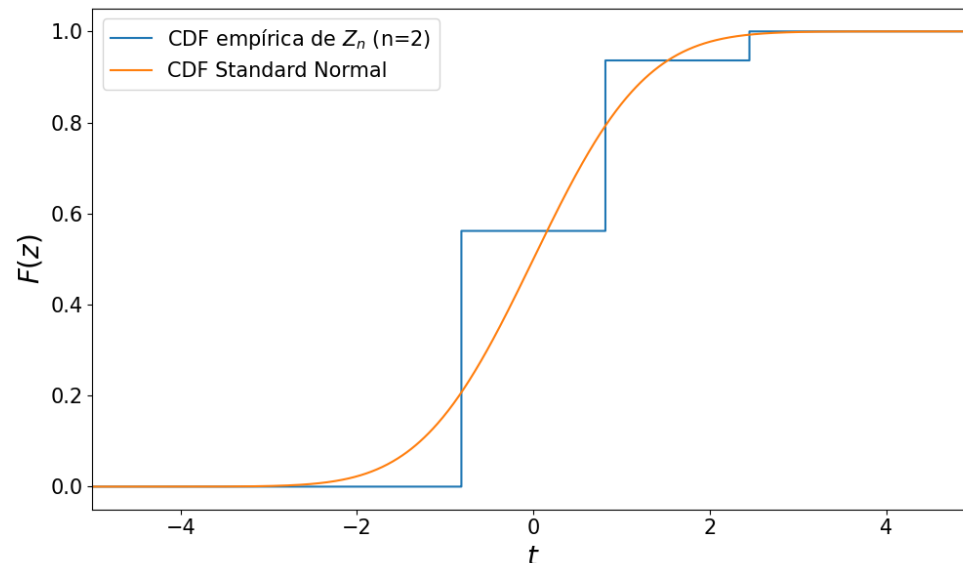
$$P(Z_n \in A) \longrightarrow P(Z \in A)$$

We denote: $Z_n \xrightarrow{d} Z$

Exemple:

$X_i \sim \text{Bernoulli}(0.25)$, i.i.d

$n = 2$



Convergence in distribution

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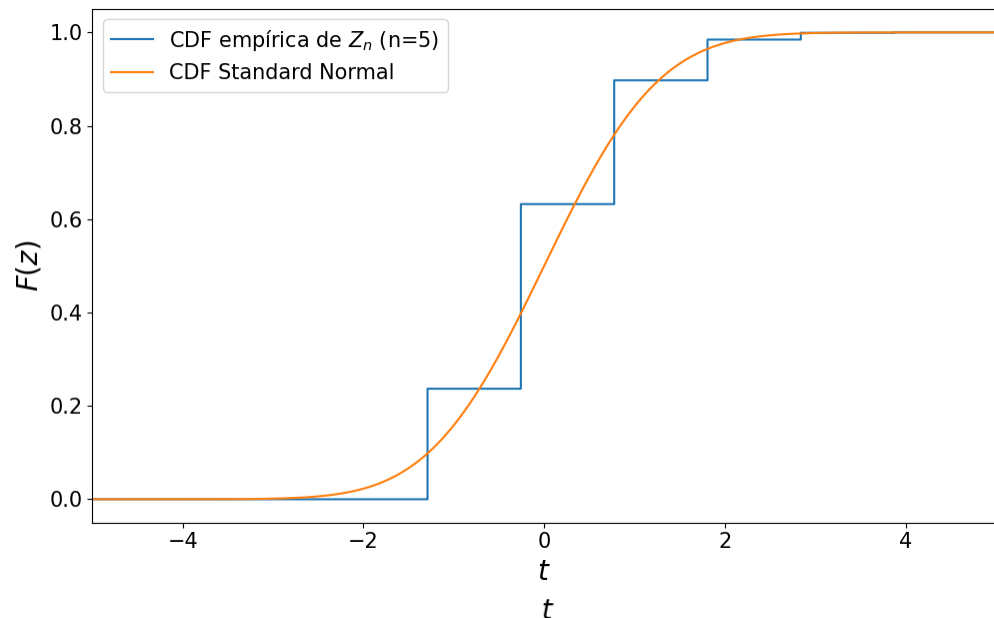
$$P(Z_n \in A) \longrightarrow P(Z \in A)$$

We denote: $Z_n \xrightarrow{d} Z$

Example:

$X_i \sim \text{Bernoulli}(0.25)$, i.i.d

$n = 5$



Convergence in distribution

Definition:

The sequence of random variables Z_n converges in distribution to $Z \sim \mathcal{N}(0, 1)$, if:

$$\lim_{n \rightarrow \infty} F_{Z_n}(t) = F_Z(t) \quad \forall t \in \mathbb{R}$$

In this case for all set $A \subset \mathbb{R}$ « sufficiently regular »:

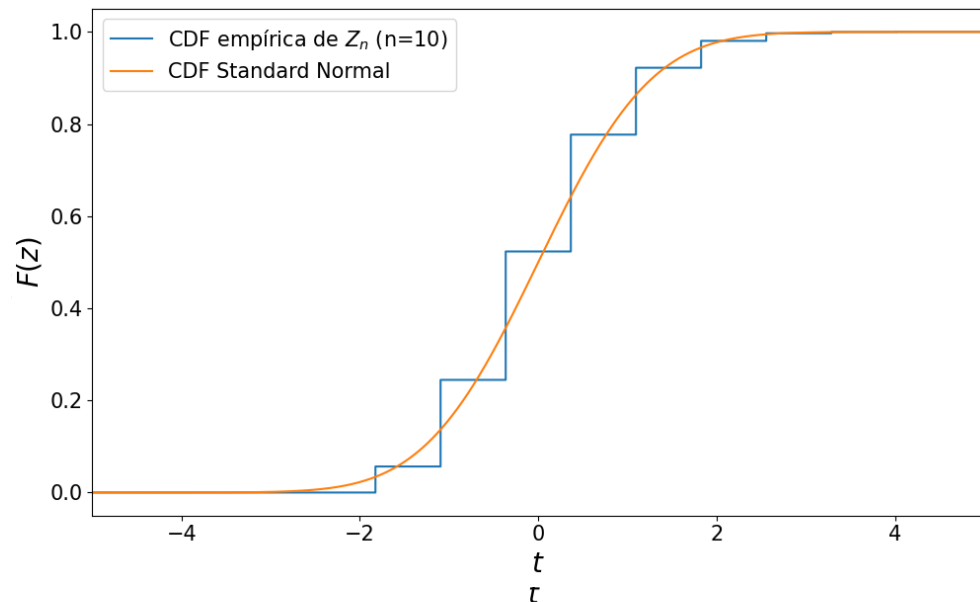
$$P(Z_n \in A) \longrightarrow P(Z \in A)$$

We denote: $Z_n \xrightarrow{d} Z$

Example:

$X_i \sim \text{Bernoulli}(0.25)$, i.i.d

$n = 10$



Convergence in distribution

Definition:

The sequence of random variables Z_n converges in distribution to $Z \sim \mathcal{N}(0, 1)$, if:

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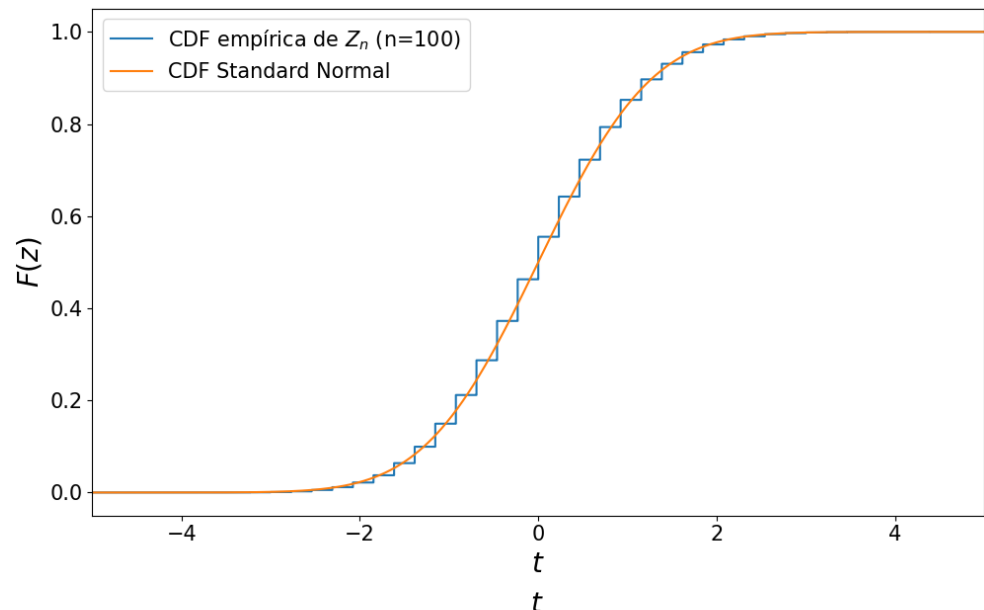
$$P(Z_n \in A) \longrightarrow P(Z \in A)$$

We denote: $Z_n \xrightarrow{d} Z$

Example:

$X_i \sim \text{Bernoulli}(0.25)$, i.i.d

$n = 100$



Confidence intervals

Practical application: All empirical science!

All empirical sciences (physics, biology, climatology) aim to infer something about an unknown reality from data. The natural next question is: given the data I have, what are the plausible values of the true underlying quantity?

Election pool

A change in perspective: So far, we have visualized and analyzed some useful measures to understand data. But what if these data represent not an entire population, but only a sample?

Example: Election pool



Candidate	Number of respondents
Candidate A	362
Candidate B	638

Table 2: Election poll results with 1,000 respondents.

Statistical question: How to estimate the real result?

Model for election pool

Probabilistic model

Let p be the **true** (unknown) proportion of people who vote for candidate A, and $1-p$ the proportion who vote for candidate B. Based on the survey, we aim to **estimate p** .

Therefore, we have a Bernoulli random variable with parameter p :

$$X = \begin{cases} 1, & \text{if the person votes for candidate A,} \\ 0, & \text{if the person votes for candidate B.} \end{cases}$$

Data

The survey is viewed as a **realization or outcome** $(x_1, x_2, \dots, x_{1000})$ of the random vector $(X_1, X_2, \dots, X_{1000})$, where $X_i \sim \text{Bernoulli}(p), i \in \{1, \dots, 1000\}$

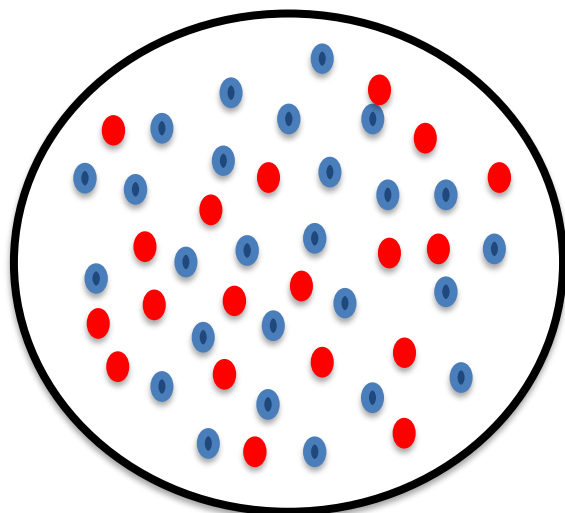
What are we assuming with this model?

- All respondents are independent
- The value of p is fixed during the survey
- Each respondent is chosen at random



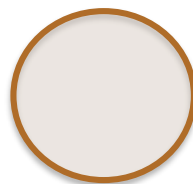
Model for election pool

Population



● Candidate A
● Candidate B

Survey



Model

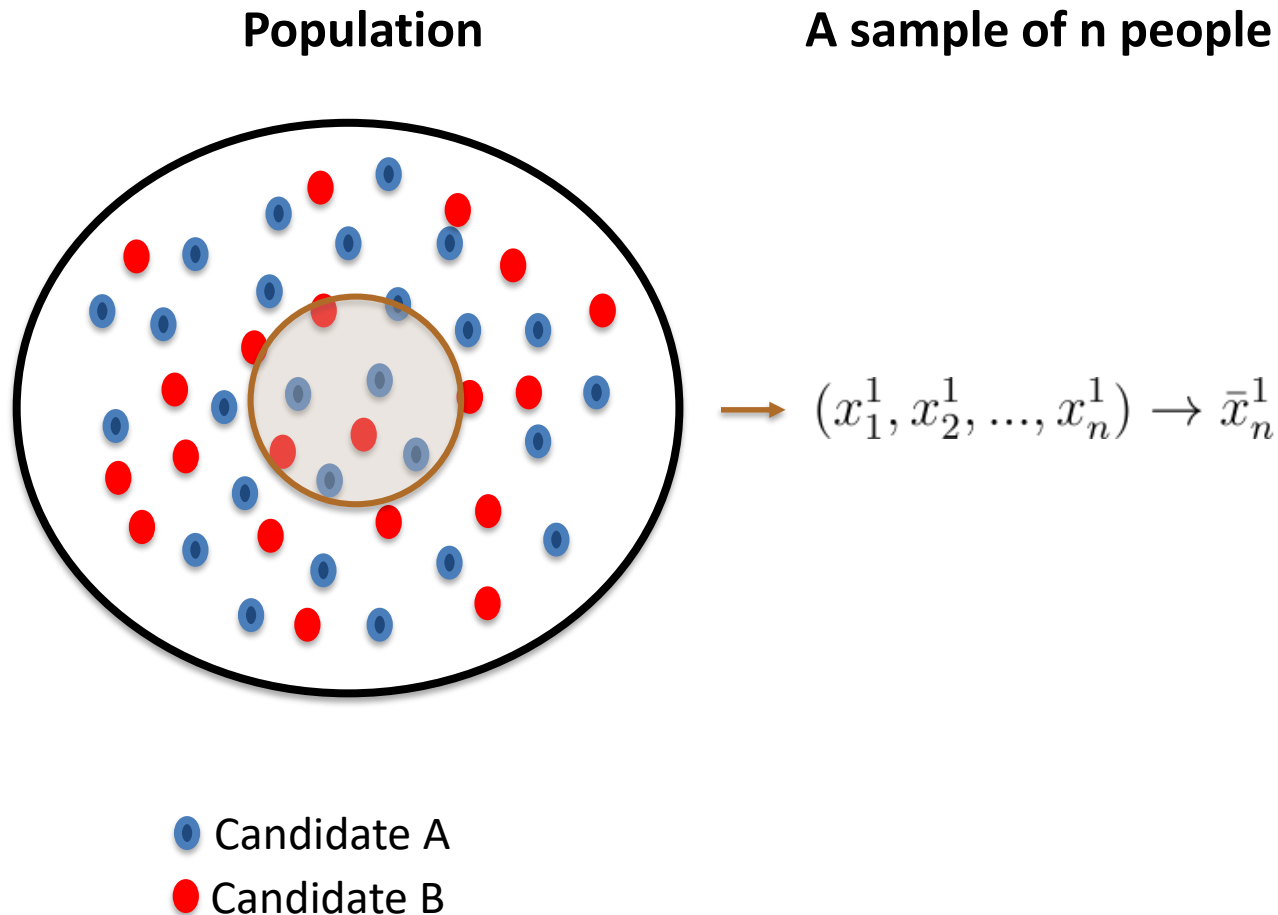
$$X = \begin{cases} 1, & \text{Candidate A,} \\ 0, & \text{Candidate B.} \end{cases}$$

$$X_i \sim \text{Bernoulli}(p), \quad i \in \{1, \dots, n\}$$

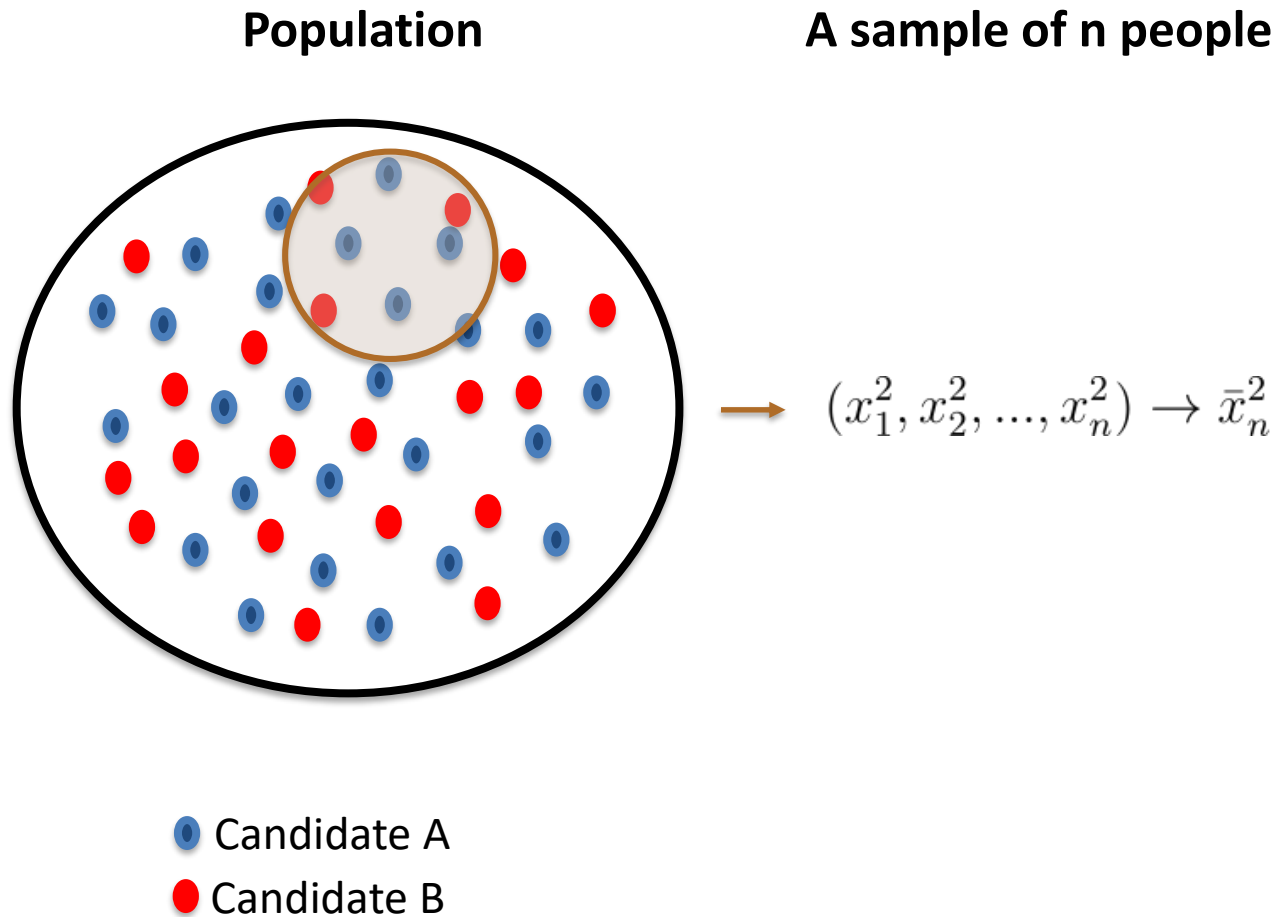
Data: $(x_1, \dots, x_n)$

Data mean: $\bar{x}_n$

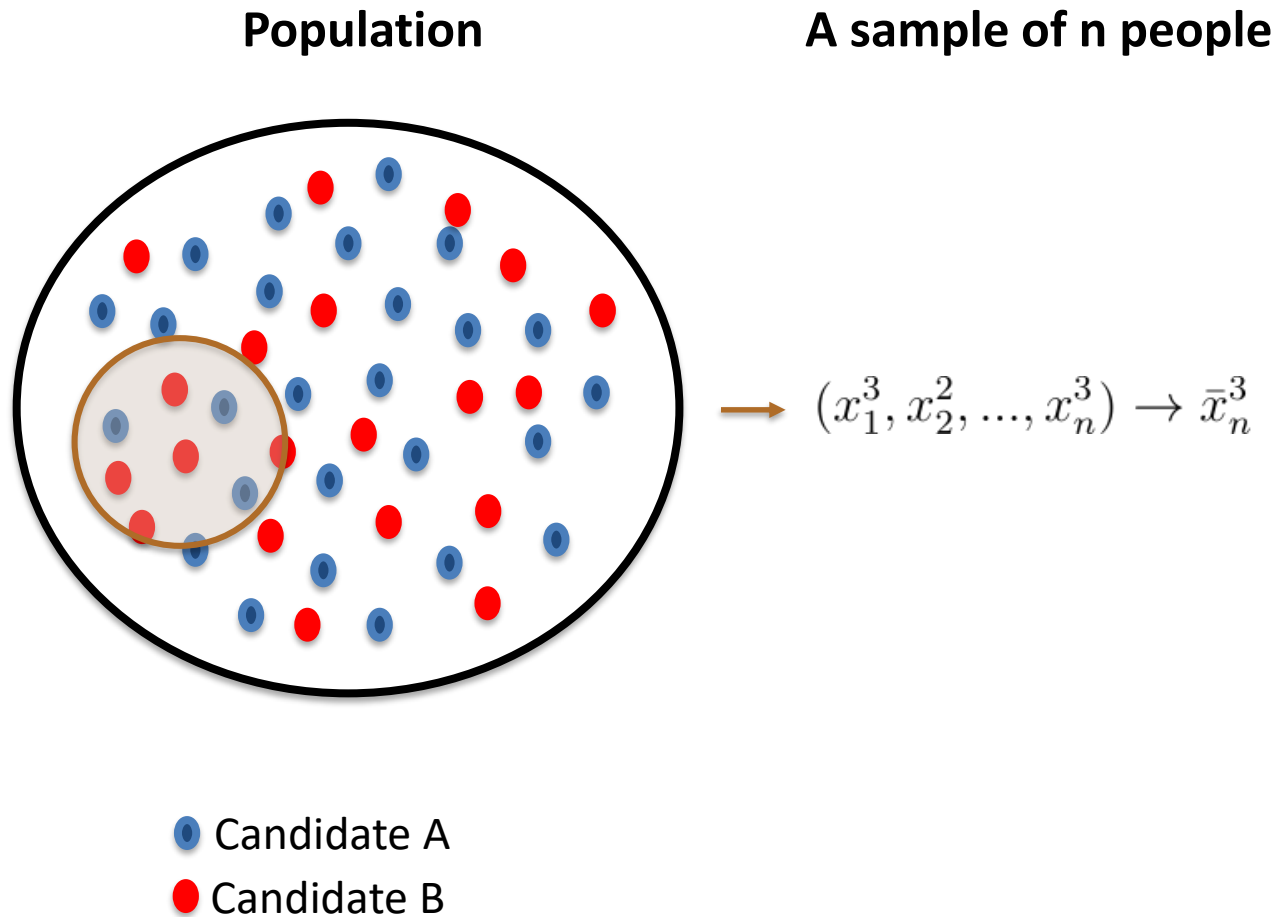
Possible outcomes of the mean estimator



Possible outcomes of the mean estimator



Possible outcomes of the mean estimator



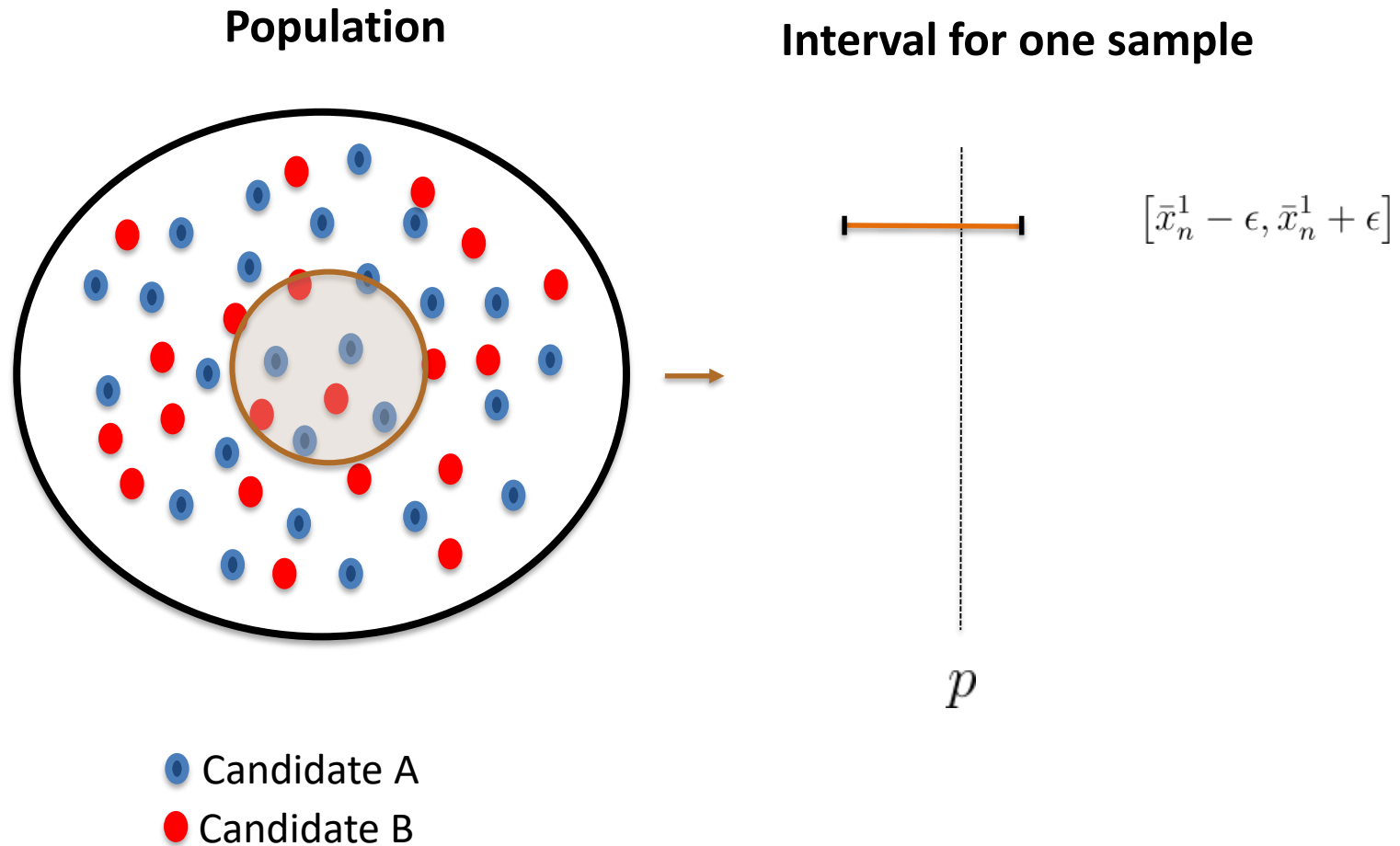
A random interval containing p

- $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$ is a random variable
- Our survey is one realization (or outcome) of $\bar{X}_n$
- We can not expect to estimate p as a fixed number
- We would like to consider a margin of error for our survey

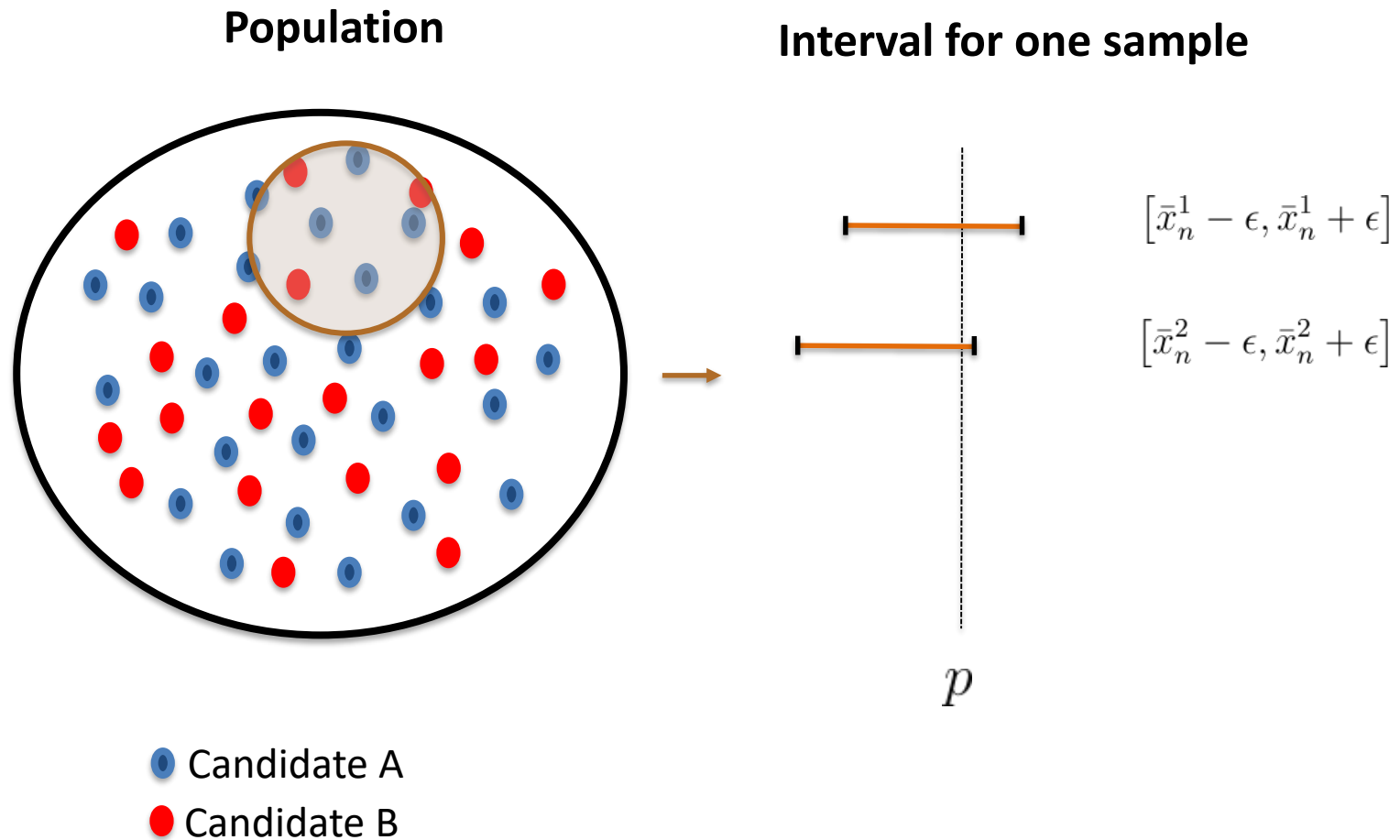
Suppose that we want to find a margin of error $\epsilon > 0$ such that:

$$p \in [\bar{x}_n - \epsilon, \bar{x}_n + \epsilon]$$

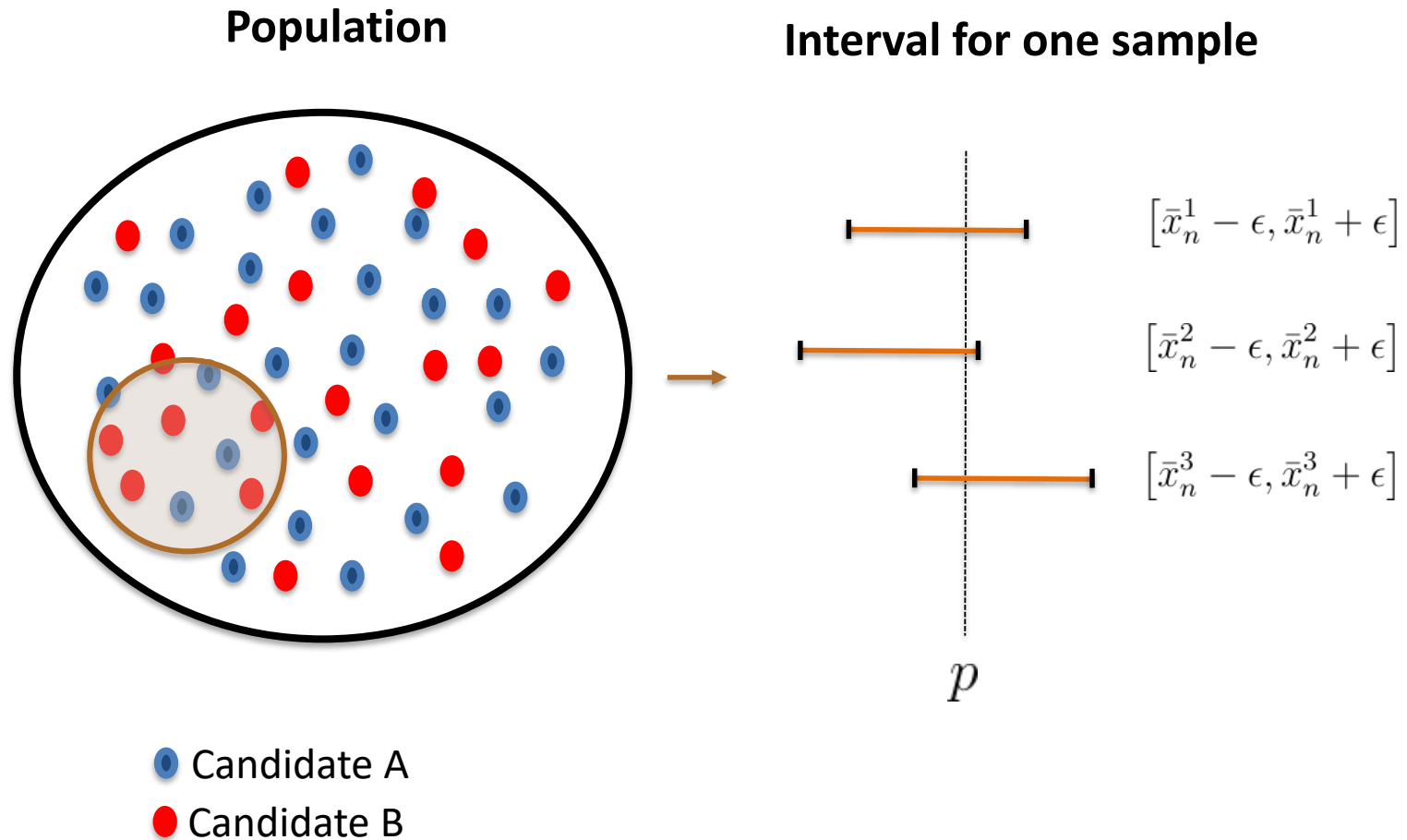
A random interval containing p



A random interval containing p



A random interval containing p



A real situation

- Let the population size be 1.000.000
- Let the real p (that we want to estimate) be 0.5
- We choose $\epsilon = 0.03$ (3%)

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Possible situation

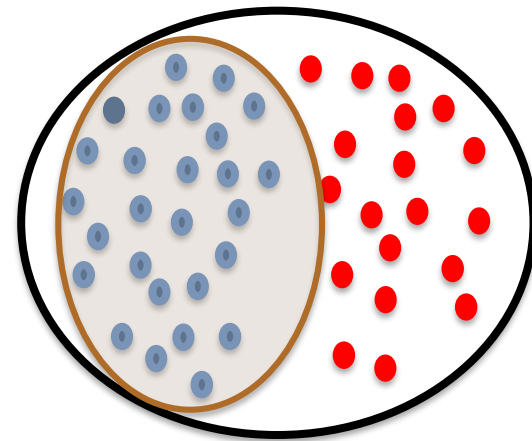
- We interview $n = 500.000$ people (that's a lot!)
- All respondents vote for Candidate A
- Then out outcome is:

$$(x_1, \dots, x_n) = (1, \dots, 1) \rightarrow \bar{x}_n = 1$$

- Then:

$$p \notin [1 - 0.03, 1 + 0.03]$$

Conclusion: Even with a 50% sample,
the margin of error is still large!



● Candidate A
● Candidate B

A probabilistic definition

- Let the population size be 1.000.000
- Let the real p (that we want to estimate) be 0.5
- We choose $\epsilon = 0.03$ (3%)

Change of perspective

Let the margin of error be defined such that, for the chosen $\epsilon > 0$:

$$P\left(p \in \underbrace{[\bar{X}_n - \epsilon, \bar{X}_n + \epsilon]}_{\text{Random confidence interval}}\right) = 0.95 \quad (*)$$

We call 0.95 a *level of confidence*.

(*) reads:

If we repeat the survey 100 times, about 95 of the resulting confidence intervals contain the **real p** .

A probabilistic definition

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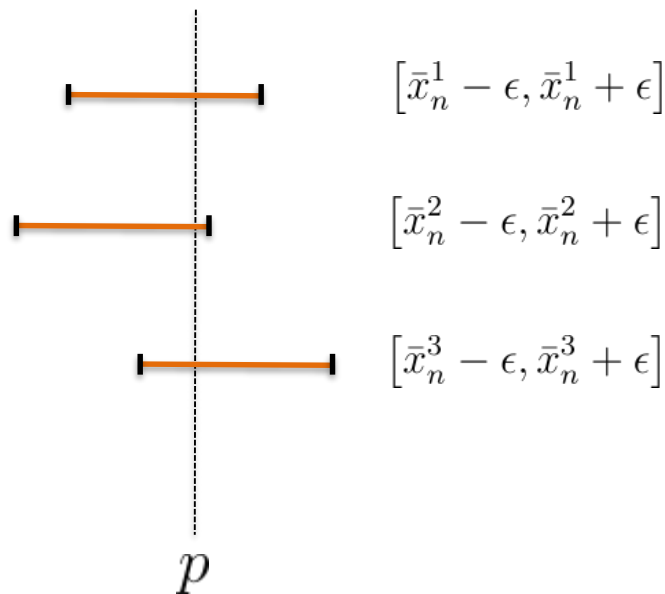
Spoiler of next slides

We will see that, for a confidence level of 95%, we need to interview fewer than 1.000 people in a population of 1.000.000 to have a margin of error $\epsilon = 0.03$ (3%).

Motivation: How reliable?

Goal: Given a **sample size** n and a level of confidence $0 \leq \alpha \leq 1$, find the margin of error $\epsilon > 0$, such that:

$$P(p \in [\bar{X}_n - \epsilon, \bar{X}_n + \epsilon]) = \text{given confidence level} = 1 - \alpha$$

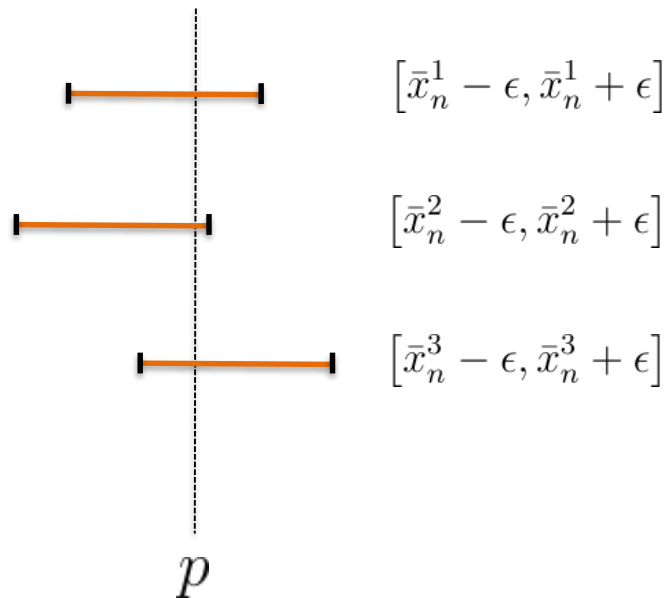


With probability $1 - \alpha$ the interval around the mean estimator that contains the real p

Motivation: How much data?

Goal: Given a **confidence level** $1 - \alpha$, for $0 \leq \alpha \leq 1$ and a margin of error $\epsilon > 0$. Find a sample size n such that:

$$P(p \in [\bar{X}_n - \epsilon, \bar{X}_n + \epsilon]) = \text{given confidence level} = 1 - \alpha$$



With probability $1 - \alpha$ the interval around the mean estimator that contains the real p

Using the symmetry of the normal

Consider $Z \sim N(0, 1)$ a normal random variable. Consider the notation:

$$q_{(1-\frac{\alpha}{2})} := Q\left(1 - \frac{\alpha}{2}\right), \quad P\left(Z \leq q_{(1-\frac{\alpha}{2})}\right) = 1 - \frac{\alpha}{2}$$

Let's compute:

$$P\left(-q_{(1-\frac{\alpha}{2})} \leq Z \leq q_{(1-\frac{\alpha}{2})}\right)$$

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Let's compute:

$$P\left(-q_{(1-\frac{\alpha}{2})} \leq Z \leq q_{(1-\frac{\alpha}{2})}\right)$$

$$P\left(-q_{(1-\frac{\alpha}{2})} \leq Z \leq q_{(1-\frac{\alpha}{2})}\right) = \underbrace{P\left(Z \leq q_{(1-\frac{\alpha}{2})}\right)}_{\text{Simple equality of events}} - \underbrace{P\left(Z \leq -q_{(1-\frac{\alpha}{2})}\right)}_{\text{Definition of } q_{(1-\frac{\alpha}{2})}} = 1 - \frac{\alpha}{2} - P\left(Z \leq -q_{(1-\frac{\alpha}{2})}\right)$$

Simple equality of events

Definition of $q_{(1-\frac{\alpha}{2})}$

$$= \underbrace{1 - \frac{\alpha}{2}}_{\text{Symmetry of normal}} - \underbrace{P\left(Z \geq q_{(1-\frac{\alpha}{2})}\right)}_{\text{Sum of complementary events equals 1}} = 1 - \frac{\alpha}{2} - \left(1 - P\left(Z \leq q_{(1-\frac{\alpha}{2})}\right)\right)$$

Symmetry of normal

Sum of complementary events equals 1

$$= 1 - \frac{\alpha}{2} - \left(1 - \left(1 - \frac{\alpha}{2}\right)\right) = 1 - \alpha$$

Confidence interval

Then, we conclude that for $Z \sim N(0, 1)$:

$$P\left(-q_{(1-\frac{\alpha}{2})} \leq Z \leq q_{(1-\frac{\alpha}{2})}\right) = 1 - \alpha, \text{ for any } 0 \leq \alpha \leq 1$$

Consider X_i identically distributed such that $\mathbb{E}[X_1] = \mu$ and $\text{Var}(X_1) = \sigma^2$. Because of the Central limit theorem, for **n large**:

$$P\left(-q_{(1-\frac{\alpha}{2})} \leq \frac{\sum_{i=1}^n X_i - n\mu}{\sigma\sqrt{n}} \leq q_{(1-\frac{\alpha}{2})}\right) \approx 1 - \alpha, \text{ for any } 0 \leq \alpha \leq 1$$

Rearranging the inequality to isolate μ :

$$P\left(\bar{X}_n - q_{(1-\frac{\alpha}{2})} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X}_n + q_{(1-\frac{\alpha}{2})} \frac{\sigma}{\sqrt{n}}\right) \approx 1 - \alpha, \text{ for any } 0 \leq \alpha \leq 1$$

Or, in the form of interval:

$$P\left(\mu \in \underbrace{\left[\bar{X}_n - q_{(1-\frac{\alpha}{2})} \frac{\sigma}{\sqrt{n}}, \bar{X}_n + q_{(1-\frac{\alpha}{2})} \frac{\sigma}{\sqrt{n}}\right]}_{\text{Random confidence interval}}\right) \approx 1 - \alpha, \text{ for any } 0 \leq \alpha \leq 1$$

Which reads:

Considering all possible outcomes of the random variable $\bar{X}_n$, the probability that the confidence interval contains μ is $1 - \alpha$.

Notation for confidence intervals

Confidence interval

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i, \quad \lim_{n \rightarrow \infty} P \left(\mu \in \underbrace{\left[\bar{X}_n - q_{(1-\frac{\alpha}{2})} \frac{\sigma}{\sqrt{n}}, \bar{X}_n + q_{(1-\frac{\alpha}{2})} \frac{\sigma}{\sqrt{n}} \right]}_{\text{Random confidence interval}} \right) = 1 - \alpha$$

Or for “large” n :

$$P \left(\mu \in \left[\bar{X}_n - q_{(1-\frac{\alpha}{2})} \frac{\sigma}{\sqrt{n}}, \bar{X}_n + q_{(1-\frac{\alpha}{2})} \frac{\sigma}{\sqrt{n}} \right] \right) \approx 1 - \alpha, \text{ for any } 0 \leq \alpha \leq 1 \quad (*)$$

(*) reads:

Considering all possible outcomes of the random variable $\bar{X}_n$, the probability that the confidence interval contains μ is $1 - \alpha$.

Notation

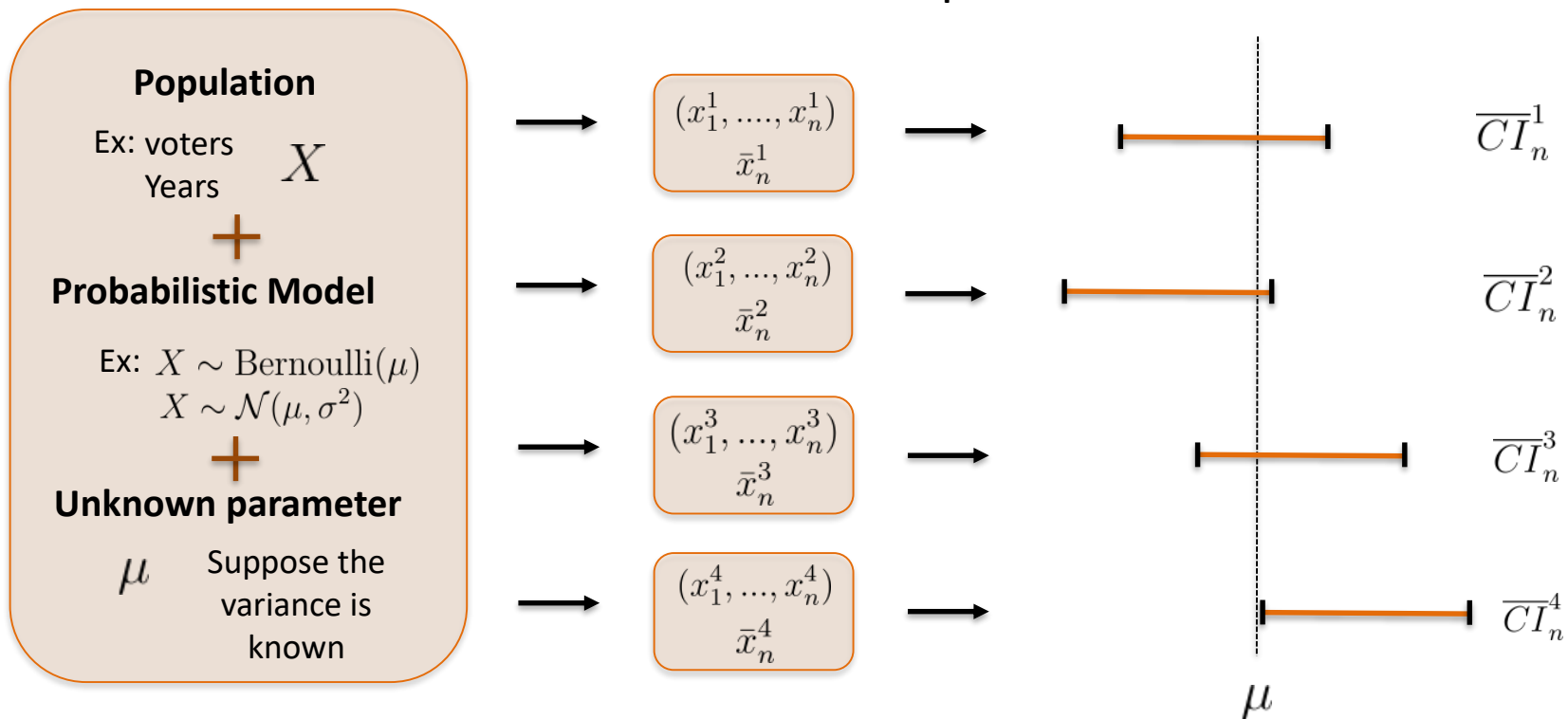
Let $(x_1, \dots, x_n)$ be an outcome of $(X_1, \dots, X_n)$, denote:

$$CI_n = \left[\bar{X}_n - q_{(1-\frac{\alpha}{2})} \frac{\sigma}{\sqrt{n}}, \bar{X}_n + q_{(1-\frac{\alpha}{2})} \frac{\sigma}{\sqrt{n}} \right] \quad \text{and} \quad \overline{CI}_n = \left[\bar{x}_n - q_{(1-\frac{\alpha}{2})} \frac{\sigma}{\sqrt{n}}, \bar{x}_n + q_{(1-\frac{\alpha}{2})} \frac{\sigma}{\sqrt{n}} \right]$$

Inference pipeline

Notation: $\overline{CI}_n = \left[\bar{x}_n - q_{(1-\frac{\alpha}{2})} \frac{\sigma}{\sqrt{n}}, \bar{x}_n + q_{(1-\frac{\alpha}{2})} \frac{\sigma}{\sqrt{n}} \right]$

Possible outcomes with n samples



Example: Calibrated measuring device

We **measure** a fixed physical quantity (for example, **temperature**) using a calibrated instrument.

The measurement error is well understood from prior calibration experiments carried out by the manufacturer. So the noise **variance** $\sigma^2 = 0.16^\circ\text{C}$ is known.

$$X_i = \text{i-th temperature measurement}$$
$$X_i = \mu + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, \sigma^2), \quad i \in \{1, \dots, n\}$$

However, the **true value μ of the temperature being measured is unknown**.

Question: Why is the normal distribution widely used to model errors?

Statistical model

Each measurement X_i , $i \in \{1, \dots, n\}$ are independent normal random variables:

$$X_i \sim \mathcal{N}(\mu, 0.16), \quad i \in \{1, \dots, n\}$$

We want to estimate the true temperature μ

Example: Calibrated measuring device

Statistical model

Each measurement X_i , $i \in \{1, \dots, n\}$ is a normal random variable:

$$X_i \sim \mathcal{N}(\mu, 0.01), \quad i \in \{1, \dots, n\}$$

We want to estimate the true temperature μ

Data

We take $n = 25$ independent temperature measurements and obtain:

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n} = 25^\circ\text{C}$$

Question: What is a 95% confidence interval for the true temperature μ ?

Use that, for the standard normal: $q_{0.975} = 1.96$.

Solution: Calibrated measuring device

$$X_1, \dots, X_n \stackrel{iid}{\sim} \mathcal{N}(\mu, \sigma^2), \quad \sigma = 0.4, \quad n = 25$$

We want a **95% confidence** interval, then: $1 - \alpha = 0.95$, and $\alpha = 0.05$

We have seen that for a standard normal:

$$P\left(-q_{(1-\frac{\alpha}{2})} \leq Z \leq q_{(1-\frac{\alpha}{2})}\right) = 1 - \alpha, \text{ for any } 0 \leq \alpha \leq 1$$

Admitting the approximation (**is it an approximation in this case?**) of the CLT, we have:

$$P\left(-q_{0.975} \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq q_{0.975}\right) = 0.95, \quad q_{0.975} = 1.96.$$

We have seen that this results in a confidence interval:

$$\overline{CI}_n = \left[\bar{x}_n - q_{(1-\frac{\alpha}{2})} \frac{\sigma}{\sqrt{n}}, \bar{x}_n + q_{(1-\frac{\alpha}{2})} \frac{\sigma}{\sqrt{n}} \right]$$

Using our data:

$$\overline{CI}_{25} = \left[25 - \frac{1.96 \times 0.4}{5}, 25 + \frac{1.96 \times 0.4}{5} \right] = [24.84, 25.15]$$

With a probability of 0.95 the true temperature is in the interval $\overline{CI}_{25}$.

Simulation: Calibrated measuring device

Pseudo-code:

```
mu      = 25
sigma   = 0.4
n       = 25
z       = 1.96

#number of simulations
m       = 100

halfWidth = z * sigma / sqrt(n)
CI_left = mu - halfWidth
CI_right = mu + halfWidth

count = 0

for b = 1 to m:
    generate X_1, ..., X_n as Normal(mu, sigma^2)
    Xbar = (1/n) * sum_{i=1}^n X_i

    if CI_left <= Xbar <= CI_right:
        count = count + 1

proportion = count / m
```

Simulation: Calibrated measuring device

Python code:

```
import numpy as np
np.random.seed(8)

# Parameters
mu = 25.0
sigma = 0.4
n = 25
alpha = 0.05
z = 1.96 # z_{0.025}

# Confidence interval that we have computed
half_width = z * sigma / np.sqrt(n)
ci_left = mu - half_width
ci_right = mu + half_width

#number of simulations of barX
m = 100
# Simulation
inside = 0 #start with 0 times barX inside the interval
for i in range(m):
    X = np.random.normal(mu, sigma, size=n)
    barX = np.mean(X)

    if (barX >= ci_left) & (barX <= ci_right):
        inside = inside + 1

prop_inside = inside/m

print("CI:", ci)
print("Half-width:", half_width)
print("Count inside:", inside)
print("Proportion inside:", prop_inside)
```

Result:

```
CI: (24.8432, 25.1568)
Half-width: 0.1568
Count inside: 96
Proportion inside: 0.96
```

Once again, back to the election poll

Election pool

A change in perspective: So far, we have visualized and analyzed some useful measures to understand data. But what if these data represent not an entire population, but only a sample?

Example: Election pool



Candidate	Number of respondents
Candidate A	362
Candidate B	638

Table 2: Election poll results with 1,000 respondents.

Statistical question: What is the probability that candidate B will win the election?

Model for election pool

Probabilistic model

Let p be the **true** (unknown) proportion of people who vote for candidate A, and $1-p$ the proportion who vote for candidate B. Based on the survey, we aim to **estimate** p .

Therefore, we have a Bernoulli random variable with parameter p :

$$X = \begin{cases} 1, & \text{if the person votes for candidate A,} \\ 0, & \text{if the person votes for candidate B.} \end{cases}$$

Data

The survey is viewed as a **realization** or **outcome** $(x_1, x_2, \dots, x_{1000})$ of the random vector $(X_1, X_2, \dots, X_{1000})$, where $X_i \sim \text{Bernoulli}(p), i \in \{1, \dots, 1000\}$

What are we assuming with this model?

- All respondents are independent
- The value of p is fixed during the survey
- Each respondent is chosen at random



Estimating p

Given a result to the election pool: $X_i \sim \text{Bernoulli}(p)$, i.i.d

$(x_1, x_2, \dots, x_n)$ which is an outcome of $(X_1, X_2, \dots, X_n)$
data set

Goal: Find the confidence interval for $n = 1000$ and a confidence level of 90%.

Information: $q_{0.95} = 1.66$

We have seen that:

$$\overline{CI}_n = \left[\bar{x}_n - q_{(1-\frac{\alpha}{2})} \frac{\sigma}{\sqrt{n}}, \bar{x}_n + q_{(1-\frac{\alpha}{2})} \frac{\sigma}{\sqrt{n}} \right]$$

Problem: The value of σ is not known and depends on p !

Confidence interval for p

Given a result to the election pool: $X_i \sim \text{Bernoulli}(p)$, i.i.d

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Problem: The value of σ is not known and depends on p !

Solution: Replace σ^2 by a variance estimator:

$$\underbrace{\sigma^2 \approx \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2}_{\text{Variance estimator}} \longrightarrow \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x}_n)^2$$

Outcome from the survey

Central limit theorem + Slutsky

The theorem:

Let $X_1, X_2, \dots$ be independent and identically distributed random variables with mean $\mu := \mathbb{E}[X_1]$ and variance estimator:

$$\hat{\sigma}_n^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 \longrightarrow \text{empirical variance}$$

Define the standardized sum:

$$Z_n^* = \frac{\sum_{i=1}^n X_i - n\mu}{\hat{\sigma}_n \sqrt{n}}$$

Then, as $n \rightarrow \infty$, the distribution of Z_n converges to the standard normal distribution:

$$Z_n^* \xrightarrow{d} \mathcal{N}(0, 1)$$

Consequence:

Let:

$$CI_n^* = \left[\bar{X}_n - q_{(1-\frac{\alpha}{2})} \frac{\hat{\sigma}_n}{\sqrt{n}}, \bar{X}_n + q_{(1-\frac{\alpha}{2})} \frac{\hat{\sigma}_n}{\sqrt{n}} \right]$$

Then:

$$\lim_{n \rightarrow \infty} P(CI_n^*) = 1 - \alpha, \text{ and for large values of } n: P(CI_n^*) \approx 1 - \alpha$$

Last time, back to the election poll

Confidence interval for p

Given a result to the election pool: $X_i \sim \text{Bernoulli}(p)$, i.i.d

$(x_1, x_2, \dots, x_n)$ which is an outcome of $(X_1, X_2, \dots, X_n)$
data set

Goal: Find the confidence interval for $n = 1000$ and a confidence level of 90%.
($q_{0.95} = 1.66$).

We have seen that:

$$\overline{CI}_n = \left[\bar{x}_n - q_{(1-\frac{\alpha}{2})} \frac{\sigma}{\sqrt{n}}, \bar{x}_n + q_{(1-\frac{\alpha}{2})} \frac{\sigma}{\sqrt{n}} \right]$$

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Outcome from the survey

Confidence interval for p

Goal: Find the confidence interval for $n = 1000$ and a confidence level of 90%.

$$q_{0.95} = 1.66$$

$$CI_n^* = \left[\bar{X}_n - q_{(1-\frac{\alpha}{2})} \frac{\hat{\sigma}_n}{\sqrt{n}}, \bar{X}_n + q_{(1-\frac{\alpha}{2})} \frac{\hat{\sigma}_n}{\sqrt{n}} \right]$$

- Confidence level: $1 - \alpha = 0.9, \quad \alpha = 0.1$
- Quantile: $1 - \frac{\alpha}{2} = 0.95, \quad q_{0.95} = 1.66$
- Mean estimator: $\bar{X}_n = \frac{362}{1000} = 0.362$
- Variance estimator: $\sigma^2 \approx \frac{1}{999} ((1 - 0.362)^2 \times 362 + (-0.362)^2 \times 638) = 0.231$
 $\Rightarrow \sigma \approx 0.48$
- Confidence interval: $\left[0.362 - 1.66 \frac{0.48}{\sqrt{1000}}, 0.362 + 1.66 \frac{0.48}{\sqrt{1000}} \right] = [0.33, 0.38]$

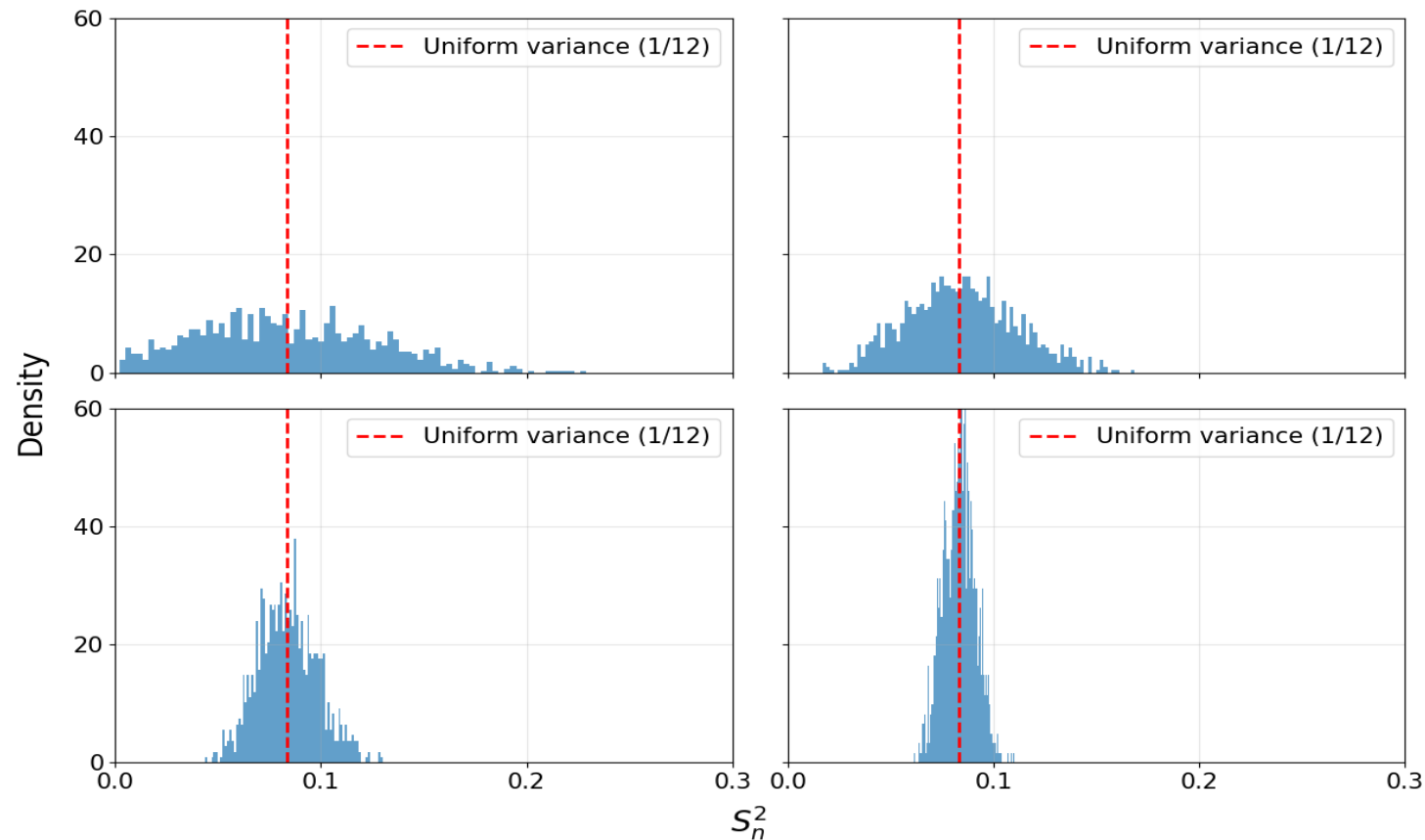

Confidence interval

Beyond the mean

Variance estimator

Example: $S_{100}^2 = \frac{1}{99} \sum_{i=1}^{100} (X_i - \bar{X})^2$, $\bar{X} = \frac{1}{n} \sum_{i=1}^{100} X_i$, for $X \sim \mathcal{U}(0, 1)$.

Observe the empirical distribution with a sample size of 1000:



Variance estimator

Example: $S_{100}^2 = \frac{1}{99} \sum_{i=1}^{100} (X_i - \bar{X})^2$, where: $\bar{X} = \frac{1}{100} \sum_{i=1}^{100} X_i$, for $X \sim \mathcal{U}(0, 1)$.

Discussion: Why the standard definition of this estimator has denominator $n-1$ instead of n ?

Variance estimator

Example: $S_{100}^2 = \frac{1}{99} \sum_{i=1}^{100} (X_i - \bar{X})^2$, where: $\bar{X} = \frac{1}{100} \sum_{i=1}^{100} X_i$, for $X \sim \mathcal{U}(0, 1)$.

Discussion: Why the standard definition of this estimator has denominator $n-1$ instead of n ?

$$\begin{aligned} \sum_{i=1}^n (X_i - \bar{X})^2 &= \sum_{i=1}^n (X_i - \mu)^2 - n(\bar{X} - \mu)^2 \\ \mathbb{E} \left[\sum_{i=1}^n (X_i - \bar{X})^2 \right] &= \sum_{i=1}^n \mathbb{E}[(X_i - \mu)^2] - n \underbrace{\mathbb{E}[(\bar{X} - \mu)^2]}_{\text{Variance of the mean}} \\ &= n\sigma^2 - n \cdot \frac{\sigma^2}{n} \\ &= (n-1)\sigma^2 \end{aligned}$$

Thank you for your attention and
participation!